Introduction

Anandha Gopalan
(with thanks to Teo Yong Meng)

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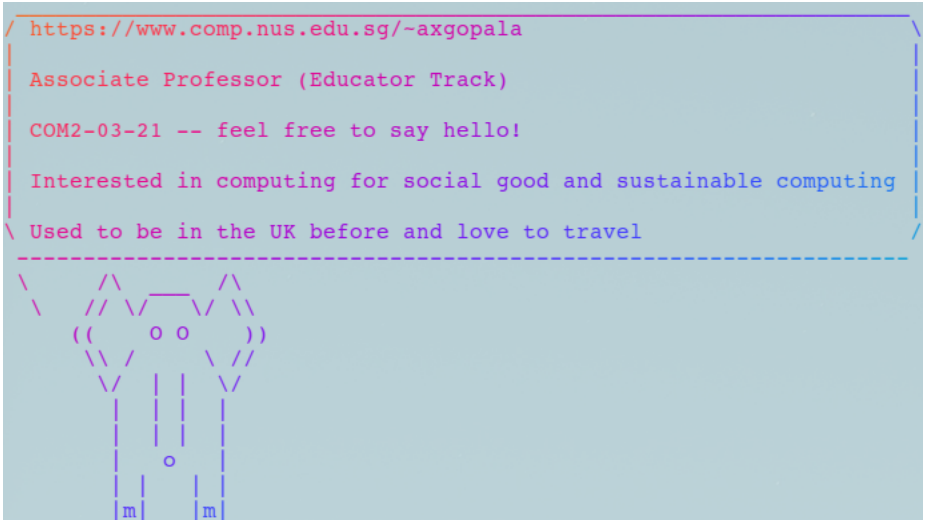
Who am I?


https://www.comp.nus.edu.sg/~axgopala
Associate Professor (Educator Track)
COM2-03-21 -- feel free to say hello!
Interested in computing for social good and sustainable computing
Used to be in the UK before and love to travel

Course Information

Canvas will be used as the LMS

- Lecture slides (available at the start of the week)
- Quizzes
- Assessments
- Link to webcast of lectures available on Panopto

EdStem is preferred for Q & A

- Join using: `https://edstem.org/us/join/CfQZdu`
- Allows creation of threads anonymously, though not anonymous to the teaching staff
- Allows creation of private threads – visible only to teaching staff

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Course TAs

Name: Tanuj Sur
PhD student

Name: Fu Yihao
PhD student

Name: Zeng Lingze
PhD student

Name: Wu Bo-Cheng
PhD student

Name: Shivansh Srivastava
MComp student

Teaching Style

Interactive, so please do engage ☺

Feel free to ask any questions ☺

No separate tutorial or lab sessions

You learn by doing in this course

Lectures provide the theory, while labs and group project help augment your learning

Topic	Lecture hours
Principles of Cloud Computing	6
Technologies behind Cloud Computing	4
Applications and Programming	8
Cloud Management	2

Detailed schedule available on *Canvas*

Assessments

Assessment	Weightage
Quizzes	10%
Labs	20%
Group Project	35%
Test	35%

Test

- Held on 03/11/26 (in lecture slot)
- 90 minutes
- **Open book** $\Rightarrow$ no access to Internet
- No make-up scheduled

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Quizzes

Ensure you keep up with the module

Review key concepts

Each quiz has 10 questions in total

Each quiz takes best out of 3 attempts

Four quizzes in total

Labs

To reinforce what you have learnt in a practical manner

Two labs in total

Knowledge gained helps towards the group project

Can use **any** cloud platform $\rightarrow$ available free tier should be more than enough

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Group Project

Develop a SaaS product in an area of interest ⇒ working prototype is **required**

Teams of 5 or 6

- **Recommendation**: Have team members from different cohorts

Activity	**Deadline**
Team formation	07/09/26
Preliminary report	28/09/26
Final submissions	13/11/26

Can use **any** cloud platform → available free tier should be more than enough

Intra-group – everyone needs to contribute

Inter-group – you are **not** to share your work with another group, even if you have discussions

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SoC Term Project Showcase

Great event to showcase your group project to industry guests – prizes also available ☺

Students have won prizes in the last few semesters ☺

Date: **11/11/26** in the **afternoon**

Last semester's event: `https://uvents.nus.edu.sg/event/28th-steps`

More details will follow after the recess week

AI and Plagiarism I

Use of AI tools **is allowed** $\Rightarrow$ ensure you follow the college guidelines on this and appropriately cite its use

Zero-tolerance for plagiarism

- Students will be reported to the university for disciplinary action for plagiarism/cheating offence
- Resources
  - `https://www.comp.nus.edu.sg/cug/plagiarism/`
  - `https://www.nus.edu.sg/celc/statements-and-e-resources-on-plagiarism/`

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AI and Plagiarism II

https://www.comp.nus.edu.sg/cug/plagiarism/

> **Plagiarism is generally defined as the practice of taking someone else's work or ideas and passing them off as one's own** *(The New Oxford Dictionary of English).*
>
> *All students share the responsibility of protecting the academic standards and reputation of the University. This responsibility can extend beyond each student's own conduct, and can include reporting incidents of suspected academic dishonesty through the appropriate channels. Students who have reasonable grounds to suspect academic dishonesty should raise their concerns directly to the relevant Head of Department, Dean of Faculty, Registrar, Vice Provost or Provost. The University does not condone plagiarism.*

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AI and Plagiarism III

https://www.nus.edu.sg/celc/statements-and-e-resources-on-plagiarism/

> *Students should adopt this rule* – **You have the obligation to make clear to the assessor which is your own work, and which is the work of others.** *Otherwise, your assessor is entitled to assume that everything being presented for assessment is being presented as entirely your own work. This is a minimum standard.*
>
> *A student may not knowingly intend to plagiarise, but that should not be used as an excuse for plagiarism.* **Students should seek clarification from their instructors or supervisors if they are unsure whether or not they are plagiarising the work of another person.**

Resources

Lots of free online resources here:
`https://github.com/Ismael-Codes/Free-Books-Cloud`

Cloud Computing: Concepts, Technology & Architecture, Thomas Erl, et al., Prentice-Hall, 2013, [chapters 3, 4, 5, 11, 15, 16]

The Datacenter as a Computer – Designing Warehouse-Scale Machines, 3^{rd} Edition, Morgan & Claypool Publishers, 2019 (available online) [chapters 1, 2, 3, 4, 6]

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Previously on CS5224 . . .

Test in week 13 with a lot of other course deadlines and project deadlines

- Was primarily because one class was missed due to a public holiday
- Now in week 12 – week 13 is free and for project consultations (if needed)

More hands on activities requested

- Access to AWS Academy courses from week 2 onwards (AWS Academy Cloud Foundations, AWS Academy Cloud Developing, AWS Academy Cloud Operations)
- Lots of resources from CS3219 for students to explore (SE toolbox)

Project has a business focus

- Moving to the cloud requires a good business case
- Project brief will make it more clear about this reasoning

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What is Cloud Computing?

Gartner Report (`https://www.gartner.com/en/infrastructure-and-it-operations-leaders/topics/cloud-computing`)

- "... a style of computing in which scalable and elastic IT-enabled capabilities are delivered as a service using Internet technologies."

NIST 2011 (`https://csrc.nist.gov/pubs/sp/800/145/final`)

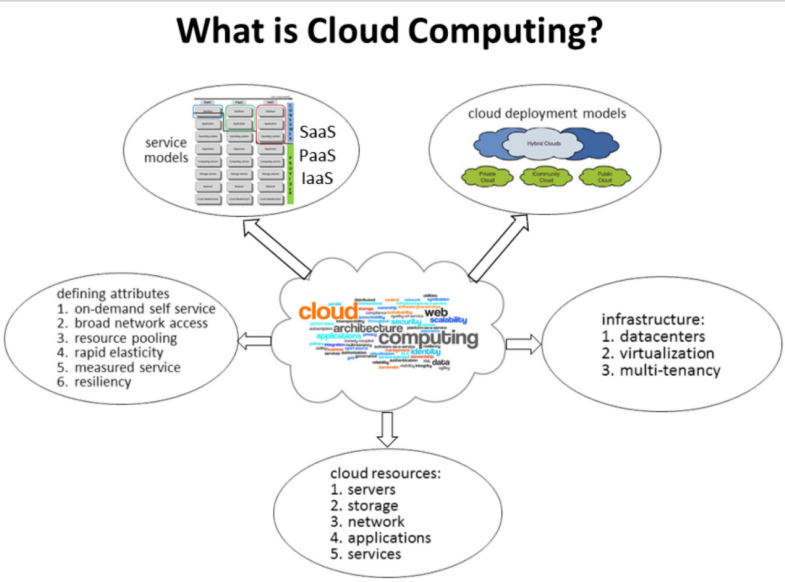
- "Cloud computing is a model for enabling *ubiquitous*, *convenient*, on-demand *network access* to a shared pool of configurable computing resources (e.g., *networks*, *servers*, *storage*, *applications*, *and services*) that can be rapidly provisioned and released with minimal management effort **or service provider interaction**."

on-demand service + elastic resource

Program is an Internet (cloud) **service**

and

platform are datacenters


What is Cloud Computing?
service models
SaaS
PaaS
IaaS
cloud deployment models
Hybrid Clouds
Private Cloud
Community Cloud
Public Cloud
defining attributes
1. on-demand self service
2. broad network access
3. resource pooling
4. rapid elasticity
5. measured service
6. resiliency
cloud
computing
architecture
web
scalability
security
infrastructure:
1. datacenters
2. virtualization
3. multi-tenancy
cloud resources:
1. servers
2. storage
3. network
4. applications
5. services

History I

1961 – "computing may someday be organized as a *public utility*", John McCarthy (1927 – 2011)

1996 – "cloud computing" was coined at Compaq Computer (MIT Technology Review)

1999 – **Salesforce.com** pioneered the concept of delivering enterprise applications via a simple website

Jul 2002 – **Amazon Web Services (AWS)** provided a suite of cloud-based services including computation, storage, and even human intelligence through *Amazon Mechanical Turk*

May 2006 – **Amazon Simple Storage Service (S3)**, a "pay-per-use" storage

History II

Aug 2006 – **Amazon Elastic Compute Cloud (EC2)**, a commercial web service (IaaS) that allows small companies and individuals to rent computers to run their own applications

Apr 2008 – **Google App Engine**, Google's PaaS, Bigtable and GFS for storage, MapReduce, etc.

Nov 2009 – **Microsoft Windows Azure**, an operating environment "designed to manage extremely large pools of computational resources". Customers run Windows-based applications over the Internet using Microsoft's data centers, while Azure organizes resources and handles spikes in demand

2011 onwards – many many many cloud providers!

Why Cloud Computing?

Technology (cloud-enabled platforms and services) for current/future innovations and disruptions

Next wave of cloud disruption delivers advanced capability around AI, IoT, among others!

Reduces business cost

- Improves match between elastic resource demand and elastic computing resource

Improves **availability** and **elasticity**

Reduces cost

However . . .

What caused the AWS outage - and why did it make the internet fall apart?

21 October 2025

Share Save

Zoe Kleinman

Technology editor

https://www.bbc.com/news/articles/cev1en9077ro

Cloudflare apologises for outage which took down X and ChatGPT

18 November 2025

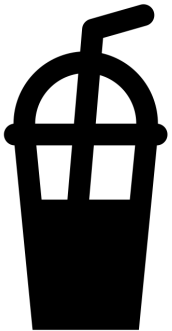
Share Save

Liv McMahon

Technology reporter

https://www.bbc.com/news/articles/c629pny4gl7o

Take a Break



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To help deliver **computing as-a-service**

- Parallel and Distributed Computing
- Programming Models
- Utility Computing
- Virtualization
- Web Technologies
- Storage and Network Technologies
- Internet

Key Terms

1. Elastic resource
2. Availability
3. Capacity planning (resource provisioning)
4. Scaling (horizontal, vertical)
5. Cloud-based IT resources
6. Cloud service
7. Trust boundary

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Key Business Drivers!

Capacity planning

Cost reduction

Organisation agility

3. Capacity Planning

Process of determining and fulfilling future demands of an organisation's IT resources, products and services

Challenges: usage/demand fluctuations, peak usage, cost of resource provisioning, ...

Strategies

- Lead strategy: add capacity in anticipation of demand
- Lag strategy: add capacity when resources reach its full capacity
- Match strategy: add capacity in small increments as demand increases

Should ideally avoid under-provisioning and over-provisioning

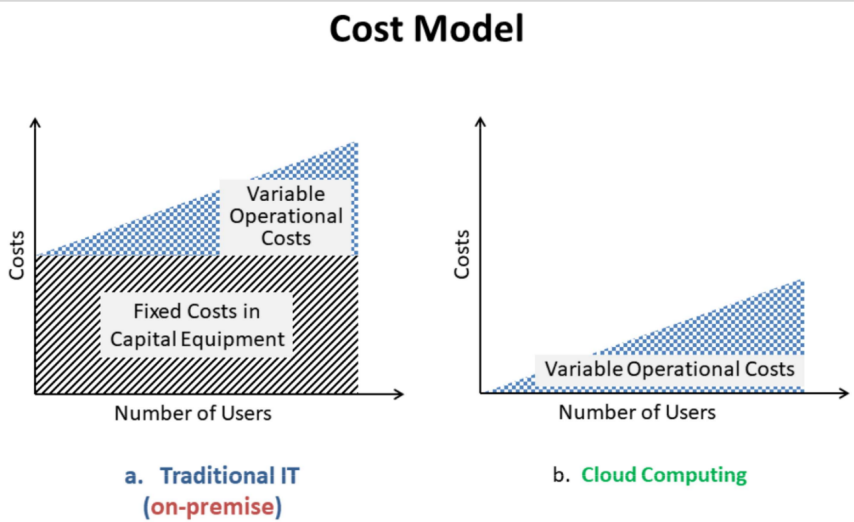
Cost Reduction

Difficult to directly align IT costs and business performance

On-premise system – IT department is a **big cost**

- Up-front investment costs
- Operational costs: technical staff (manpower), utility bills (power and cooling), security and access control to protect infrastructure, admin/account cost (software licences, etc.)

Cloud computing typically offers cost efficiency at scale


Cost Model
Costs
Variable Operational Costs
Fixed Costs in Capital Equipment
Number of Users
a. Traditional IT (on-premise)
Costs
Variable Operational Costs
Number of Users
b. Cloud Computing

Hwang, P199

Organisational Agility

A measure of an organisation's responsiveness to change

On-premise: Up-front investment and infrastructure ownership costs may be prohibitive

Cloud

- Elastic IT resources to respond to business cycles beyond what was previously predicted or planned
- Software fixes/updates: Update datacenters vs millions of clients
- Datacenter allows faster introduction of new hardware innovations

Key Terms

1. Elastic resource
2. Availability
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5. Scaling (horizontal, vertical)
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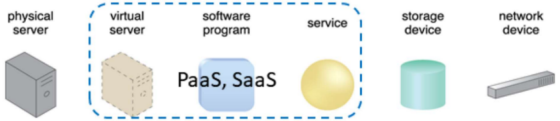
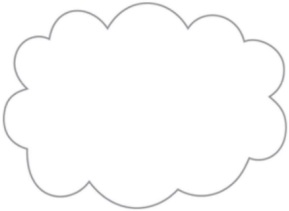
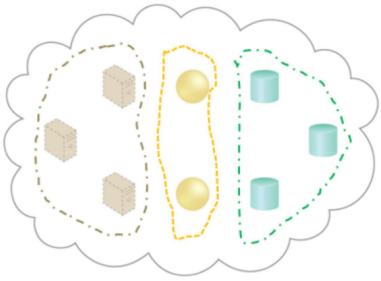
More Concepts and Terminology



Examples of common Cloud-based IT resources and their corresponding symbols



Symbol denotes the Boundary of a Cloud Environment



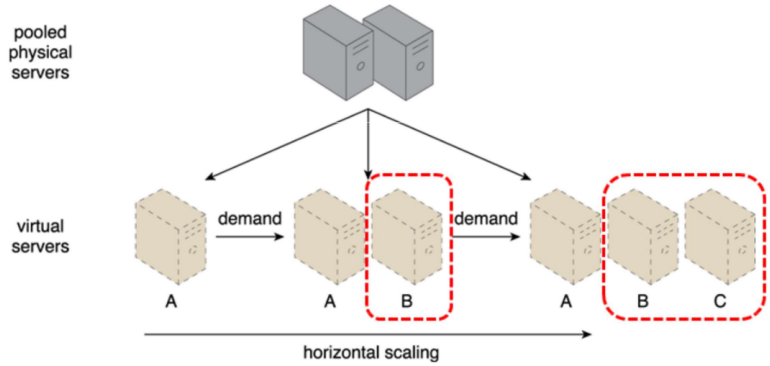
Example: A cloud hosting 8 IT resources: 3 virtual servers, 2 **cloud services**, and 3 storage devices

Ability of the IT resource to handle increased or decreased usage demands

Horizontal Scaling (out or in) – add same resource type

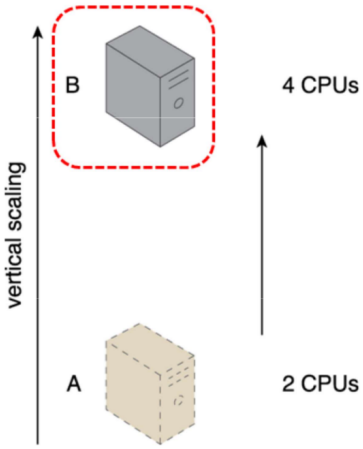
Vertical Scaling (up or down) – replace resource with higher or lower capacity or add resources to a single node

Horizontal Scaling



An IT resource (Virtual Server A) is **scaled out** by adding more of the same resource (Virtual Servers B and C)

Vertical Scaling



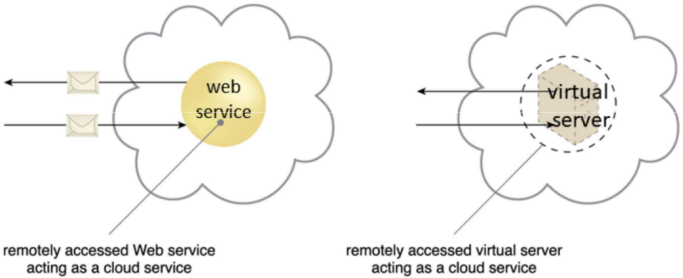
An IT resource (Virtual Server A) with 2 CPUs is **scaled up** by replacing it with a more powerful resource (Server B with 4 CPUs)

Scaling Comparison

	Horizontal	Vertical
Cost	Less expensive using commodity hardware	More expensive using specialized hardware
Availability	Resources instantly available	Resources normally instantly available
Ease of setup	Resource replication and automated scaling	May need additional setup
Hardware capacity	Not limited by hardware capacity	Limited by maximum hardware capacity

6. Cloud Service

Any IT resource made remotely accessible via a cloud



Multitude of service usage models – next lecture

- IaaS (Infrastructure-as-a-Service)
- PaaS (Platform-as-a-Service)
- SaaS (Software-as-a-Service)

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Technical Challenges

Software Development – different cloud platforms and services across cloud providers

Tools are continuously evolving

Moving large data is still expensive

Security

Internet dependence

Quality of service

Energy costs

...

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Non-Technical Challenges

Increased security vulnerabilities

Reduced operational governance control

Privacy/legal issues: data localization, multi-regional compliances

Vendor lock-in

Service level agreement

...

Increased Security Vulnerabilities

Responsibility over data security becomes shared with cloud providers

Expansion of trust boundary introduces new vulnerabilities

Shared IT resources across different cloud consumers introduce **overlapping trust boundaries** – opportunities for malicious cloud consumers to steal or damage data

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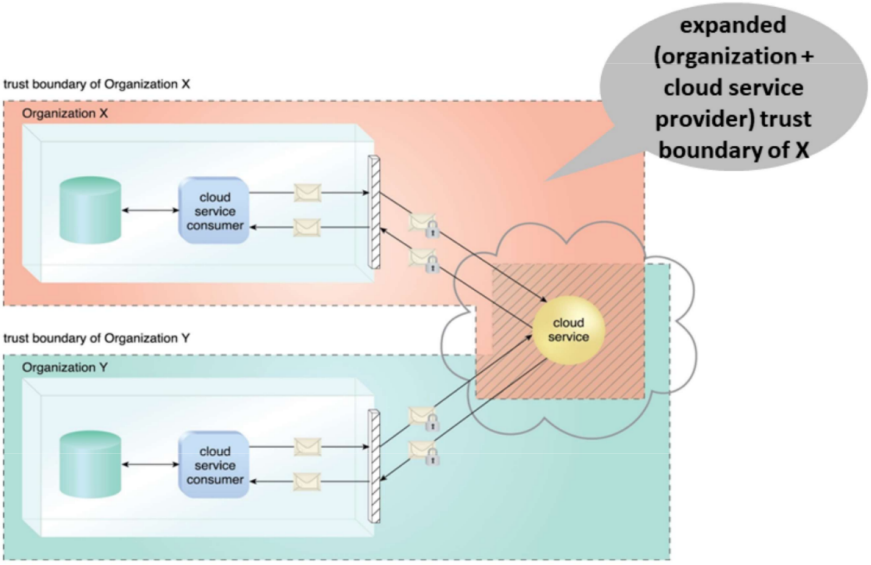
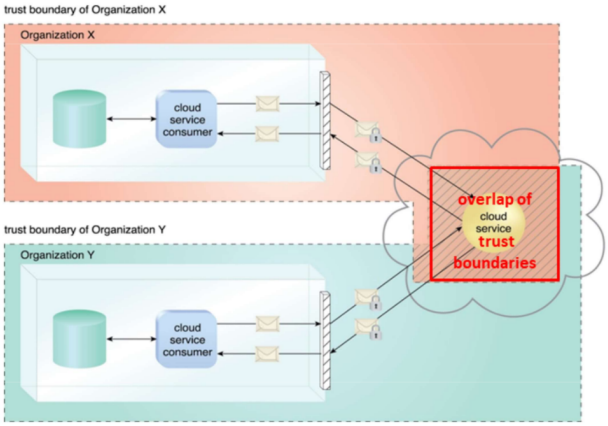
Expanded Trust Boundaries



Overlapping Trust Boundaries



Shaded area with diagonal lines indicates the overlap of trust boundaries between Organisation X and Organisation Y

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Reduced Operational Governance Control

For cloud consumers – different levels of IT resource governance between on-premise and cloud

Unreliable cloud provider may not maintain/meet SLA guarantees

Physical distance between provider and consumer introduces additional latency and bandwidth constraints

Privacy/Legal Issues

Data centers usually set up in affordable/convenient locations

Data Localization/Residency – Industry or government regulations on data privacy or storage policies (very important!)

- E.g. Data belonging to Singapore citizens to be kept within Singapore

Legal issues on accessibility and disclosure of data

- E.g., laws require data to be disclosed appropriately

Tension between personal rights (privacy) and society

- E.g., location, proximity data, etc.

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Note of Caution

The Rise Of Cloud Repatriation: Why Companies Are Bringing Data In-House

Forbes | Technology Council

Marcin Zgola Forbes Councils Member

Forbes Technology Council COUNCIL POST | Membership (Fee-Based)

https://www.forbes.com/councils/forbestechcouncil/2023/04/18/the-rise-of-cloud-repatriation-why-companies-are-bringing-data-in-house/

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Next Week

Things get serious!

Concepts and Models

References

Chapter 3, Cloud Computing: Concepts, Technology and Architecture, Thomas Erl, Zaigham Mahmood and Ricardo Puttini, Prentice-Hall, 2013.

Chapter 1, The Datacenter as a Computer – Designing Warehouse-Scale Machines, 3^{rd} Edition, Morgan and Claypool Publishers, 2019 (available online)

Above the Clouds: A Berkeley View of Cloud Computing, 2009

The NIST Definition of Cloud Computing, NIST Report, 2011

Continuous Feedback

https://forms.cloud.microsoft/r/ZJxTf2wFjr

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Concepts and Models

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(with thanks to Teo Yong Meng)

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Outline

Key Components

Cloud Service Models

Conceptual Reference Architecture

Cloud Deployment Models

Key Components – NIST 2011 I

Five Essential Characteristics

Four Deployment Models

Three Service Models

Key Components – NIST 2011 II

Essential Characteristics

1. On-demand Self-service – via a service portal
2. Broad Network Access – ubiquitous access
3. Resource Pooling – location independent, multi-tenancy ⇐
4. Rapid Elasticity – time to market/fast deployment
5. Measured Service – pay-per-use

Key Components – NIST 2011 III

Deployment Models

1. Public Cloud
2. Private Cloud
3. Community Cloud
4. Hybrid Cloud

Service Models

1. Software-as-a-Service (SaaS)
2. Platform-as-a-Service (PaaS)
3. Infrastructure-as-a-Service (IaaS)

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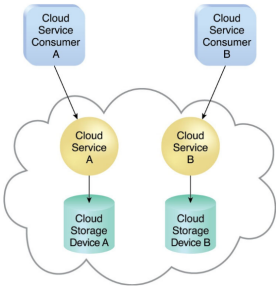
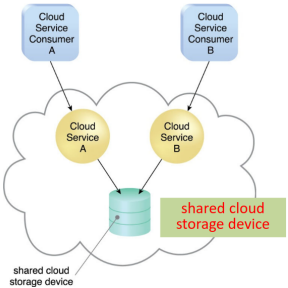


Single Tenant

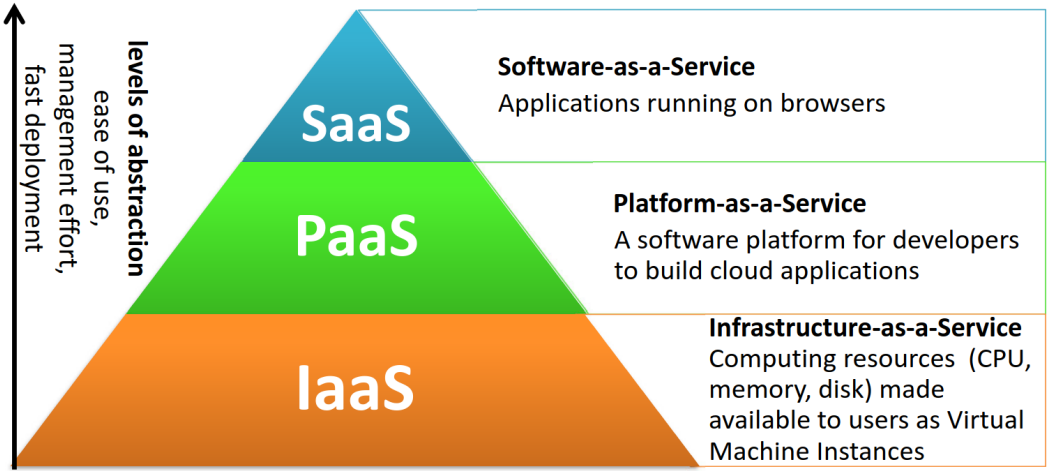
- Each cloud consumer has a **separate** IT resource instance



Multi-Tenancy

- Cloud provider pools its IT resources to serve multiple cloud service consumers
- **An instance** of the program serves multiple cloud consumers (tenants)

Cloud Service Models



Cloud Service Models

Software-as-a-Service (SaaS)

Use provider's applications over a network
E.g.: Salesforce.com, Google apps, Microsoft 365, ...

Platform-as-a-Service (PaaS)

Deploy customer-related applications to a cloud
E.g.: Google's app engine, Heroku, Microsoft Azure App Service, ...

Infrastructure-as-a-Service (IaaS)

Rent processing, storage, network capacity, etc.
E.g.: Amazon EC2, DigitalOcean, Linode, Alibaba ECS, ...

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Cloud Service Models

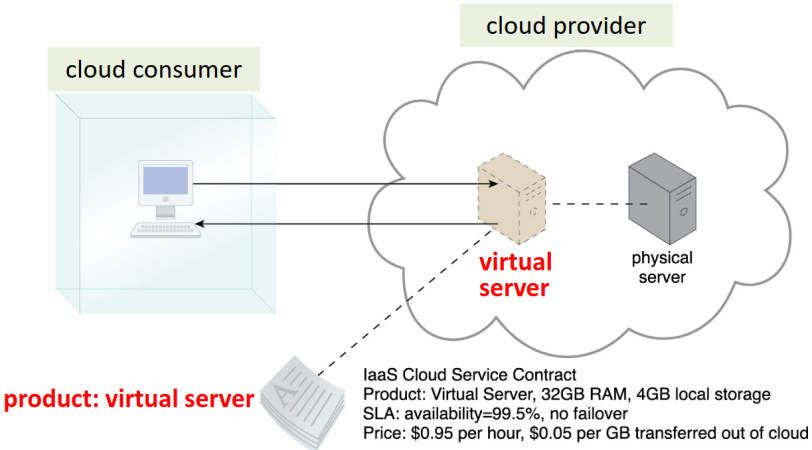
Why different models?

Different levels of abstraction and management – use depending on need!

Infrastructure-as-a-Service (IaaS)

Lowest level of abstraction

Physical resources are abstracted as virtual resources



Servers

Storage

Networks

Applications & Middleware

...

Platform-as-a-Service (PaaS)

Two parts: Platform Software, and Computing Resources needed to run the platform software

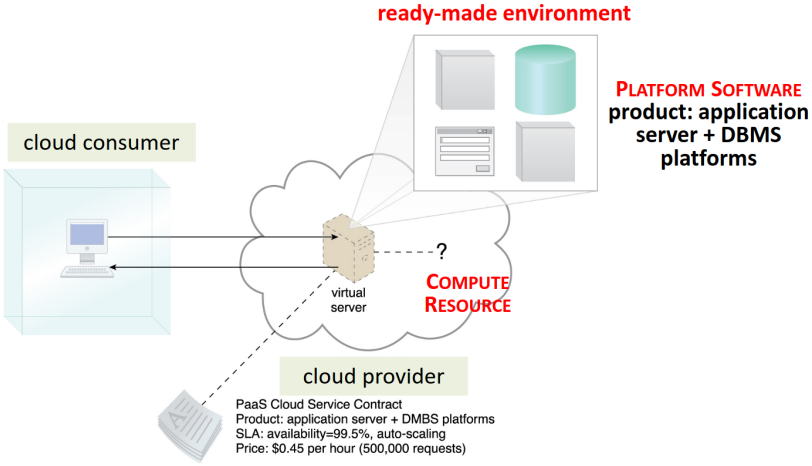
PaaS Provider

Develops the platform software and decides where to host/run the PaaS

PaaS Consumer

Incorporates the ready-made PaaS into the application being developed

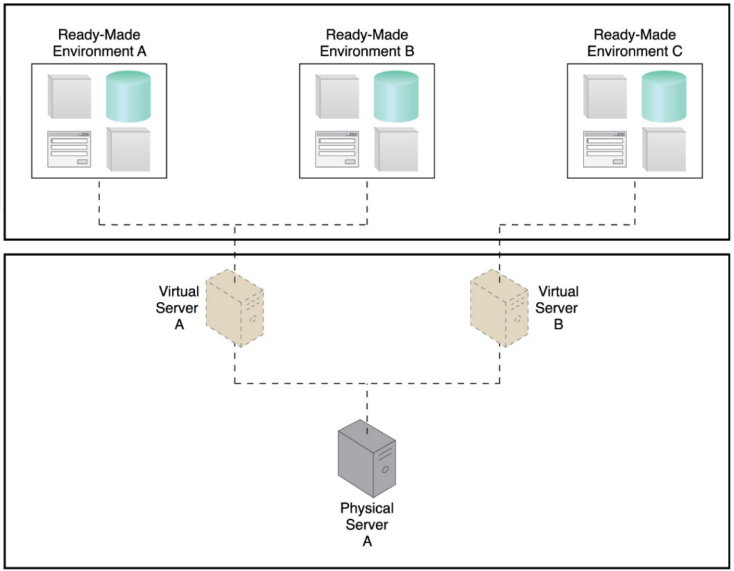
Platform-as-a-Service (PaaS)



Combining IaaS and PaaS

PaaS platform software

IaaS compute resource runs platform software



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Example: Combining IaaS and PaaS

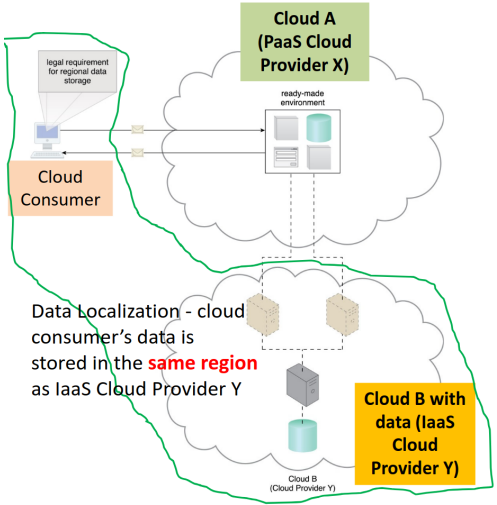
Meeting Legal Requirements for Data Storage

Cloud consumer subscribes to PaaS offered by provider **X (Cloud A)**

To meet requirement

Services offered by provider **X** are physically hosted on provider **Y (Cloud B)**

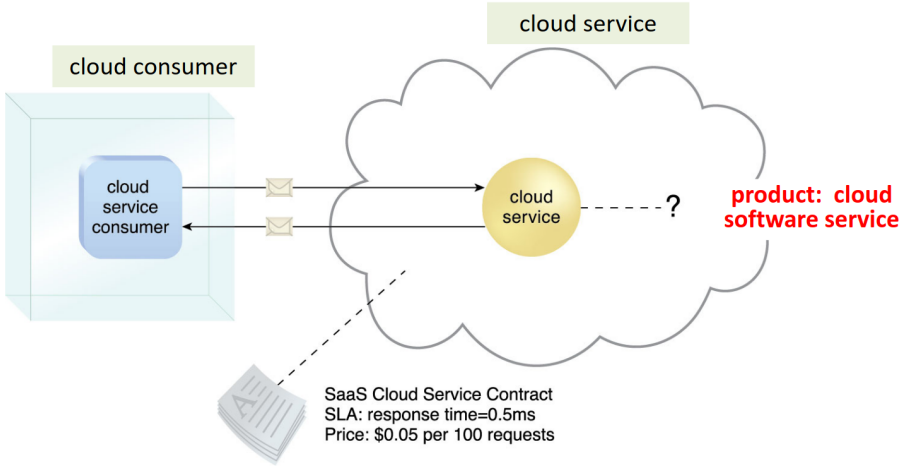
⇒ **Cloud B** is physically located in the same region as the cloud consumer



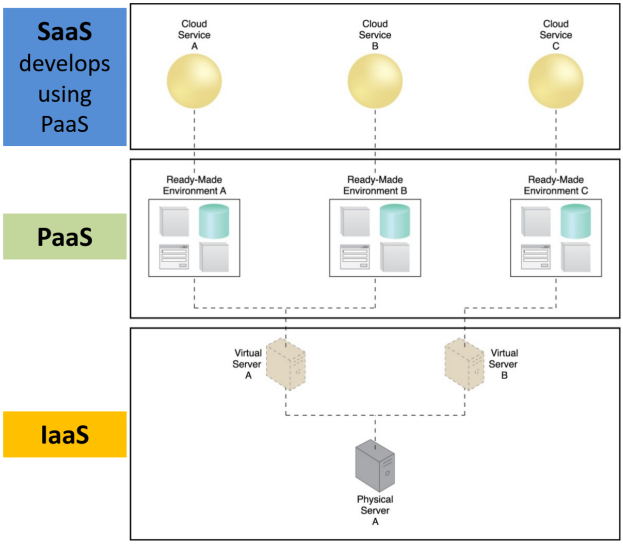
Software-as-a-Service (SaaS)

SaaS provider – installs and maintains software

SaaS consumer – remotely runs the software on the cloud



Combining IaaS, PaaS, and SaaS



Variants of Service Models

Storage-as-a-Service

Database-as-a-Service

Security-as-a-Service

Communication-as-a-Service

Integration-as-a-Service

Testing–as-a-Service

Function-as-a-Service – serverless computing

...

Everything-as-a-Service

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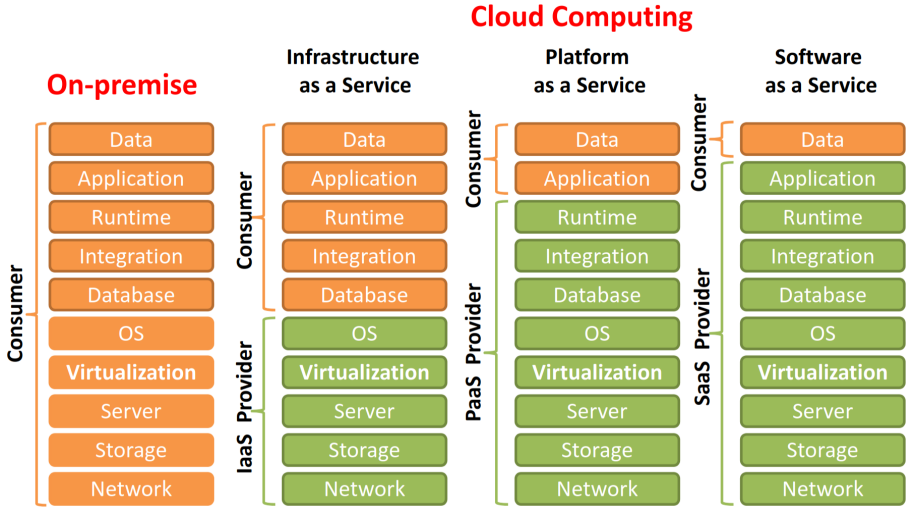
Comparing Service Models

Responsibility of Consumer vs Provider



Service Models	Pros	Cons
On-premise	Highest flexibility	**Long** time-to-market
IaaS	Scalability, No H/W procurement	Privacy
PaaS	DB, Frameworks, Middleware ready	Vendor lock-in
SaaS	**Reduce** time-to-market	Lowest flexibility

Reference Model – NIST 2011

Focuses on the what (cloud services provide) and not the how (design solution and implementation)

Does not represent the system architecture of a specific cloud

Provides a common reference framework to describe, discuss and develop a system-specific architecture

Vendor and Implementation Independent

https://doi.org/10.6028/NIST.SP.500-292

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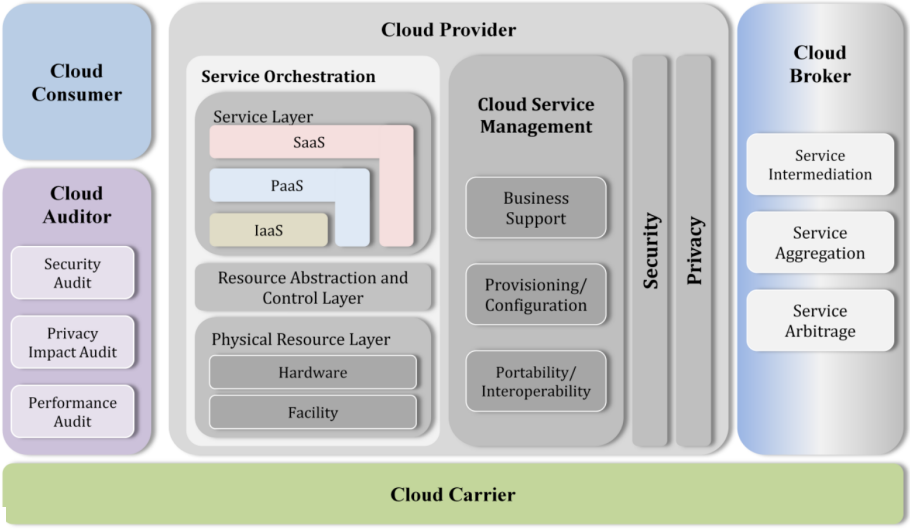
Cloud Computing Reference Architecture



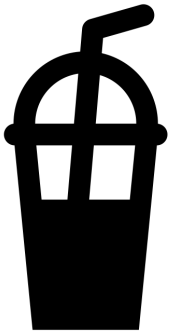
Actor

Entity (person/organisation) that participates in a transaction or process and/or performs tasks in cloud computing

Five major actors

1. **Cloud Consumer**
2. **Cloud Provider**
3. **Cloud Auditor**
4. **Cloud Broker**
5. **Cloud Carrier**

Take a Break



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Actor Roles

1. **Cloud Consumer** – maintains a business relationship with, and uses services from, **Cloud Providers**
2. **Cloud Provider** – offers a cloud service to **Cloud Consumers**
3. **Cloud Auditor** – conducts independent assessment of cloud services, system operations, performance and security of the cloud implementation
4. **Cloud Broker** – manages the use, performance and delivery of cloud services, and negotiates the relationships between **Cloud Providers** and **Cloud Consumers**
5. **Cloud Carrier** – provides connectivity and transport of cloud services from **Cloud Providers** to **Cloud Consumers**

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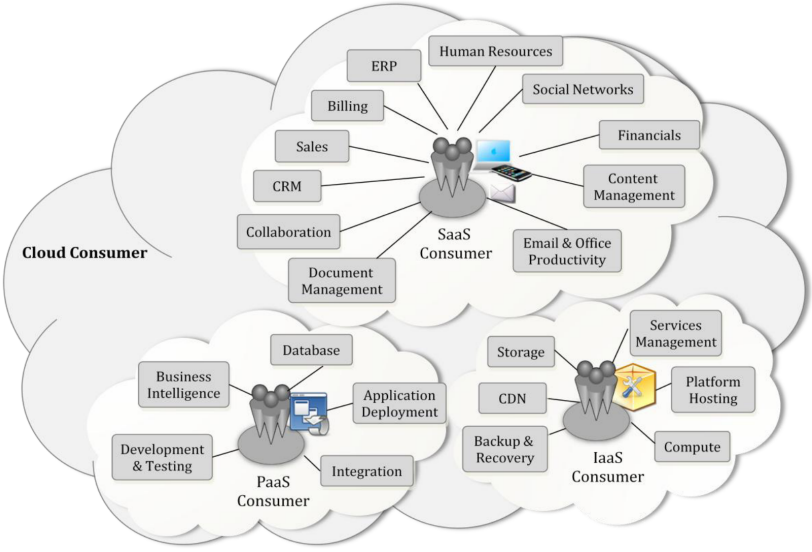
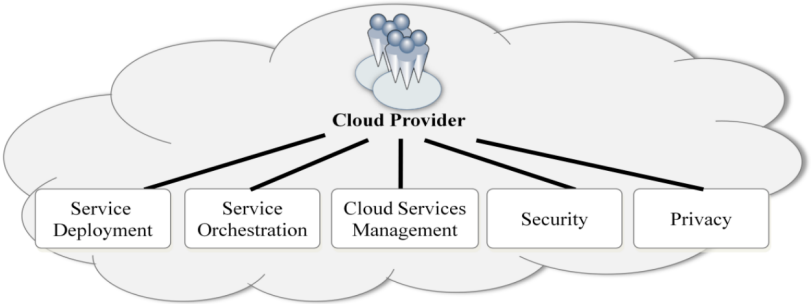
Cloud Consumer



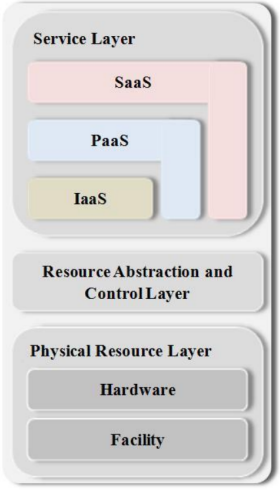
Cloud Provider



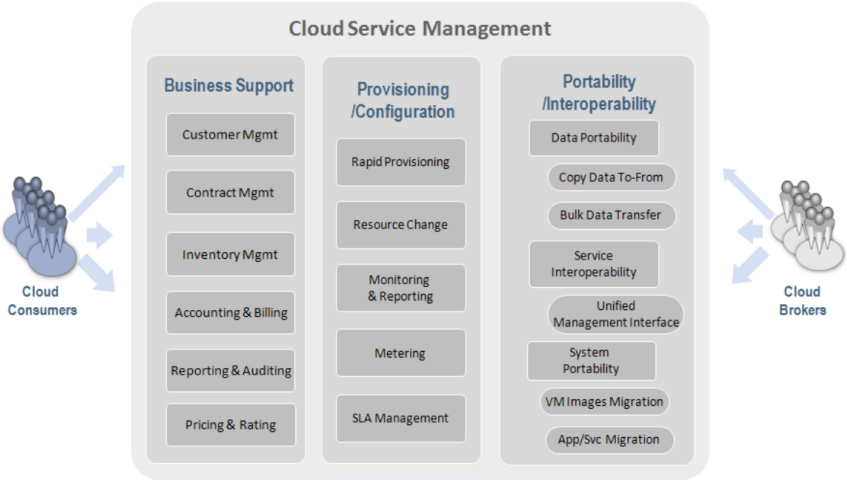
Depending on the service models, three types of cloud providers

1. SaaS cloud provider
2. PaaS cloud provider
3. IaaS cloud provider

Compose system components to support the cloud providers activities in arrangement, coordination and management of computing resources to provide cloud services to cloud consumers



Cloud Services Management



Service-related functions for the **management and operation** of the services provided to the cloud consumers

Cloud Service Management

Business Support – set of business-related services to deal with clients (consumers) and supporting processes such as accounting & billing, pricing & rating, ...

Provisioning/Configuration

- Rapid provisioning – automatically deploying cloud systems based on the requested services/resources/capabilities
- Resource change – adjusting configuration/resource assignment for repairs, upgrades and joining new nodes into the cloud
- ...

Portability/Interoperability

- Should provide mechanisms to support data portability, service interoperability, and system portability

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Security and Privacy

Security

- Cross-cutting and spans across all layers of the reference model
- Concerns all relevant actors!

Privacy

For Cloud Providers

- Personal Information (PI)
- Personally Identifiable Information (PII)

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Trust Boundary

Logical perimeter that represents which IT resources are trusted by an organisation

Implications of Cloud

- Increased Security Vulnerabilities
- Reduced Operational Governance Control
- Multi-Tenancy: Overlapping Trust Boundaries

Cloud Auditor

Independent examination of cloud service controls

- Security Controls
- Privacy Impact
- Performance
- ...

Verify compliance with regulation and security policy, and conformance to standards through review of objective evidence

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Cloud Broker

To help cloud consumers manage integration of cloud services

Three Main Services

1. Service Intermediation – provides value-added services to cloud consumers
2. Service Aggregation
   - Combines and integrates multiple services into one or more new services
   - Provides data integration
   - Ensures secure data movement between cloud consumers and multiple cloud providers
3. Service Arbitrage – flexibility to choose services from multiple agencies

Cloud Carrier

Provides connectivity and transport of cloud services between cloud consumers and providers

Cloud providers set up SLAs with a cloud carrier to provide consistent services to their consumers

May be required to provide dedicated and secured connections between cloud providers and consumers

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Cloud Deployment Models

Public Cloud

Available to the general public, i.e., **shared** by all consumers

Private Cloud

Solely for use by an organisation – suitable for enterprises with large scale IT

Community Cloud

Shared by multiple organisations based on common operational and regulatory requirements

Hybrid (Federated) Cloud

Two or more public and private clouds that inter operate

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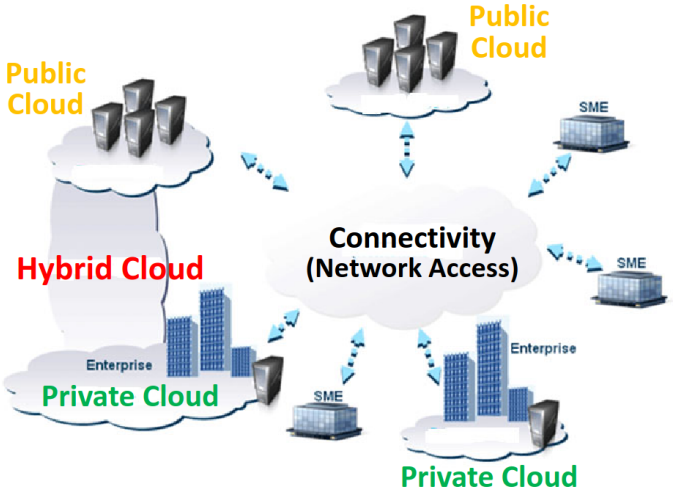
Cloud Deployment Models



Public Cloud

Most commonly used cloud infrastructure

Organisations act as cloud consumers while accessing IT resources and services made available by different cloud providers

Organisational Needs

- Low upfront costs with potentially a pay-as-you-go model
- Available anytime, anywhere
- Maintenance is handled by the provider

Cloud Providers

Amazon, Microsoft, Google, Oracle, IBM, Alibaba, Zoho, DigitalOcean, ...

Private Cloud

Cloud consumer's organisation has exclusive access and usage of cloud resources

Organisational Needs

- High level of control – especially for sensitive data
- Restricted access – often limited to organisation
- Maintenance is handled predominantly in-house

Can be managed on-site or by a third-party (out-sourced private cloud)

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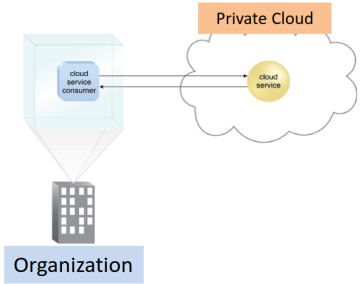
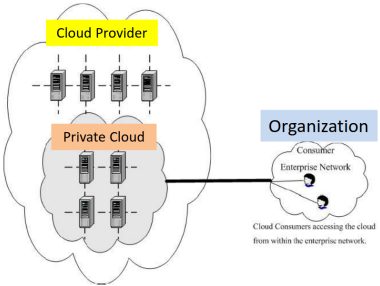
Private Cloud



On-Site Private Cloud



Out-Sourced Private Cloud

Security? On-site vs out-sourced

Dedicated IT resources hosted by a third-party provider

Virtual Private Cloud – secure, isolated environment within a public cloud that is customisable by you

Private Cloud

Concerns

Limited SLA, Reliability, Security, Availability, Trust, ...

Cloud Providers

Amazon Virtual Private Cloud

Cisco Private Cloud Solutions

Virtual Private Cloud – Google Cloud

Azure Dedicated Host – Private Cloud

...

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Community Cloud

Positioned between public and private clouds

Organisational Needs

- Cost of private cloud is prohibitive
- Shared needs and regulatory compliance with other organisations
- Restricted access – often limited to participating organisations

Need a clear governance structure for management

Examples

Government departments, Educational Institutions, Health Care Industry, ...

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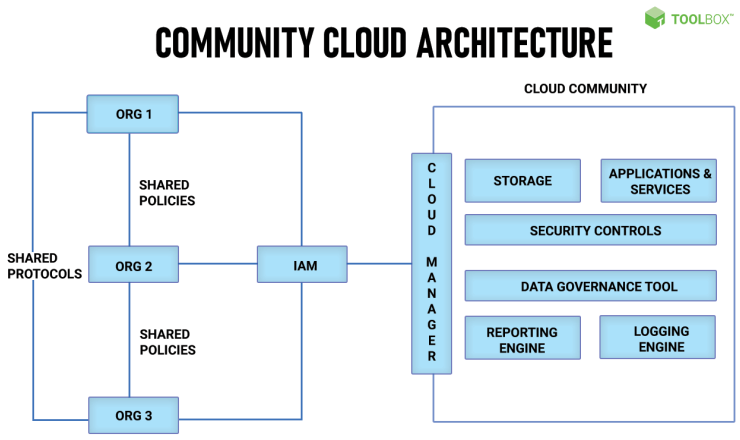
Community Cloud



https://www.spiceworks.com/tech/cloud/articles/what-is-community-cloud/

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Hybrid Cloud

Consists of two or more clouds (on-site private, on-site community, off-site private off-site community or public) that remain as distinct entities

Connected together to allow for data and application interoperability

Organisational Needs

- Balance control and cost
- Flexible approach depending on data and applications, e.g., sensitive data on private cloud while application on public cloud

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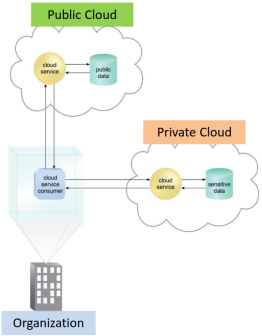
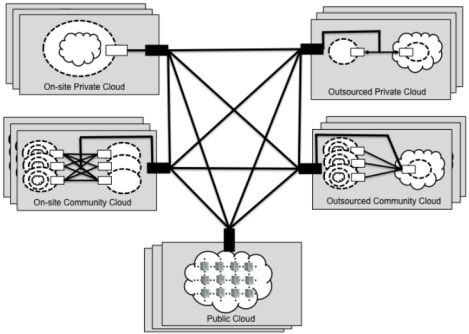
Hybrid Cloud



2 different clouds



5 different clouds

Sovereign Cloud

Why?

- Meet regulatory and compliance needs of a country or jurisdiction
- Stores data (including metadata) within national borders

Organisational Needs

- Adhere to local data protection laws
- Prevents foreign access to data under all circumstances

Cloud Providers

Microsoft Sovereign Cloud, Sovereign Cloud Oracle, SAP Sovereign Cloud, ...

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Next Week

Cloud Architectures

References

NIST Cloud Computing Reference Architecture, NIST Report 2011, https://doi.org/10.6028/NIST.SP.500-292

The NIST Definition of Cloud Computing, NIST, 2011, https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-145.pdf

Chapter 4, Cloud Computing: Concepts, Technology & Architecture, Thomas Erl, Zaigham Mahmood and Ricardo Puttini, Prentice-Hall, 2013

Continuous Feedback

https://forms.cloud.microsoft/r/ZJxTf2wFjr

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Cloud Architectures

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resource organization

Outline

Resource Organisation

Recall: Essential Characteristics NIST 2011

1. On-demand Self-service
2. Broad Network Access
3. **Resource Pooling**
4. **Rapid Elasticity**
5. Measured Service

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In Response to ...

- Fluctuation in cloud consumer's demand over time – **number of jobs**, and **resource demand within a job**
- Fluctuation in available cloud resources – cloud provider provides services to many cloud customers

Achieve elasticity or scaling

Achieve balanced utilisation of cloud resources

How is this done in practice?

Workload Distribution

Service Load Balancing

Resource Pooling

Dynamic Scalability

Elastic Resource Capacity

application level

resource level

Workload Distribution Architecture

Aim

Distribute consumer's workload over available cloud resources – use horizontal scaling

Key Component

Load Balancer – distributes workload to reduce over-utilisation or under-utilisation of resources

horizontal scaling

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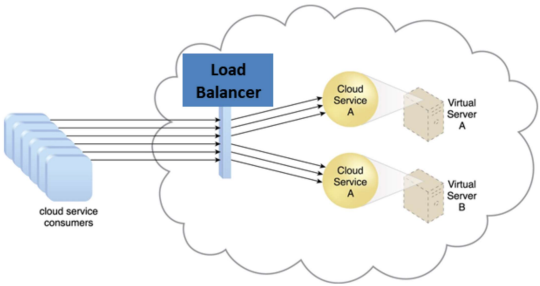
Workload Distribution Architecture

E.g., Amazon Elastic Load Balancing Service



Automatically distributes incoming application traffic

Achieve greater levels of fault tolerance for applications

Seamlessly provides the required amount of load balancing capacity needed

Load Balancer

AWS

Uses elastic load balancing ⇒ automatically distributes incoming traffic across multiple targets

Routes traffic only to healthy targets

Different Load Balancers

- **Application Load Balancer**
- Network Load Balancer
- Gateway Load Balancer
- Classic Load Balancer

network level - like roads and cars

gateway level - you want the right traffic on your roads

classic - for backward compatibility

Serves as a single point of contact for clients ⇔ contains one or more **listeners**

Listener

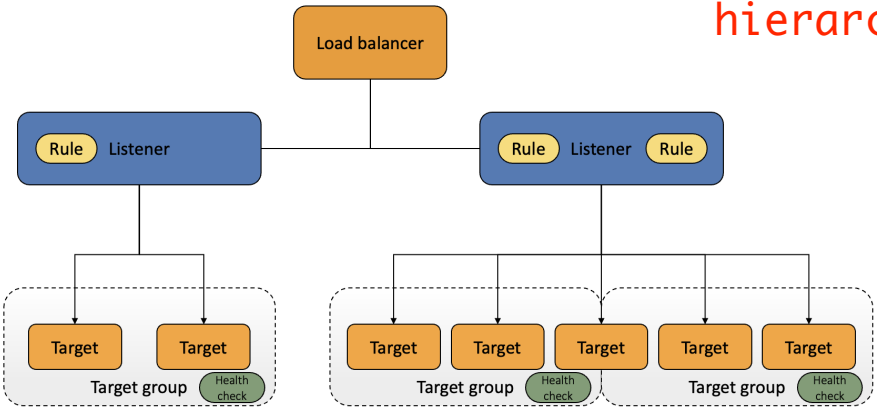
- Checks for requests from clients using the protocol and port that you configure
- User-defined rules routes requests to its registered targets (in target groups)
- Rules consist of **priority**, one or more **actions**, and one or more **conditions**
- A default rule is **necessary**

Steps

1. Load Balancer receives a request
2. Request is evaluated and appropriate rule is applied
3. Select a target from the target group for the rule action

protocol - the kind of request
port - where is the listener waiting

hierarchy



https://docs.aws.amazon.com/elasticloadbalancing/latest/application/introduction.html

Service Load Balancing Architecture

Specialised variation of the workload distribution architecture

Key Feature

Redundant cloud service deployments are created with a load balancer to dynamically distribute workloads

Resource pool $\Leftrightarrow$ **Duplicate** cloud service

Load Balancers

Independent of cloud services and their host servers

Built-In to cloud service as part of application or server environment

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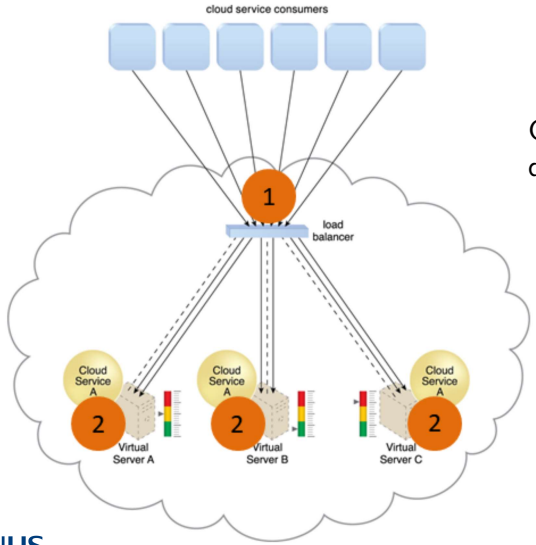
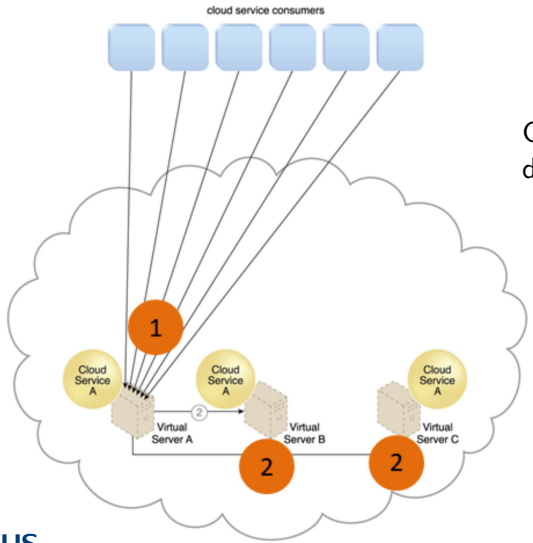
Independent Load Balancer



Cloud service A is **replicated** and runs on different virtual servers

1. Load Balancer intercepts requests from cloud consumers and forwards them to the virtual servers
2. Workload processing is horizontally scaled

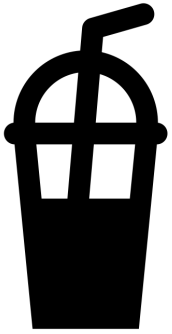
Built-In Load Balancer



Cloud service A is **replicated** and runs on different virtual servers

1. Cloud consumer's requests are sent to cloud service A on virtual server A
2. Built-in load balancing logic means that requests are distributed to virtual servers B and C

Take a Break



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Resource Pooling Architecture

Aim

Aggregate cloud resources of different types to serve diverse needs of cloud consumers

Key Feature

Automatically maintain identical IT resources to ensure they remain **synchronised**

Common Resource Pools

- Server pool (Physical/Virtual)
- CPU pool
- Memory pool
- Storage pool
- Network pool

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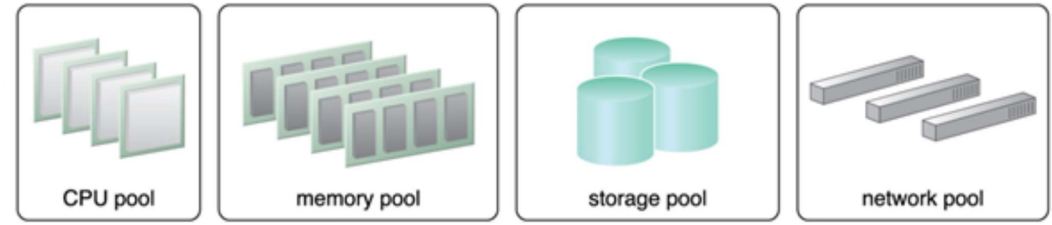
Resource Pooling Architecture

Dedicated Pools



Example

Dedicated pool with 4 **sub-pools** with CPUs, memory, storage, and network devices

Issue!

Becomes complex $\Rightarrow$ multiple pools for specific consumer or application

Resource Pooling Architecture

Create hierarchical structure of parent, sibling, and, nested pools to meet diverse requirements

Sibling Resource Pool

Are drawn from physically grouped IT resources and not spread out across data centers

co-located resources

Isolated from one another so each cloud consumer is given access to its respective pool

Nested Resource Pool

Larger pools divided into smaller pools with same type of IT resources

Used for assigning resource pools to different departments and groups within the same organisation

template

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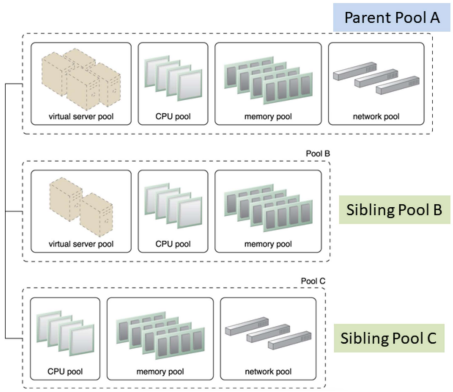
Sibling Resource Pools

Example

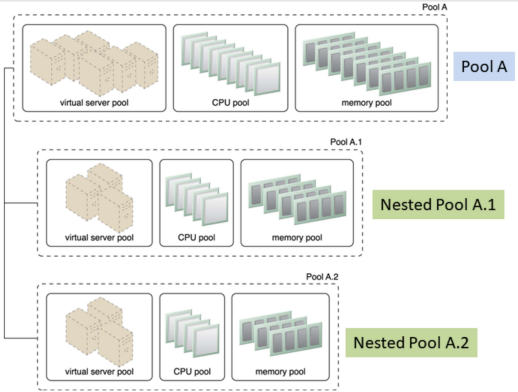


Pools B and C are sibling pools of Pool A

from same datacenter

Nested Resource Pools

Example



Pools A.1 and A.2 are nested pools of Pool A

same template, only different numbers

Dynamic Scalability Architecture I

Aim

Enable variable resource utilisation to meet **usage demand fluctuations**

Key Feature

Dynamic allocation of resources from resource pools based on predefined scaling conditions

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Dynamic Scalability Architecture II

Common Methods

- Dynamic **Horizontal** scaling – **same** resource instance scaled dynamically with fluctuating workloads (e.g., replicating resources)
- Dynamic **Vertical** scaling – scale processing capacity of a **single IT resource** (e.g., increase memory dynamically when virtual server is loaded)
- Dynamic **Relocation** – resource relocated to host with larger capacity (e.g., faster storage device with better I/O capacity)

```
relocation -> move to entirely new machine
```

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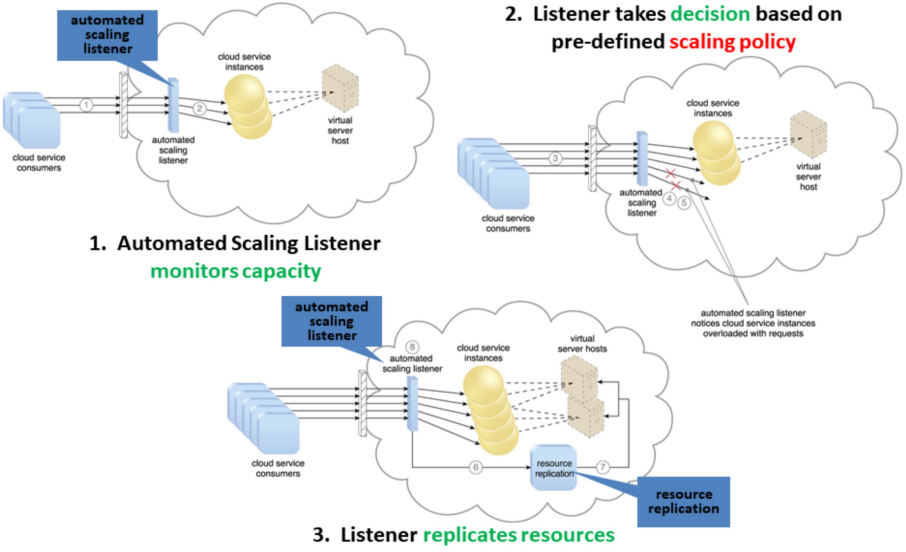
Horizontal Dynamic Scalability

Example



Elastic Resource Capacity Architecture

Aim

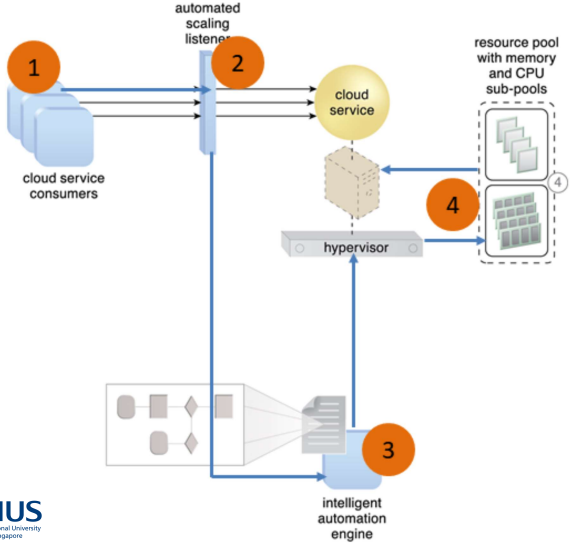
Dynamic provisioning of virtual servers

Key Features

- Fluctuating processing requirements
- Allocates and reclaims CPUs, RAMs, etc.
- Run-time virtual server is monitored
- Additional resources leveraged before capacity threshold reached
- Virtual server and IT resources are vertically scaled
- Uses Intelligence Automation Engine

Elastic Resource Capacity Architecture

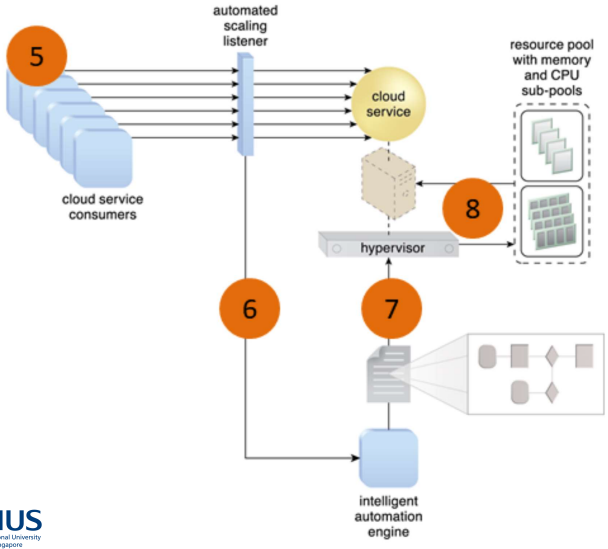
Example



1. Cloud consumers send requests to a cloud service
2. Requests are monitored by an Automated Scaling Listener
3. Intelligent Automation Engine deployed with workflow logic
4. Workflow logic is capable of notifying the resource pool using allocation requests

Elastic Resource Capacity Architecture

Example



5. Cloud consumer's requests increase dramatically
6. Automated scaling listener signals the Intelligent Automation Engine to execute the script
7. Script runs workflow logic that signals the hypervisor to allocate more resources from the resource pools
8. Hypervisor allocates additional resources to handle the increased workload

Scaling On-Premise Provisioning

Aim

Scale on-premise IT resources into the cloud whenever capacity thresholds have been reached

Example

Cloud Bursting Architecture

Cloud Bursting Architecture

Definition

Burst Out – to meet higher usage demand, dynamic scaling of on-premise resources to cloud-based resources

Definition

Burst In – on lower demand, revert back to on-premise and release cloud resources

Cloud services are redundant, pre-deployed and inactive until cloud bursting

Features Required

Automated Scaling Listener

Resource replication – resources on cloud and on-premise are **synchronised**

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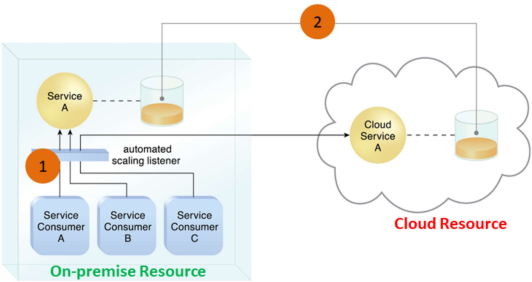
Cloud Bursting Architecture

Example



Automated Scaling Listener monitors the usage of on-premise service A

1. When usage threshold is exceeded, Service Consumer C is redirected to Service A's redundant implementation on the cloud
2. Resource replication system ensures the state management databases are synchronised

Next Week

Resource Hosting and Datacenter

References

Chapter 11, Cloud Computing: Concepts, Technology & Architecture, Thomas Erl, Zaigham Mahmood and Ricardo Puttini, Prentice-Hall, 2013

Continuous Feedback

https://forms.cloud.microsoft/r/ZJxTf2wFjr

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Resource Hosting & Datacenter

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(with thanks to Teo Yong Meng)

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Outline

Resource Hosting

Main Components in a Datacenter

Datacenter Configuration

Resource Hosting

On-Premise

- Organisations are responsible for deploying, operating and maintaining internet connectivity
- Have complete control over Quality of Service (QoS)
- Providing security

Cloud-Based

- Use multiple cloud carriers – QoS dependent on carriers
- Easier to adopt for applications with relaxed bandwidth and latency
- Security concerns

not inherently insecure
just more actors

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Terminologies

Latency

Time it takes a packet to travel from one data node to another – important when response time is key for applications

Bandwidth

Number of bits transferred per unit time – important when a substantial amount of data transfer is required by the application

`easier to increase bandwidth than latency (physical property)`

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On-Premise vs Cloud-Based

Time Scales of Computing Latencies

Event	Latency	≈ Scaled Time
1 CPU cycle	0.3 ns	1 second
Level 1 cache access	0.9 ns	3 seconds
Main memory access (DRAM, from CPU)	120 ns	6 minutes
Solid-state disk I/O (Flash memory)	50 – 150 μs	2 – 6 days
Rotational disk I/O	1 – 10 ms	1 – 12 months
Internet: Singapore to Hong Kong (2555 Km)	40 ms	4 years
Internet: Singapore to London (10856 Km)	155 ms	16 years
Flight: Singapore to Tokyo (5322 Km)	7 hrs	2.5 million years!

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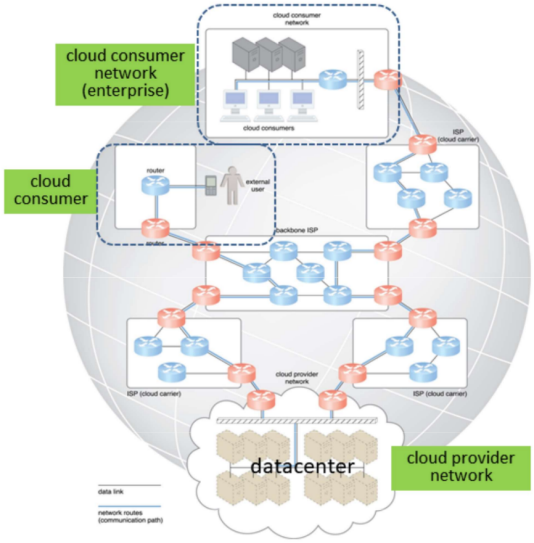
Cloud Connectivity



importance of cloud carriers or ISPs

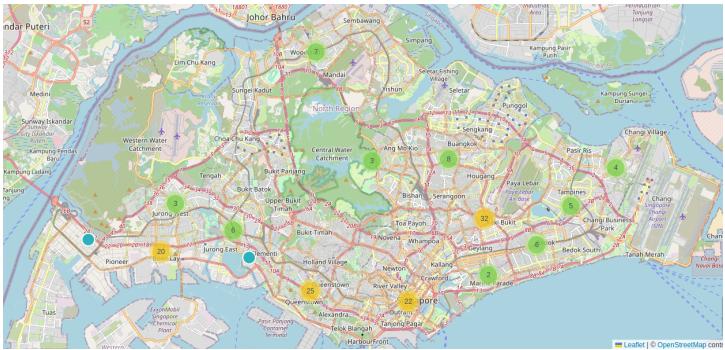
CDN:
have versions of the content closer to the consumers

Singapore Data Centers Locations

How Many?

145 Locations

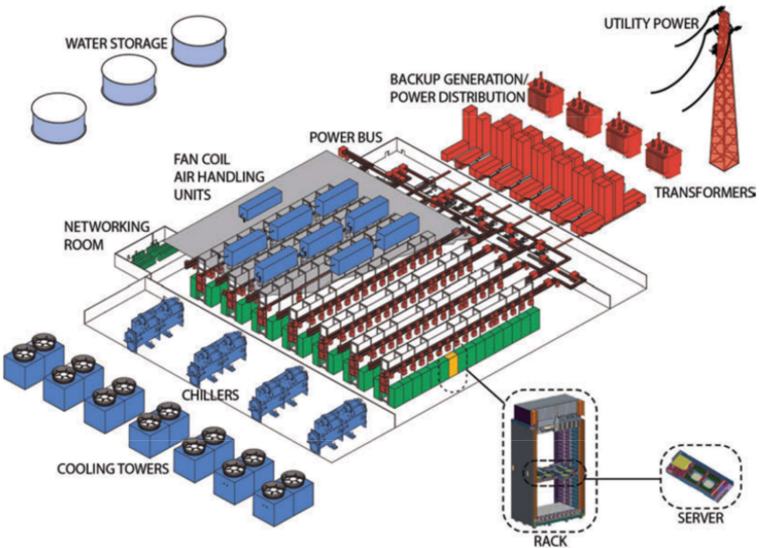
50 Providers



https://www.datacenters.com/locations/singapore

Datacenter I

Components

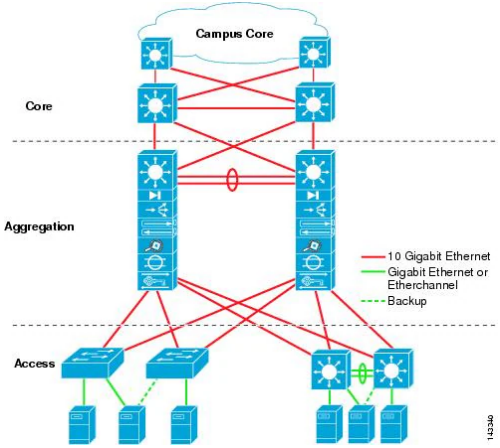


Main server hall – compute, storage, and networking equipment

Mechanical yard – hosts cooling systems

Electrical yard – generators and power distribution centers

Datacenter Servers I



organization: layering

https://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Data_Center/DC_Infra2_5/DCI_SRND_2_5a_book/DCInfra_1a.html

Datacenter Servers II

like lego -> building blocks

Uses commodity hardware with a modular architecture

Servers are modularized and packed as a cluster in server racks

Server racks are replicated as scaled up computational resources to form a datacenter

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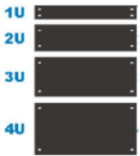
Datacenter Servers III

a. server

1U Server (1.75 inches or 44.45mm)

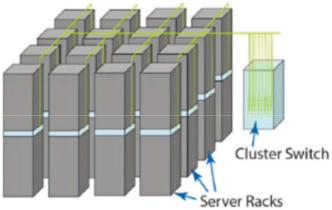


1 rack unit
Standardized amount of space

b. cluster

42U Server Rack

c. Multiple Clusters



d. datacenter

Datacenter Servers

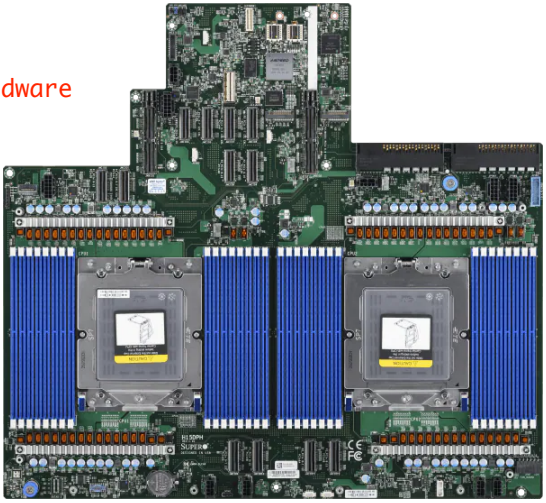
Example – Supermicro AS -1127H7-N — 1U Hyper Server

https://www.supermicro.com/en/products/system/hyper/1u/as-1127h7-n

Datacenter Servers

Example – Supermicro AS -1127H7-N — 1U Hyper Server

Actually generic hardware
Not specialized



2 CPUs
Connected by buses

Blue -> memory slots

https://www.supermicro.com/en/products/system/hyper/1u/as-1127h7-n

Dual processor(s) AMD EPYC™ 9006 SP7 Series Processors

- Up to 512C/1024T 512 cores/ 1024 threads

32 DIMM slots memory

3 PCIe expansion slots for networking and storage

Flexible Networking via AIOM (Advanced I/O Module)

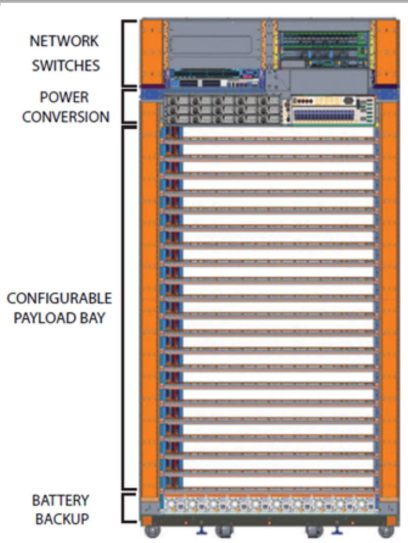
...

Supports servers, storage and networking equipment

Has provision for power, battery backup and rack-level switch (e.g., Top Of Rack – TOR)

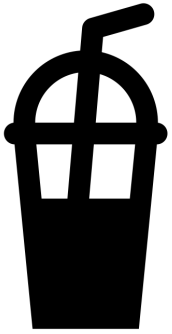
Rack-level switch – connects IT resources within rack and also connects the rack to the datacenter network

Typical commodity rack can house 42 1U servers, each of 1.75in height




```
rack -> connections
reduce latency -> keep workloads local
```

Take a Break



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Two **categories**

Private to **individual running tasks** – resides in local DRAM or disk

Part of shared state of **distributed workload** – distributed storage network

Storage Hierarchy

One server – caches, DRAM memory, Disks, Flash Disks

Each rack – local and distributed memory

Across racks – local and distributed memory

Datacenter Network

Carrier Interconnection

Between datacenter (LAN) and cloud consumers (WAN)

Local Area Network (LAN)

Within a rack

Redundant connectivity

Storage Area Network (SAN)

Between servers and storage systems

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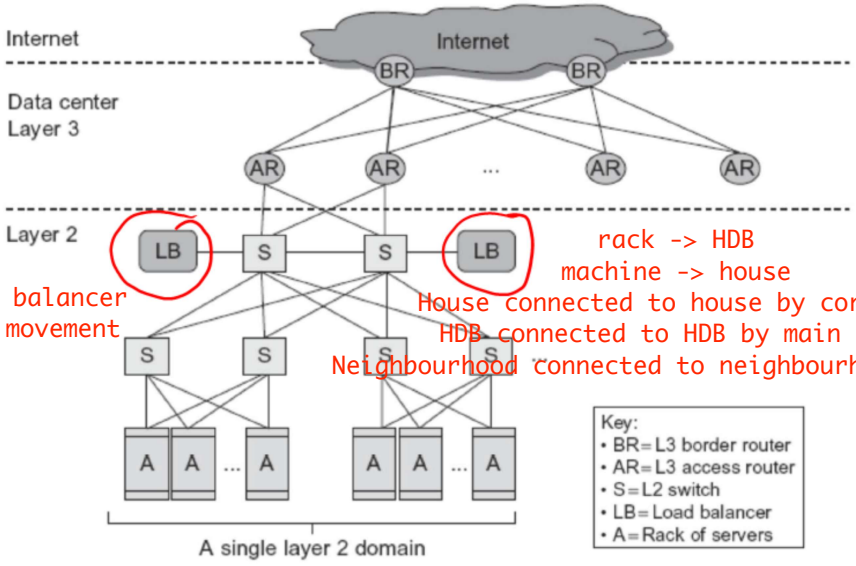
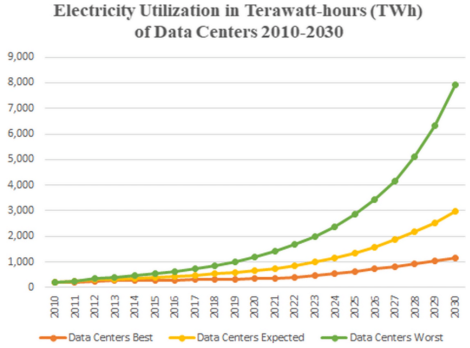
Datacenter Network



network load balancer
manage data movement

rack -> HDB
machine -> house
House connected to house by corridor
HDB connected to HDB by main road
Neighbourhood connected to neighbourhood by road

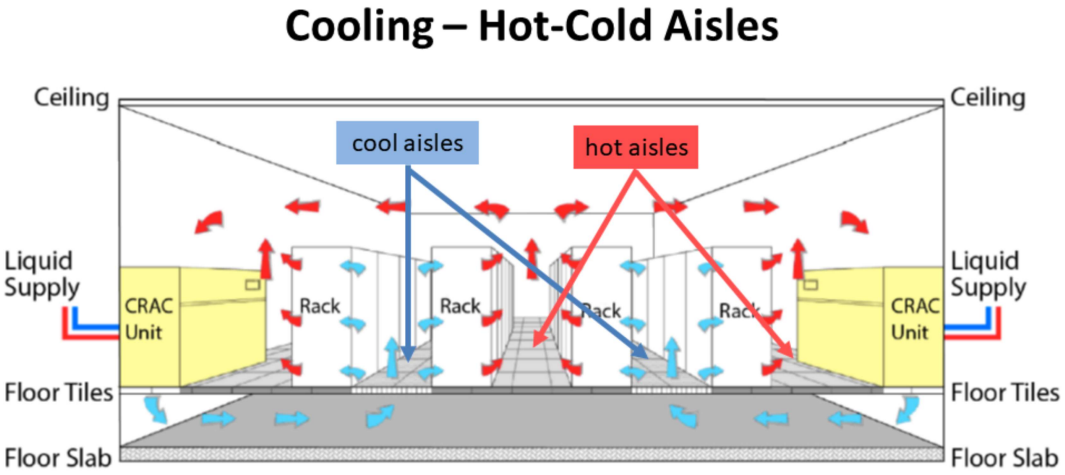
Datacenter Electricity Demand



https://link.springer.com/article/10.1007/s10586-021-03431-z

Datacenters consume a **lot** of energy

Most of this is converted into heat – cooling is a big issue!



Cooling system in a raised-floor datacenter with hot-cold air circulation support water heat exchange facilities

Datacenter Cooling

From air to liquid (water) cooling – Water is 1000x more efficient than air to remove heat

From portable water to seawater

- Planned Singapore Floating Datacenter Park (`https://www.keppeldatacentres.com/innovations/floating-data-centre/`)

In the sea?

- Absolutely! (Microsoft Project Natick was the first Proof of Concept – `https://www.microsoft.com/en-us/research/project/natick/`)
- The 'Cloud' in the Ocean ☺

Water Usage!

Very high –
`https://www.eesi.org/articles/view/data-centers-and-water-consumption`

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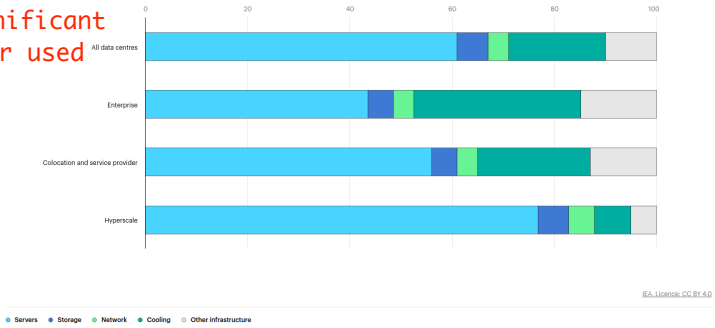
data center in space? possible

Datacenter

Power Usage Efficiency

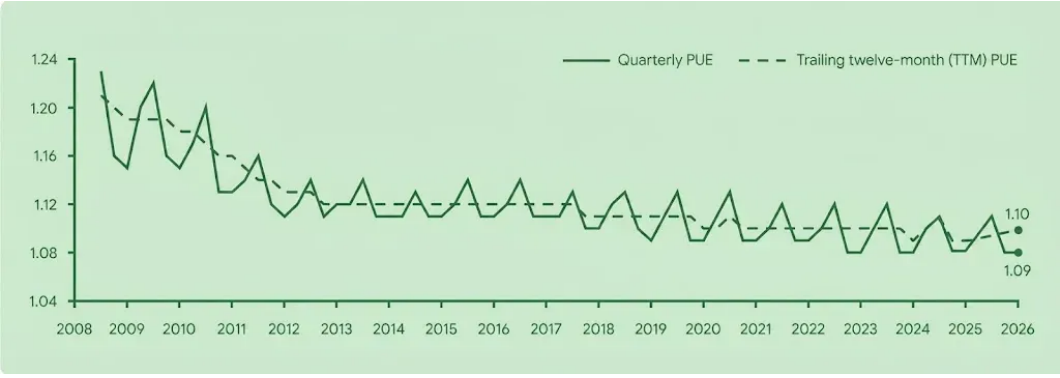
cooling is a significant portion of power used



https://www.iea.org/data-and-statistics/charts/share-of-electricity-consumption-by-data-centre-and-equipment-type-2024

$$PUE = \frac{total\ power\ used\ by\ datacenter}{power\ used\ by\ IT\ Equipment} \Rightarrow \text{in ideal case PUE is 1}$$

Average PUE for all data centers



https://datacenters.google/efficiency/

Energy Proportional Systems

Energy Efficiency is not a linear function of load – idle system uses 50% power

An energy proportional system consumes almost no power when idle, very little power under light load, and, gradually, more power as the load increases

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Provides a ranking system to indicate the reliability of a datacenter

Tier	Description	Expected Uptime
1	Single path for power distribution, USP, and cooling distribution, no redundant components	99.671%
2	Partial redundancy for power and cooling	99.741%
3	Multiple paths for power and cooling, and each path with redundant components, concurrently maintainable	99.982%
4	Redundancy for every component	99.995%

Availability & Cost

Tier 1 < Tier 2 < Tier 3 < Tier 4

Tier 3 is typically industry standard
Tier 4 might be overkill

SoC Datacenter Visit

22/09/26 from **2pm**

24/09/26 from **7pm**

Register your interest: `https://forms.cloud.microsoft/r/dC2vmB0f3W`
(available from: **01/09/26 at 20:00 – 08/09/26 at 20:30**)

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Next Week

Virtualisation and Multitenancy

References

Chapter 5 and Appendix D1, Cloud Computing: Concepts, Technology & Architecture, Thomas Erl, Zaigham Mahmood and Ricardo Puttini, Prentice-Hall, 2013

Chapters 1 & 4, The Datacenter as a Computer: Designing Warehouse-Scale Machines, 3rd Edition, Morgan & Claypool Publishers, 2019

Continuous Feedback

https://forms.cloud.microsoft/r/ZJxTf2wFjr

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Virtualisation & Multitenancy

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(with thanks to Teo Yong Meng)

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Outline

Overview

Examples

Virtualisation Approaches

Containers

Multitenancy

Virtualisation

Definition

Enables a **single physical** infrastructure to function as **multiple logical** infrastructure or resource

Many OSs can run and share a single hardware (one-to-many improves utilisation)

E.g., processor, memory, storage, network

Why?

Improve resource utilisation through sharing – lower cost, improve scalability, etc.

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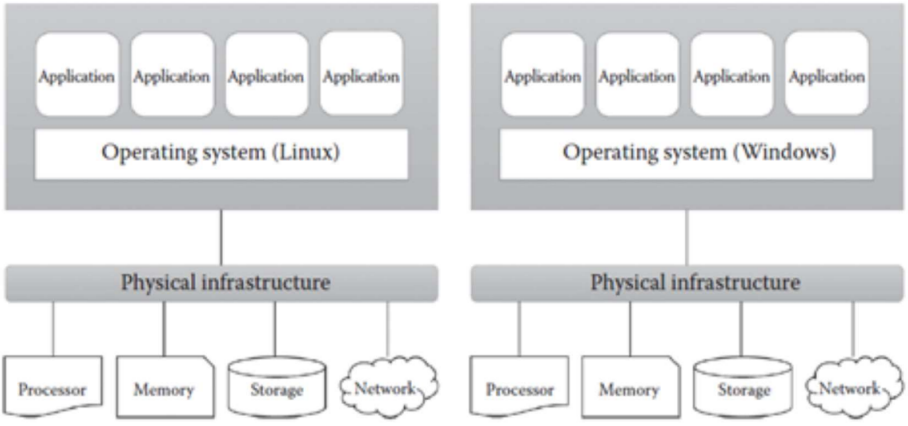
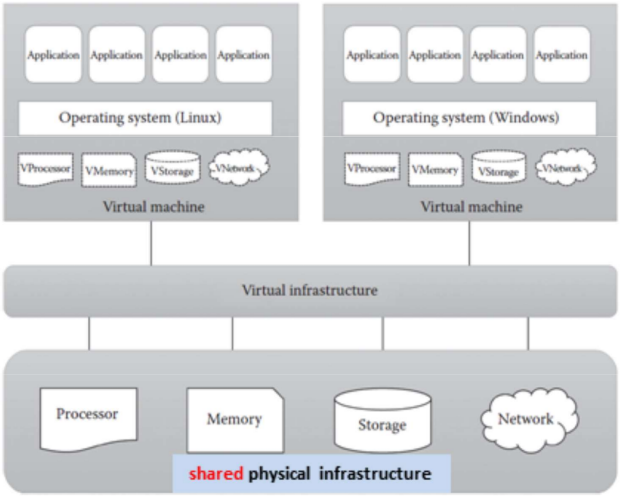
Before Virtualisation



Physical Infrastructure

1:1 relationship with physical infra

After Virtualisation



Virtual Infrastructure

many:1

Virtualisation

Pros

Hardware Independence – software-hardware dependencies in traditional non-virtualised environment

Resource Consolidation – increase hardware utilisation, load balancing, ...

Resource Replication – rapid scaling as resources are virtual (created as virtual disk images)

Cons

Performance overhead

Single point of failure – virtualisation software

Types of Virtualisation

Processor virtualisation

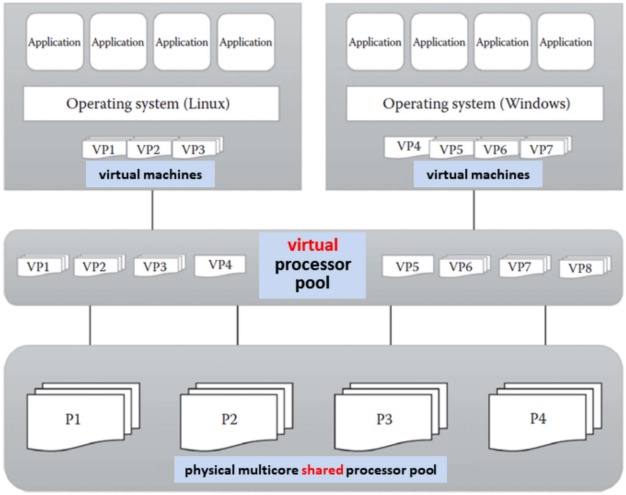
Memory virtualisation

Storage virtualisation

Network virtualisation

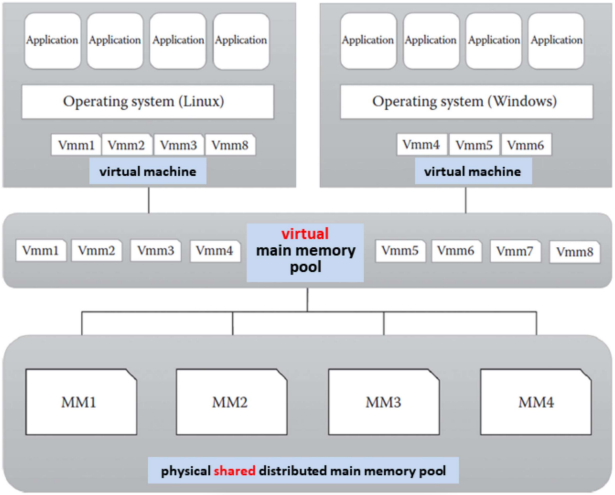
Processor Virtualisation

Abstracting physical processor to pool of virtual processors



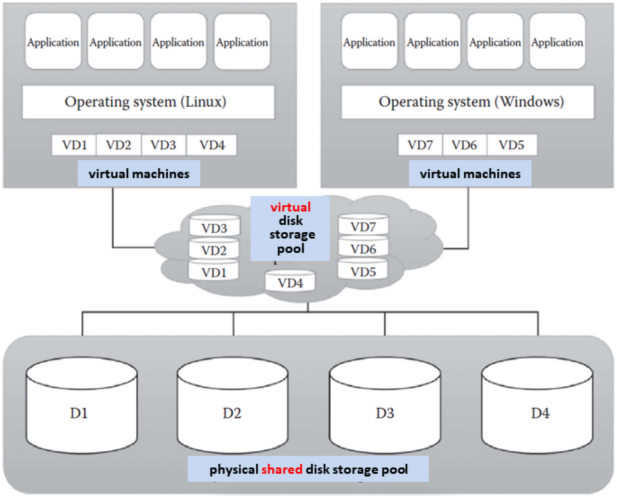
Memory Virtualisation

Providing virtual main memory to the VMs that are abstracted from the physical main memory



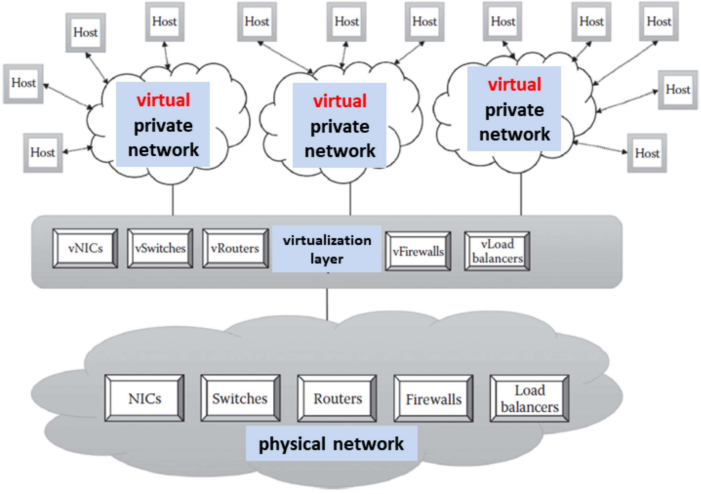
Storage Virtualisation

Providing multiple logical storage that is abstracted from multiple physical storage



Network Virtualisation

Process of abstracting physical network components to form a virtual network



Virtualisation Approaches

Virtualisation Software

Hypervisor or **Virtual Machine Manager (VMM)**

- Software that sits between VMs and physical infrastructure
- Allows IaaS provider to manage VMs and execute consumer's guide OS
- Physical infrastructure resources are shared among the consumer's VMs

Virtualisation Approaches

Full Virtualisation, Para Virtualisation, Hardware-Assisted Virtualisation

Depending on the type of virtualisation approach, the virtualisation software and guest OS will run in different privileged levels

how does the guest talk to the hardware?

only privileged instructions (OS instructions) should reach the hardware
lower the ring number higher the privilege

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Full Virtualisation

VM can run different guest OSs

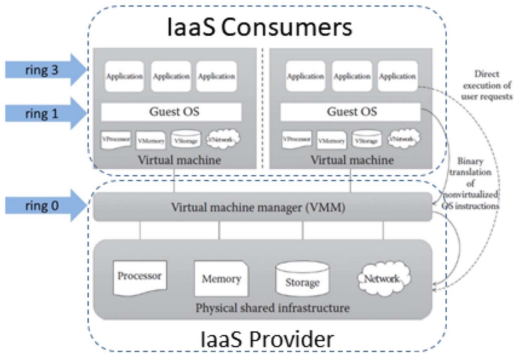
Guest OS is not aware that it is virtualised

VMM at **ring 0** provides all the virtual infrastructures needed for VMs, i.e., guest OS communicates with hardware through VMM

VMM translates guest OS instruction with instructions for the physical hardware

- **Binary translation**

software-assisted



VMM mimics the hardware
Talks to guest OS on hardware's behalf
Then translates the instruction to hardware

Full Virtualisation

Pros

Provides VMs with best isolation and security

Different OSs can run simultaneously

Virtual guest OS can easily be migrated to native hardware

Easy to install and use – no changes to guest OS

Cons

Binary translation overhead – reduced system performance

Need for correct combination of hardware and software – hardware may not have drivers for virtualisation software

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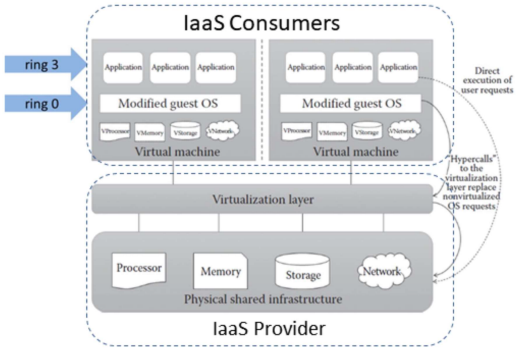
Para Virtualisation

Modified guest OS replaces OS requests with **hypercalls** – also called OS-assisted virtualisation

Hypercalls – system calls with privilege to communicate directly between OS and hypervisor without translation

Guest OS knows it is running in a virtualised environment



OS-assisted

VMM doesnt do translation
It gives guest OS an interface to talk to hardware

Para Virtualisation

Pros

Eliminates overhead of binary translation – improves performance

No need for special hardware

Cons

Overhead of guest OS kernel modifications

Modified guest OS cannot migrate to run on physical hardware

VMs lack backward compatibility, difficult to migrate to other hosts

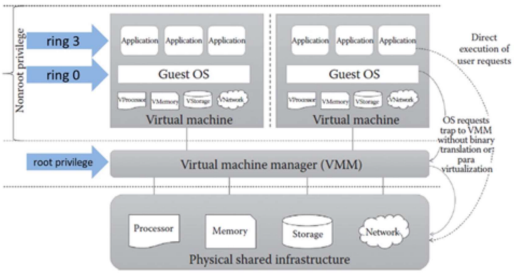
Hardware-Assisted Virtualisation

IaaS Consumer

- Hardware (processor) extension supports virtualisation – guest states are stored in virtual machine control structures
- OS requests directly trap the hypervisor – eliminates binary translation and guest OS modification

VMM has the highest root privilege, while guest OS and user applications have the same non-root privilege

native virtualization



hardware steps in

Virtualisation Approaches

Summary

from perspective of guestOS

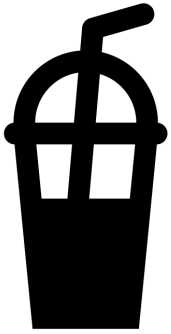
	Full	Para	Hardware-Assisted
Technique	Binary Translation and direct execution	Hypercalls	OS requests trap to VMM without binary translation or para virtualisation
Guest OS modification?	No	Yes	No
Compatibility	Excellent	Poor	Excellent
Is guest OS hypervisor independent?	Yes	No	Yes
Popular vendor(s)	VMware ESX	Xen	Microsoft, Virtual Iron, XenSource

no need guest OS modification but incur translation overhead

avoid overhead but needs to modify guest OS to make hypercalls

best of both worlds

Take a Break



next.. where does virtualization layer sit?

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runs directly on hardware

Type 1

Also called **Bare Metal** (or native) – runs on physical infrastructure without any help from the host OS

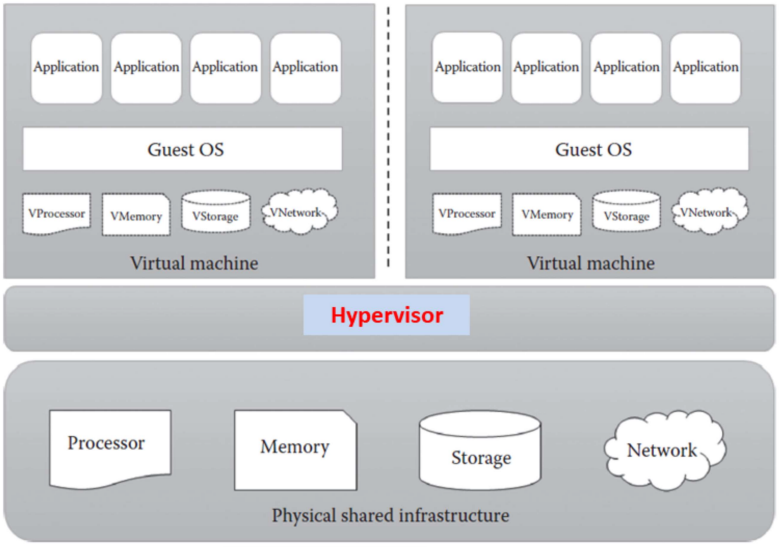
Type 2

Hosted (or embedded) – requires the help of host OS to communicate with underlying infrastructure

e.g. VirtualBox, Parallels…

from perspective of hypervisor

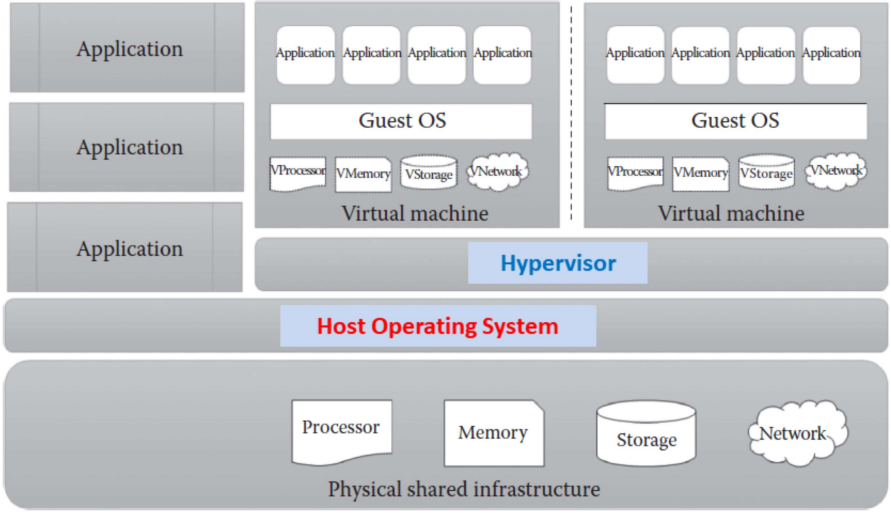
Bare Metal Hypervisor



no host OS

for larger scale applications

Hosted Hypervisor



for smaller applications

Hypervisor	Vendor	Type
Xen	University of Cambridge Computer Laboratory	Type 1
VMWare ESXi	VMware, Inc.	Type 1
Hyper-V	Microsoft	Type 1
KVM	Open virtualization alliance	Type 1
VMWare workstation	VMware, Inc.	Type 2
Oracle Virtualbox	Oracle Corporation	Type 2
Parallels Desktop	Parallels	Type 2

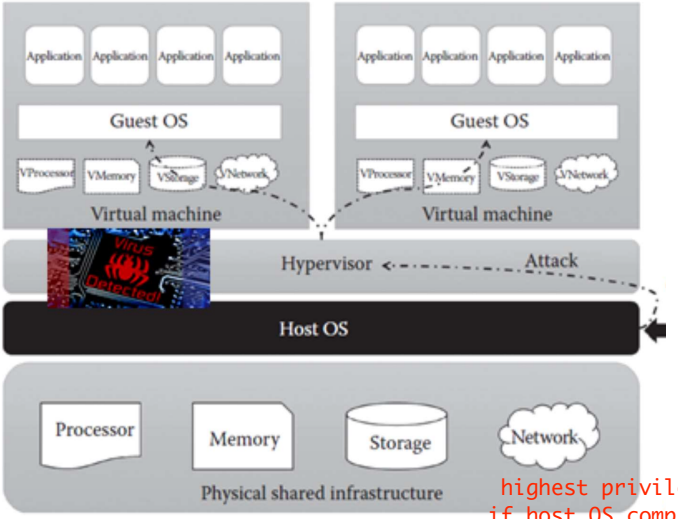
Host OS

Compromise the Host OS

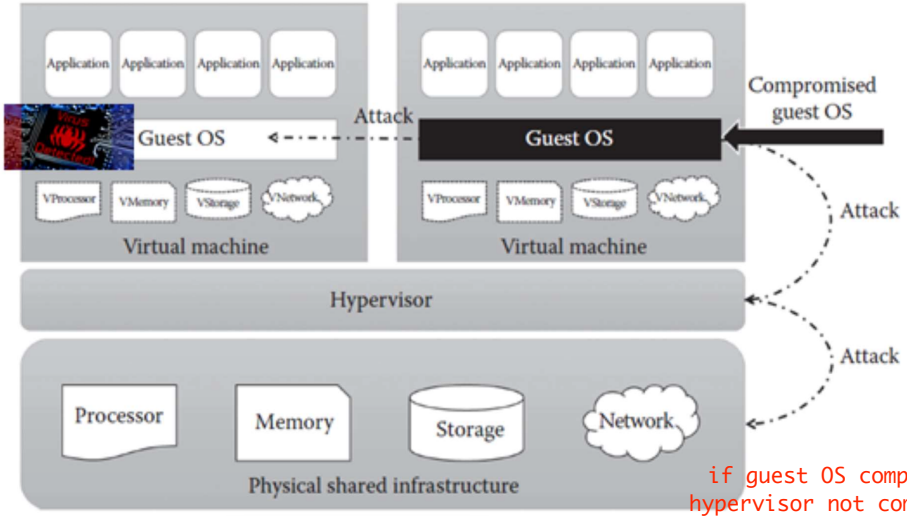
Guest OS

Compromise the Guest OS

`less likely for hypervisor to get compromised`



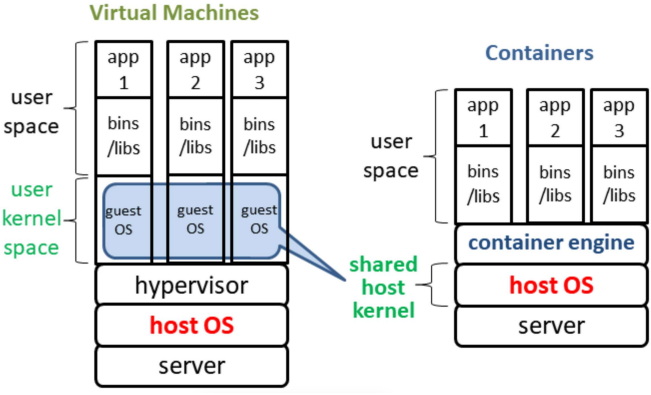
highest privilege -> hypervisor
if host OS compromised hypervisor
can compromise the guest OSes



if guest OS compromised
hypervisor not compromised
virus is isolated

Containers (vs VMs)

do we really need a full fledged VM?



Containers allow existence of multiple isolated user space instances, e.g., Docker

Containers share the host system's kernel with other containers ⇒ lightweight

quickly develop, deploy, test…

What is Docker?

Open Platform for developing, shipping and running applications

Enables you to separate your applications from your infrastructure so that you can deliver software quickly

Ability to package and run an application in a loosely isolated environment called a **container**

Containers are lightweight and contain everything needed to run the application $\Rightarrow$ portability!

Uses the idea of namespaces to provide the isolated workspace

Written in the Go programming language and takes advantage of several features of the Linux kernel

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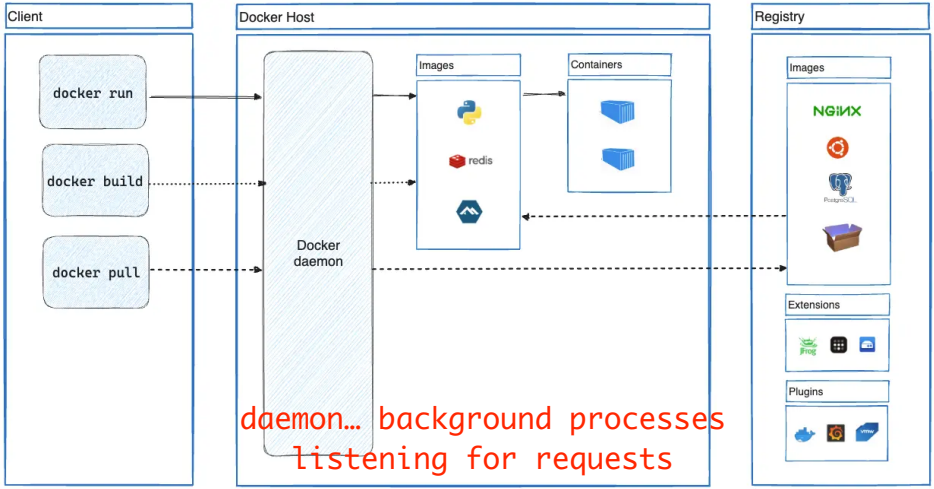
how does Docker run on Mac?… virtualization

Docker Architecture



Uses a client-server architecture

`https://docs.docker.com/get-started/docker-overview/`

Docker Components I

Docker Desktop

Easy to install application for your host OS – enables you to build and share containerised applications and microservices

Includes Docker daemon, Docker client, Docker compose, ...

Docker Components II

Docker Client

Primary way users interact with Docker

Can communicate with more than one daemons

Docker daemon

Building, running and distributing the Docker containers

Listens for Docker API requests

Manages Docker objects such as images, containers, networks, etc.

Can also communicate with other daemons to manage Docker services

Docker client and daemon communicate using a REST API, over UNIX sockets or a network interface

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Docker Components III

Docker registries

Stores images

Docker Hub is a public registry that anyone can use, and Docker looks for images on Docker Hub by default

Images

Read-only template with instructions for creating a Docker container

Often based on another image, but you can also create your own using a Dockerfile

Docker Components IV

Containers

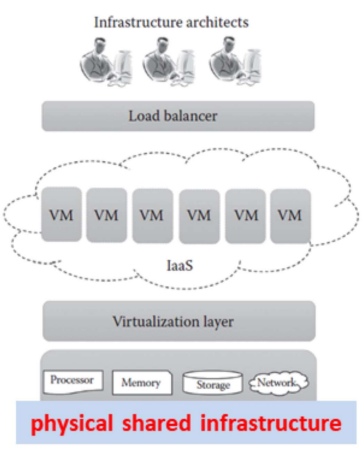
Runnable instance of an image

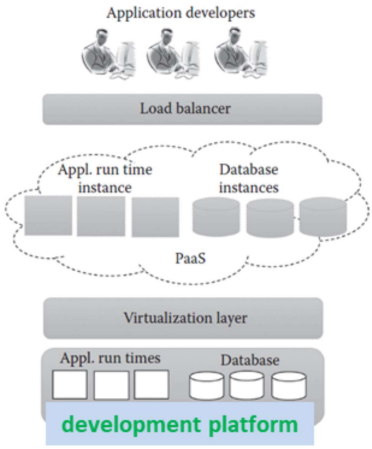
Can create, start, stop, move, or delete a container using the Docker API

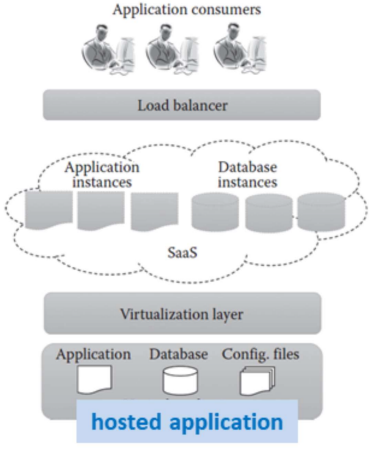
Can control how isolated a container's elements are from other containers or the host machine

`super lightweight`

Virtualisation and Cloud Service Models







IaaS **PaaS** **SaaS**

VMs docker multitenancy

Multitenancy

Cloud Providers

A single instance of *software* runs on a server and serves multiple tenants

Enable multiple users (tenants) to share the same application, e.g., cloud platforms, cloud services, etc.

`e.g. Dropbox, email`

Cloud Consumers

Dedicated instance of software with customised view

Tenants isolation ensures that tenants do not have access to data and configuration information that it does not own

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Multitenancy vs Virtualisation

Multitenancy

Dedicated consumer instance of software

Virtualisation

Abstraction of physical resources

In multitenancy, giving consumers an instance of the software (tenant usage & data isolation) is possible *because* of virtualisation

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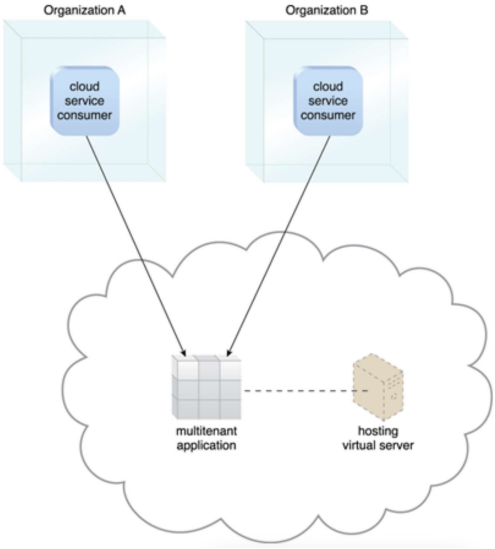
Multitenancy

Example



Multitenancy

Characteristics

Usage/Tenant Isolation – one tenant's behaviour does not affect other tenants

Data Security – separate procedures for each tenant

Recovery – backup, restore

Scalability – usage by tenants and tenant numbers can increase

Metered Usage – each tenant is charged only for features being used

Data Tier Isolation – databases, tables, etc., can be isolated or allowed as shared resources among tenants

different tenants will have different agreements, think Dropbox
e.g. you have 500mb, I have 1gb, someone has 1Tb, etc…

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Next Week

Applications and Paradigms (Part 1)

References

Chapter 5 and Appendix D, Cloud Computing: Concepts, Technology & Architecture, Thomas Erl, Zaigham Mahmood and Ricardo Puttini, Prentice-Hall, 2013

Chapter 7, Essentials of Cloud Computing, CRC Press, 2015

Continuous Feedback

https://forms.cloud.microsoft/r/ZJxTf2wFjr

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Applications & Paradigms

Part 1

Anandha Gopalan
(with thanks to Teo Yong Meng)

axgopala@nus.edu.sg

Outline

Cloud Applications

Cloud Application Development Models

Recall: Cloud Providers

Common Features

Basic and higher-level services with different levels of abstraction (IaaS, PaaS, SaaS)

Compute and storage resources – virtual servers, storage, networking, ...

Deployment management services – load balancer, auto-scaling, monitoring, message queueing, ...

for computational resources

User interface – graphical UI, CLI

...

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Cloud Applications

Focus primarily on enterprise computing

Ideal ☺

Applications which can be divided into as many independent tasks as possible

E.g., Web services, database services, search services, machine learning with large-scale models, ...

Not Ideal

Applications with a complex workflow and multiple dependencies, such as high-performance computing

e.g. part A -> part B -> part C

Applications with intense communication among concurrent instances

networking issues

Workload cannot be arbitrarily partitioned

bandwidth and latency

Key Challenges!

Cloud Consumer – scale application to accommodate a dynamic load, recover after system failure, efficiently support checkpoint/restart, ...

Cloud Provider – manage a large number of systems, provide quality of service guarantee, ...

Performance Isolation

Virtualisation shares VMs and multitenancy shares cloud application services ⇒ problem when utilisation is high!

multiple copies at different locations

Reliability

Server failures expected when a large number of servers cooperate to compute

Latency/Bandwidth Fluctuations

Due to sharing of resources

Data Logging

Necessary, but also need to limit it due to performance considerations

way to recover -> backups
how often? you'll need to interrupt your app

Reliance on Internet and web technology – high accessibility, web browser universality, ease of web-based service development

Cloud services use web technology as both the implementation medium as well as the management interface

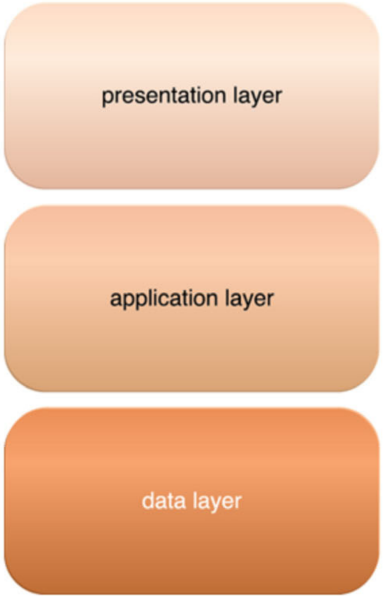
Cloud applications typically use a web-based client-server architecture – web browser client and web server

API - 2 elements are communicating
Some kind of contract between

Architecture of Web Applications I

Three-tier architecture abstraction for a web application

1. **Presentation layer** – represents user-interface
2. **Application layer** – represents implementation logic
3. **Data layer** – comprises of persistent data stores



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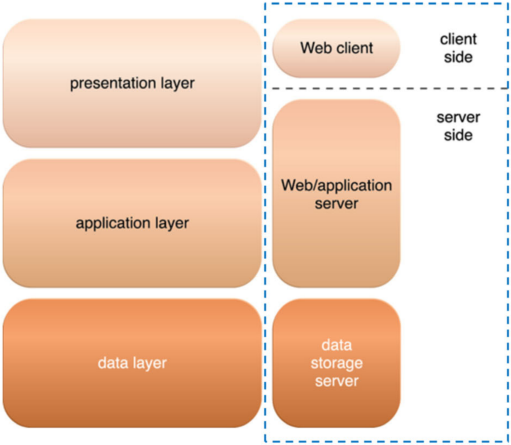
Architecture of Web Applications II

1. **Presentation layer** – sits on both the client and server side
2. **Application layer** – Web server receives client requests and retrieves the requested resources or generates web content based on application logic
3. **Data layer** – Web servers interact with application servers and the underlying database(s)



Web services technology – provides "as-a-service" cloud delivery models

Simple Object Access Protocol (SOAP)

Application protocol for web applications

Defines a common web service messaging format for request and response exchanges – based on XML and uses TCP or UDP as the transport protocols

protocol - set of rules defining communication btw 2 parties

structured format, larger enterprises, security important

Architecture of Web Applications IV

Representational State Transfer (REST)

Software Architecture for distributed hypermedia systems

Supports client communication with stateless servers

Platform and language independent

Supports data caching

lightweight option

hypermedia - different types of media

Uniform interface

Client-server decoupling

Statelessness — when you connect again, the system has no knowledge of what happened previously

Cacheability — so you can return repeated requests faster

Layered system architecture

Code on demand (optional)

Web Services for Cloud Applications

Cloud services are naturally distributed – single service request might require message passing between multiple servers

Cloud services are accessed by different clients using different programming languages and technologies (e.g., Java, Python, ...) – requires a standard interface for communication between clients and server `language agnostic`

Web service APIs implement a standard communication interface for cloud services, e.g., REST API requests/responses are transferred on HTTP using formats such as XML and JSON

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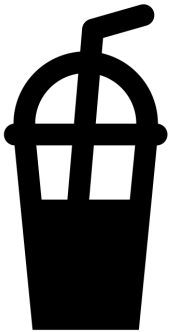
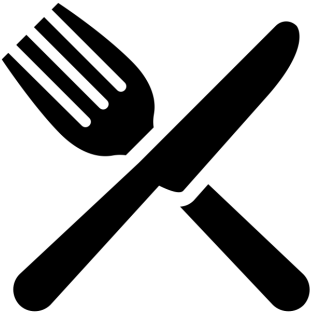
Cloud Application Development Models

Characteristics of Cloud Service Models

Example of setting up a data analytics application

Take a Break





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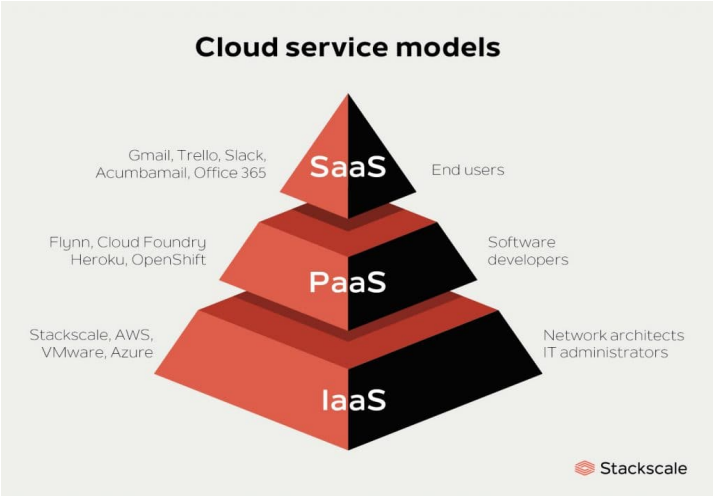
Recall: Basic Cloud Service Models

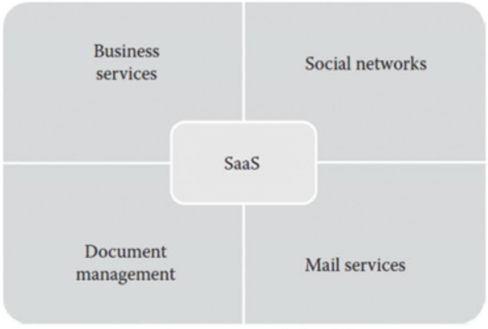


https://www.stackscale.com/wp-content/uploads/2020/04/cloud-service-models-iaas-paas-saas-stackscale-1024x711.jpg

Software as a Service

Consumers execute developed applications by supplying data and obtaining the results

Provider develops and provides the application – running of application is transparent to the consumer



Services Provided by SaaS Providers

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SaaS Providers

Characteristics

Multi-tenanted applications

Web access

Centralised management of SaaS services

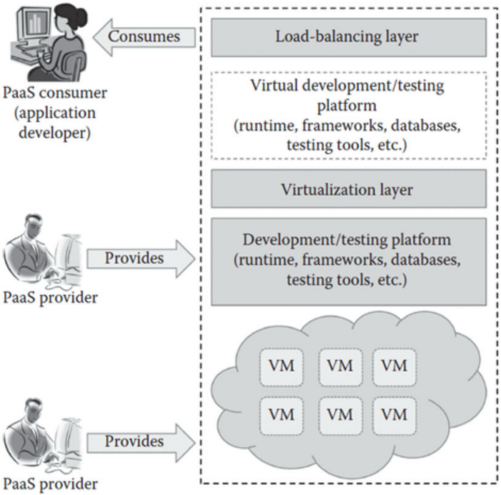
Multi-device support

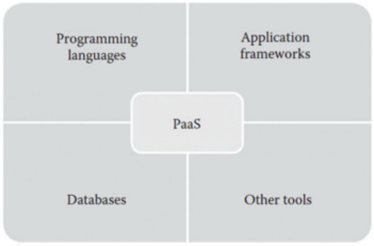
Scalability under varying loads

High availability

API integration with other software

Platform as a Service





Services Provided by PaaS Providers

used by application developers

Overview of PaaS

PaaS Providers

Characteristics

All in one – same IDE to develop, test, deploy, host, and maintain applications

Web access to development platforms

Offline access for developers

Built-in scalability

Collaborative platform for developers

Diverse client tools

examples: Salesforce, Heroku, Digital Ocean

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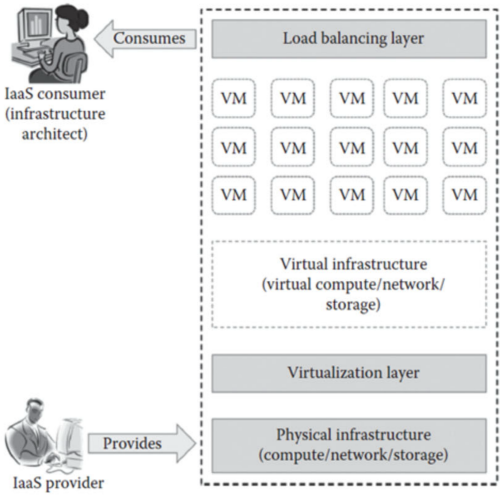
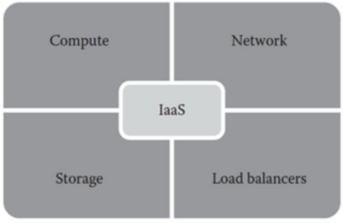
Infrastructure as a Service





Services Provided by IaaS Providers

Overview of IaaS

IaaS Providers

Characteristics

Web access to resources — `LAMP stack - packaged dev env`

Centralised physical resource management

Elastic services and dynamic scaling

Shared infrastructure across multiple users

Precofigured VMs

Metered services

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Setting up a Data Analytics Application

Key Question!

Which cloud service model to use, i.e., which "as-a-service" do you go for?

... It depends! ☺

Application is located on the cloud

Clients use a thin client (web browser) to access application

Often built on top of multi-tiered PaaS and IaaS

- Clients do not deal with underlying PaaS and IaaS
- Clients do not deal with APIs

Various pricing mechanisms – free!, pay-as-you-go, subscription, etc.

Use a data analytics application – SaaS consumer

Key Steps

1. Access your favourite application or AI ☺
2. Give it your data
3. Report on the analysis!

Setting up a Data Analytics Application

PaaS

Consumer

Creates software using tools and libraries from PaaS provider

Controls software deployment and configuration settings

Provider

Installs and maintains software, libraries and middleware `keep things up to date`

Handles the transparent infrastructure from the user

Exposes an API for PaaS consumers (programmers)

Provides integrated services of scalability, maintenance, and versioning, e.g., total application hosting, development, testing, and deployment environment

Set up a data analytics application – PaaS consumer

Key Steps — `package to build web application`

1. Access a system with MongoDB, Express, React, Node.js (MERN)
2. Build your own data analytics pipeline
3. Give it your data
4. Report on the analysis!

Set up a data analytics application – IaaS consumer

Key Steps

1. Create a compute instance running Linux
2. Install and configure MongoDB, Express, React, Node.js (MERN)
3. Build your own data analytics pipeline
4. Give it your data
5. Report on the analysis!

IaaS	PaaS	SaaS
Create a compute instance running Linux	...	...
Install and configure MongoDB, Express, React, Node.js (MERN)	Access a system with MongoDB, Express, React, Node.js (MERN)	...
Build your own data analytics pipeline	Build your own data analytics pipeline	Access your favourite application or AI
Give it your data	Give it your data	Give it your data
Report on the analysis	Report on the analysis	Report on the analysis

Next Week

Recess Week ♫

Week After

Applications and Programming (Part 2)

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References

REST APIs (`https://www.ibm.com/think/topics/rest-apis`)

Continuous Feedback

https://forms.cloud.microsoft/r/ZJxTf2wFjr

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Applications & Paradigms

Part 2

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(with thanks to Teo Yong Meng)

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Outline

MapReduce Programming Model

Function-as-a-Service

MapReduce Programming Model

Supports arbitrarily divisible workload

Supports distributed computing on large data sets on multiple machines (clusters, public or private clouds, ...)

How large is the data?

Web-scale data on the order of 100s of GBs to TBs to PBs

Input data will not likely fit on a single computer's hard drive

Needs a distributed file system, e.g. Google File System

Inspired by the map and reduce functions in functional programming languages

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MapReduce Programming Model

Key Idea!

Split data into blocks and assign each block to an instance/process for parallel execution

Merge partial results produced by individual instances after all instances have completed execution

Transforms a set of *input* <**key, value**> **pairs** into a set of *output* <**key, value**> **pairs**

SPMD (Same Program Multiple Data), where a master instance partitions the data and gathers the partial results like GPUs

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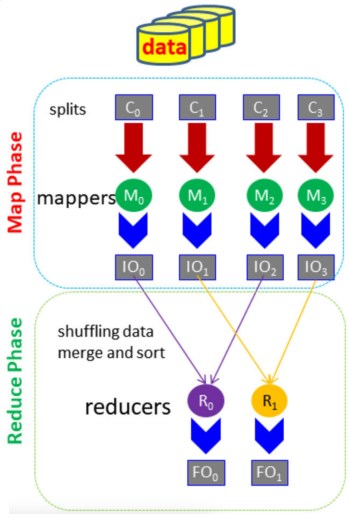
Programme Structure

1. Read (a lot of) data
2. **MAP** (extract data you need from each record)
3. *Shuffle* and *Sort* data
4. **REDUCE** (aggregate, summarise, filter, transform extracted data, etc.)
5. Write the results

imagine large scale operations on data
e.g. Google search

as a programmer
you only care about map and reduce
everything else is done internally!



you can have multiple layers of reducers
e.g. convergence for ML algos

Divided into two phases – *map* and *reduce*

Splits are processed in isolation by tasks called Mappers

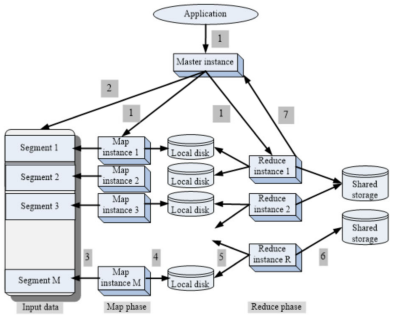
intermediary output stored on local disk

Outputs from mappers, called intermediate outputs (IOs), are brought into a second set of tasks called Reducers

Process of reading intermediate output into a set of reducers is called shuffling

Reducers produce the final outputs (FOs)

who coordinates the steps?
some kind of master node

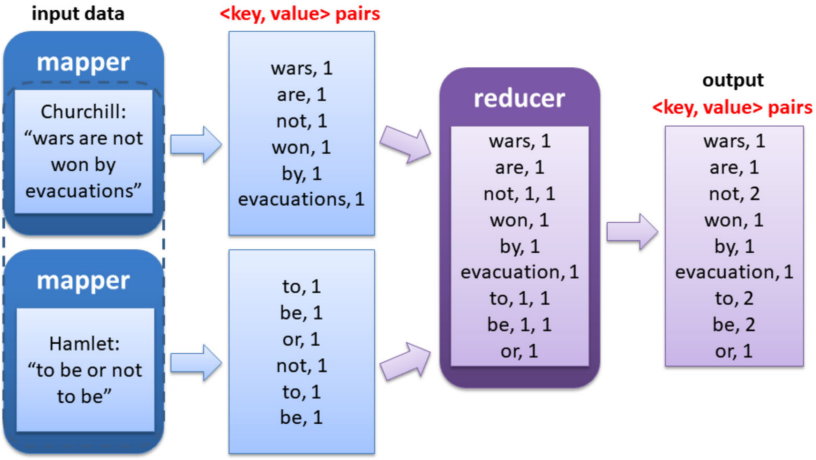


where are all these machines?
the more spread out -> more latency

1. Application starts a master instance, M worker instances for Map phase, and later R worker instances for Reduce phase
2. Master instance partitions input data into M *segments*
3. Each map instance reads its input data segment and processes the data
4. Results of the processing are stored on local disks of the servers where the map instance runs
5. When all map instances have finished processing their data, R reduce instances read the results and **merge** the partial results
6. Final results are written by the reduce instance to a shared storage server
7. Master instance monitors reduce instances and terminates application when all tasks are complete

MapReduce Example – Word Counting

Given a text file, count the number of occurrences of all words in the file



* for simplicity, we left out the sort phase

MapReduce Example – Word Counting

Map – data is partitioned into 4 smaller data files ⇒ split 0 to split 3

Shuffle

- Each smaller data file (split) is evaluated by a map task
- A map produces data (partition) or intermediate results that belongs to different reduce tasks

Merge and sort – partitions on arriving at a reduce task are merged into one file and sorted to produce a sorted file

Reduce – evaluates the sorted file to produce the final result

MapReduce programme developer codes only the map and reduce functions ⇒ shuffle, merge and sort are taken care off by the MapReduce system implementation

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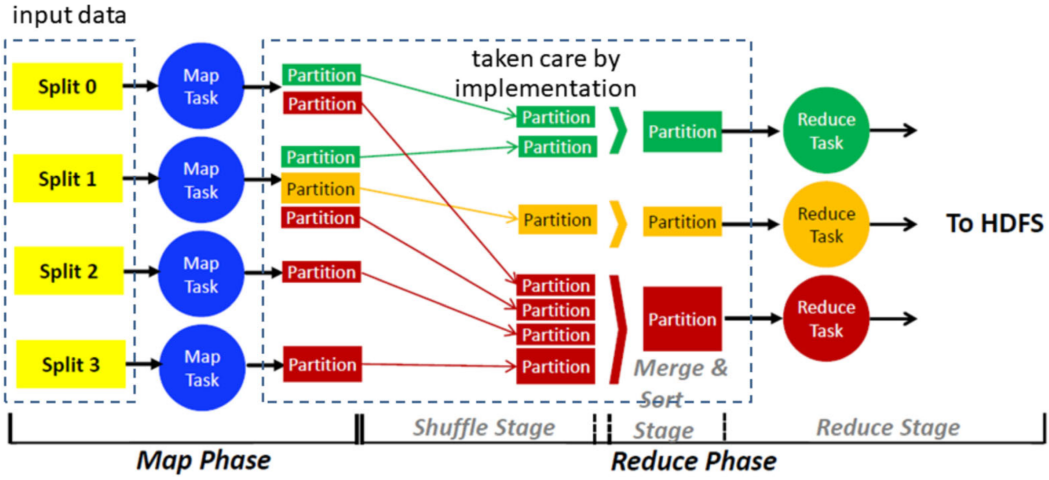
MapReduce Example – Word Counting



MapReduce Example – Word Counting

Map – gets the split data file

```
function map ()
  tokenize line;
  while (tokenizer has more tokens)
    output <key,value> pair for word
```

token = independent piece of info

Reduce – gets the word along with a list of 1s

```
function reduce ()
  int sum = 0;
  sum = total number of 1s
  output <key,value> pair for word
```

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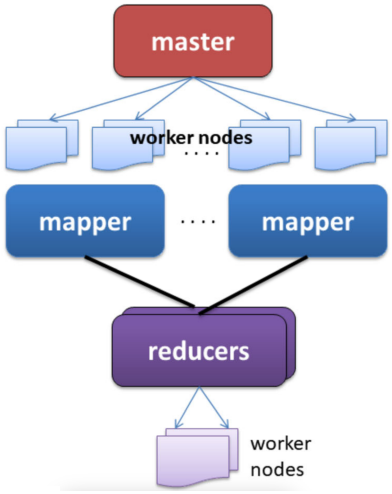


- Master program creates data-localised tasks to minimise network communication
  minimal communication -> minimum latency
- Worker nodes switch between mapper and reducer cost reduction: reuse the machine that did mapping to do reducing
- **Map** workers run in parallel
  - Each has its own data set
  - Each creates intermediate local values from each input data set
- **Reduce** workers run in parallel
  - May reduce intermediate data
  - There is an optimal number of reducers for best performance

Google

Indexing for Google search

Organising articles in Google news

Performing various user statistics

Meta

Data Mining

Optimising ads for users

Detecting spam on Facebook

Issue

Google MapReduce – closed source, developed by Google (2004) to process large amounts of raw data

Solution

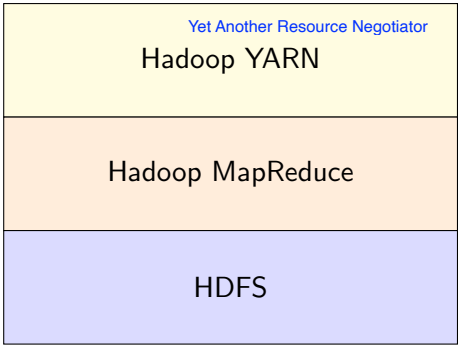
Hadoop MapReduce – open source, developed by Apache + Yahoo

large scale data processing

Presents MapReduce as an analytics engine with a distributed storage layer referred to as Hadoop Distributed File System (*HDFS*)

HDFS mimics Google File System (*GFS*)

Very popular implementation of MapReduce



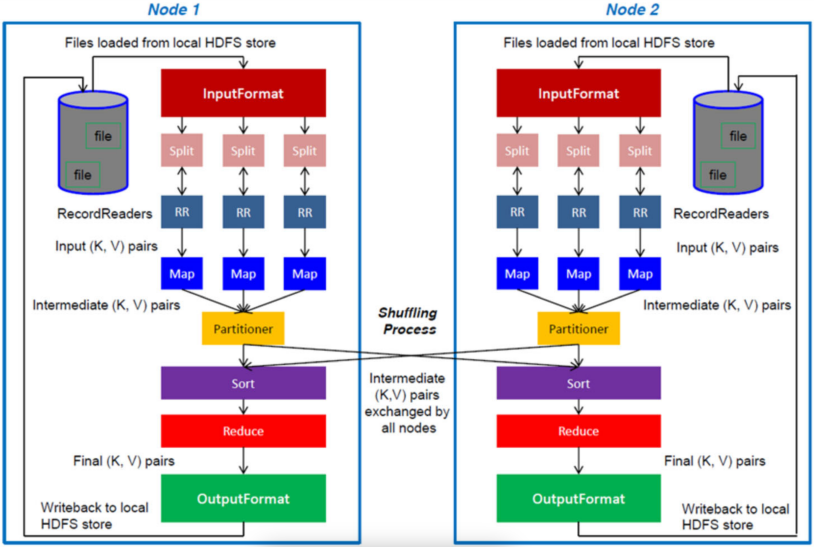
Resource Allocation

Scheduler

Computation

Storage

Hadoop on Multi-Node



where is slow?
why store things on disk?

Apache Spark

because nowaways more memory available than 5 years ago

Fast, **in memory** data processing engine

Can run independently or on top of Hadoop

Designed for real-time and advanced analytics

Feature	Apache Spark	Hadoop (MapReduce)
Batch Processing	Yes speedup	Yes
Real-time Processing	Yes	No
Machine Learning	Built-in (MLlib)	Need external tools

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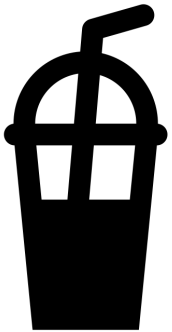
Take a Break



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Function-as-a-Service (FaaS)

Approach to build scalable, reactive, event-driven applications

there's an event you want to react to it as fast as you can

Uses functions as unit of deployment – application is divided into smaller, single-purpose (**Lambda**) functions

Event triggers a lambda function and exits after execution

No idle capacity!

e.g. fire alarm -> you just react and run

Lambda is stateless

just like REST API!

Serverless FaaS

server is taken care of for you

Build (serverless) applications without the need to provision or manage servers ⇒ consumer only pays when code is executing

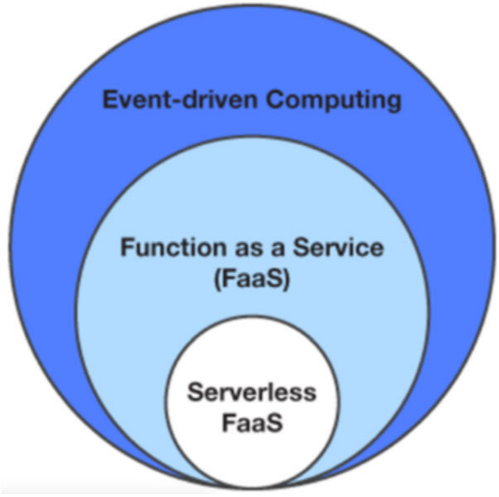
No VMs/containers in the programming model

Vendor provides provision-free scalability, i.e., scale with workload

Use Cases

Web/mobile backends, real-time data processing, chatbots, virtual assistants, ...

FaaS


Event-driven Computing
Function as a Service
(FaaS)
Serverless
FaaS

Functions should perform only one action lightweight

Functions should be self-contained ⇒ should not call other functions

Use as few libraries as possible

FaaS vs PaaS vs Containers vs VMs I

Provisioning time is much lower – ms vs minutes vs hours!

Ongoing administration is almost none

Allows for elastic scaling as each action is inherently scaled

No capacity planning required

Limited ability to persist connections and state must be kept in an external service or resource disadvantage!

stateless
just like RESTApi
cookies -> maintain state

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FaaS vs PaaS vs Containers vs VMs II

FaaS provider manages all maintenance (similar to PaaS)

High availability is inherent in the FaaS model

Resource utilisation is excellent

another advantage: easier debugging
cos functions are smaller

FaaS model has resource limits

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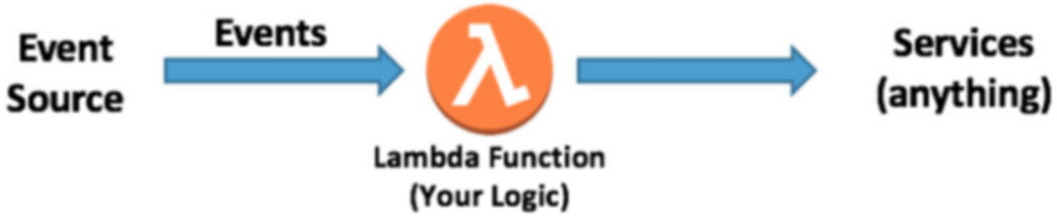
How a Lambda Function Runs



How a Lambda Function Runs

Lambda function – your application logic, i.e., code you want to execute (Java, Node.js, Python, ...)

Configuration – defines how your code is executed, i.e., configure event sources that invoke your functions

Example

Processing an event when an API gateway receives an HTTPS request

System takes care of code to integrate event source with lambda function, manage infrastructure that detects events and invoke lambda, etc.

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3 Steps

1. **Create lambda service**
   - Business logic (code)
   - Configuration – region to run, AWS credentials, etc.
   - Event – the event that triggers a lambda function
2. **Deploy service created**
3. **Invoke/test your service**

Using AWS Lambda

Web Interface

AWS Management Console
(`https://aws.amazon.com/lambda/getting-started/`)

Command Line Interface

AWS SAM CLI (`https://docs.aws.amazon.com/serverless-application-model/latest/developerguide/serverless-getting-started-hello-world.html`) Serverless Application Model

There are also some 3^{rd} party CLIs available

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Next Week

More Examples! ☺

References

What is Function as a Service (FaaS)?
(https://www.ibm.com/think/topics/faas)

MapReduce: Simplified Data Processing on Large Clusters
(https://static.googleusercontent.com/media/research.google.com/en/
/archive/mapreduce-osdi04.pdf)

Apache Hadoop (https://hadoop.apache.org/)

The Hadoop Distributed File System
(https://ieeexplore.ieee.org/document/5496972)

Continuous Feedback

https://forms.cloud.microsoft/r/ZJxTf2wFjr

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Example Applications

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Outline

Example 1

K-Means Clustering using MapReduce

Example 2

Building a Video-Sharing SaaS Cloud Application

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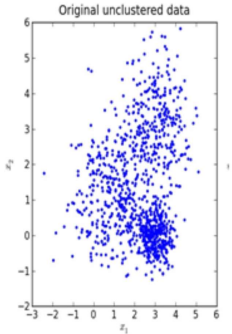
K-Means Clustering

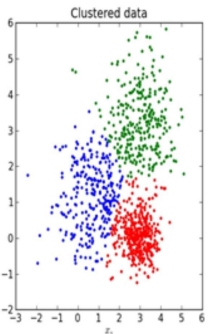
Algorithm to cluster **m objects** based on **n attributes** into **k groups**, where $\mathbf{k} < \mathbf{m}$

Minimise distances between intra-cluster points

Maximise distances between inter-cluster points

Useful method in data analysis (e.g., retail, streaming services, sports science, etc.)





Distance Metric

Three properties

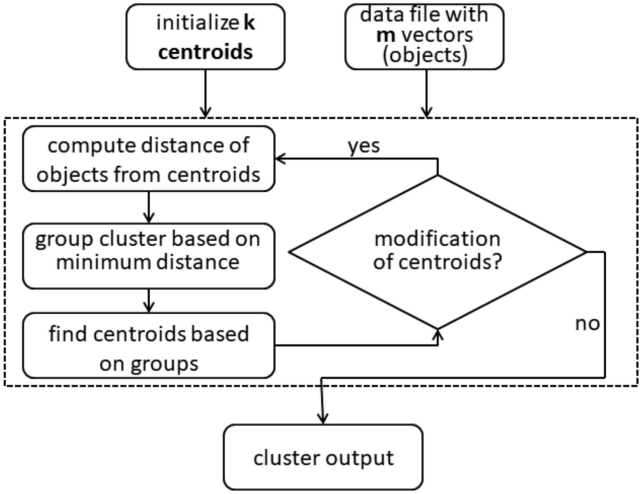
1. Non-negative
2. Equal to 0 iff $p = q$
3. Triangle inequality, i.e., for 3 points p, q and r, d (p, r) $<$ d (p, q) $+$ d (q, r)

Choice of different distance metrics

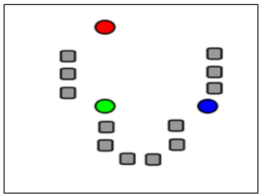
One class is the p-norm

- For any p > 0, distance between vectors q and r (each of dimension n) is defined as $((q_1 - r_1)^p + (q_2 - r_2)^p + \ldots + (q_n - r_n)^p)^{\frac{1}{p}}$
- E.g., Euclidean distance $(2 - \textit{norm distance})$

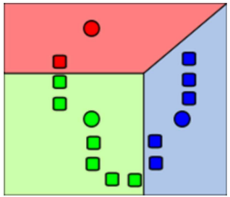
K-Means Algorithm I



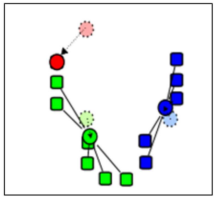
K-Means Algorithm II



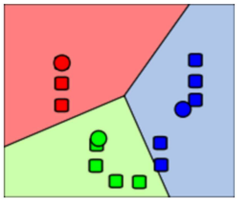
1. initialization



2. compute distance and form clusters



3. recompute centroids



4. output

K-Means Algorithm III

Steps

1. Initialise
2. Iterate
   - Compute distance and form clusters
   - Recompute centroids
3. Output results

Challenge!

What is **k**?

Dataset

Twitter dataset from `https://snap.stanford.edu/data/egonets-Twitter.html`

Aim

Apply k-means to determine how many Twitter users fall under a cluster with a centroid of X followers and Y followees

Using MapReduce

What is the **Map** phase? – compute clusters

What is the **Reduce** phase? – compute new centroids

Map Phase

Compute distances between an input point and all the previously obtained centroid of clusters

Find the cluster with the minimum distance

Add the input point into the cluster with the minimum distance

Reduce Phase

Receive all the points belonging to a particular cluster

Compute new centroid

Input Key-Value Pairs I

Important!

Pre-process the data!

Key $\rightarrow$ ID of a user (or participating point)

Value $\rightarrow$ vector of attributes of that particular ID

Example

key: 104997734; value: len: 2; 0:1.0; 1:2.0;

key: 46198765; value: len: 2; 0:5.129283016944966; 1:2.321928094887362;

"len: 2" $\Rightarrow$ two elements

num followers num followees

"0:5.12" $\Rightarrow 0^{th}$ element is 5.12 $\Leftrightarrow \log_2$ (number of followers)

"1:2.32" $\Rightarrow 1^{st}$ element is 2.32 $\Leftrightarrow \log_2$ (number of followees)

Input Key-Value Pairs II

Why log?

Deal with outliers in the data

E.g., if someone has 20,000 followers, the clusters are heavily affected and hence, common practice is to take log values to dampen this type of effect

Output Key-Value Pairs

Key → ID of a cluster

Value → vector representing centroid for the cluster and the number of points inside the cluster

Example

key: cluster0; value: 23904; len 2; 0:2.005394687254275; 1:1.3290466203187175;

key: cluster1; value: 22125; len 2; 0:5.281555290448195; 1:5.6107313297218275;

x coords y coords

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K-Means Implementation I

Steps

1. Upload data to cloud
2. Upload executable code to cloud
3. Create clusters
4. View output results on cloud
5. Repeat as necessary, to get next clusters

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K-Means Implementation II

Technology Needed!

Storage (e.g., S3)

- Amazon S3 (Simple Storage Service) is a web **storage** service

MapReduce Engine (e.g., Amazon EMR)

- Amazon EMR (previously called Amazon Elastic MapReduce) is a managed cluster platform that simplifies running big data frameworks

K-Means Implementation III

AWS Example

Data and code loaded on S3 storage

Implementation using EMR

Output results stored on S3

K-Means Example Output I

key: cluster0; value: 23904; len 2; 0:2.005394687254275; 1:1.3290466203187175;

key: cluster1; value: 22125; len 2; 0:5.281555290448195; 1:5.6107313297218275;

key: cluster2; value: 5166; len 2; 0:5.051385274987691; 1:2.7115082846455305;

key: cluster3; value: 25074; len 2; 0:2.944391259561573; 1:3.7132202620068284;

What can we deduce?

K-Means Example Output II

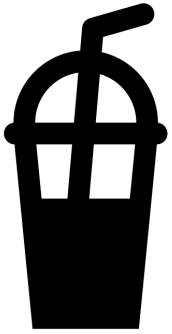
Number of points in the cluster that has a centroid with $\log_2$ (number of followers) = 5.05 and, $\log_2$ (number of followees) = 2.71 is only **5166**

Other clusters with small differences between number of followers versus number of followees contain many more points

Verdict

A user typically has a similar number of followers and followees

Take a Break



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Objective

Aim

Demonstrate the building of a SaaS video-sharing application

Users can **upload** (store) videos

Users can **stream** or **download** videos

Challenges

Videos are uploaded in different formats ⇒ convert them to streaming format

Video conversion – CPU intensive!

not every format can be streamed..

Upload of videos and streaming videos to multiple users – I/O intensive!

What about user accounts? For now, no (thankfully! ☺)

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Video-Sharing Application

Steps

1. Upload shared videos (**Upload Process**)
2. Encode uploaded videos (**Encoding Process**)
3. Stream shared videos to users (**Streaming Process**)

Upload Process I

Requirements

Cloud consumer uses a web browser

SaaS Provider

- Interface to upload videos
- Storage to store videos
- Need to balance workload

Upload Process II

Technology Needed!

Web servers – to manage client connections (e.g., EC2)

- Amazon EC2 (Elastic Compute Cloud) is a web service that enables launching and managing *server instances* in Amazon's data centers

Storage servers (e.g., S3)

- Amazon S3 (Simple Storage Service) is a web **storage** service

Additional elements could include a load-balancer, auto-scaling, message passing, database, etc.

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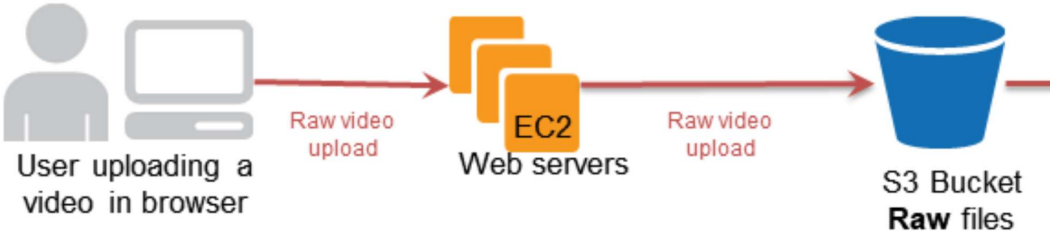
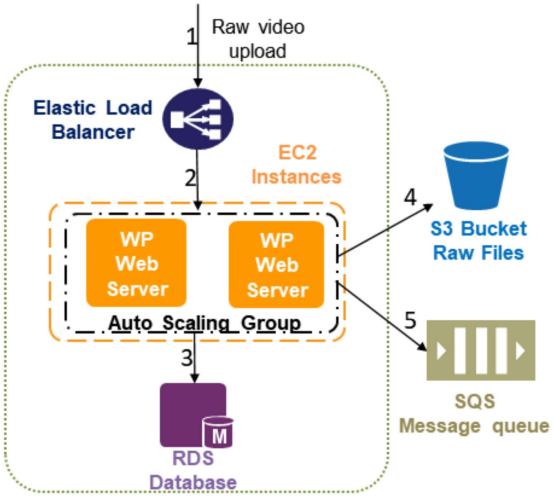
Upload Process III



Upload Process

Detailed



machine level

need to program some logic

Name	IaaS	PaaS	SaaS
EC2	Y		
S3	Y		
Auto scaling	Y		
ElasticLoadBalancer	Y		
SQS		Y	
Web Browser			Y

Encoding Process I

Requirements

SaaS Provider

- Video encoder
- Storage to store encoded videos
- Need to balance workload (because encoding is compute expensive)

Encoding Process II

Technology Needed!

Video encoder (e.g., ffmpeg)

- ffmpeg is an open source project, which allows for video transcoding

Servers – to manage the encoding (e.g., EC2)

Storage servers (e.g., S3)

Additional elements could include auto-scaling, message passing, etc.

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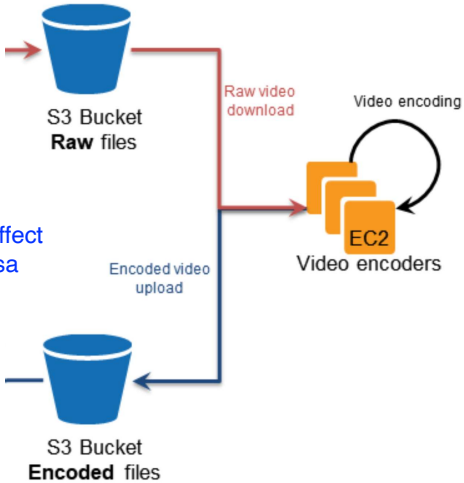
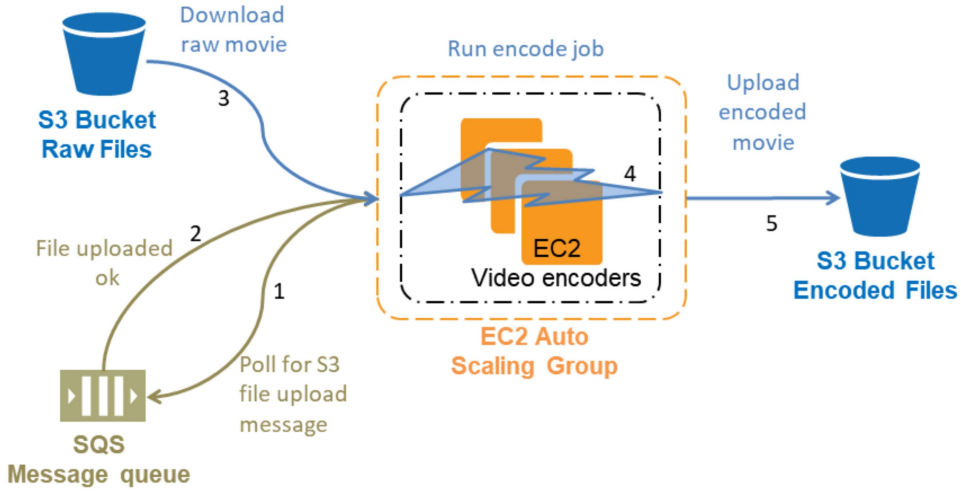
Encoding Process III



why separate?
so busy uploading wouldnt affect the encoding and vice versa

Encoding Process

Detailed



Name	IaaS	PaaS	SaaS
EC2	Y		
S3	Y		
Auto scaling	Y		
ElasticLoadBalancer	Y		
SQS		Y	
Video Encoder		Y	

internal component for dev

Streaming Process I

Requirements

Cloud consumer uses a web browser

SaaS Provider

- Video streamer
- Storage for encoded videos (shared with previous stage)
- Need to balance workload

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Streaming Process II

Technology Needed!

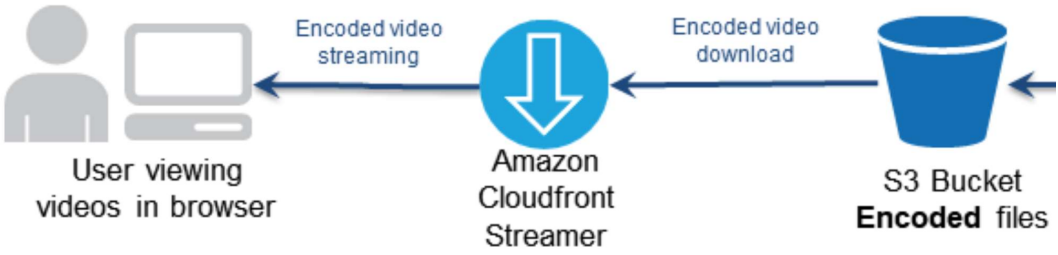
Streaming servers – to stream the videos (e.g., Amazon CloudFront Streamer)

- Amazon CloudFront is a web service that speeds up (content) distribution of static and dynamic web content through a worldwide network of edge locations

local cache

Storage servers (e.g., S3)

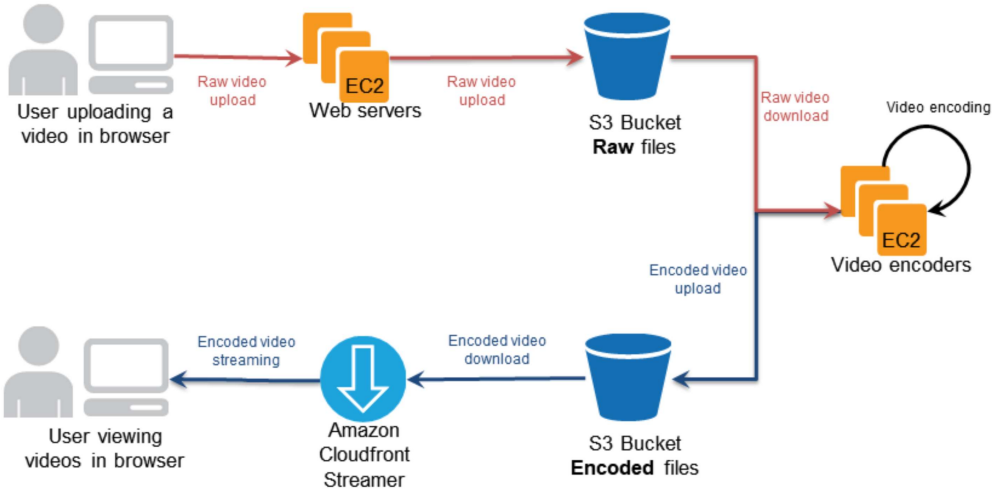
Streaming Process III



for the end user

Name	IaaS	PaaS	SaaS
S3	Y		
CloudFront			Y
Media Player			Y
Entire Application			Y

Putting It All Together



Pricing

Costs

EC2 Instances

S3 storage

CloudFront content delivery network

PaaS software – based on usage

Pricing Example

≈ 50 uploading hits per day

- Each upload is 100 MB of HD movie
- Encoding process takes 1 minute per movie
- 5 GB per day or 150 GB per month

≈ 500 streaming hits per day

- Each user streams 100 MB of HD movies (≈ 3 minutes)
- Assume same location
- 50 GB per day or 1.5 TB per month

Can use AWS Pricing Calculator – `https://calculator.aws`

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Pricing Comparisons

Traditional

Compare cloud vs on-premise

Inter-Cloud

Compare between different cloud providers

Intra-Cloud

Compare between different offerings with the same provider (e.g., different EC2 instances, etc.)

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Next Week

Infrastructure as Code

Continuous Feedback

https://forms.cloud.microsoft/r/ZJxTf2wFjr

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Infrastructure as Code

Anandha Gopalan

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Outline

Basics

Cloud Provider Tools

Independent Tools

What is Infrastructure?

Virtual Machines

Networks

Databases

Load-balancers

Other cloud resources

Challenge!

How do we manage them?

Managing Infrastructure

Graphical User Interface – click a lot of interesting looking buttons ☺

Command line - write a lot of interesting looking scripts ☺

Potentially many issues

Manual and time-consuming

Complex and error-prone

Scalability

Configuration drift

...

Idea

Express desired infrastructure through code

Infrastructure as Code

Ability to provision and support your computing infrastructure using code instead of manual processes and settings
(`https://aws.amazon.com/what-is/iac/`)

Why code?

Can use best practices from software development

- Version control
- CI/CD pipelines
- Testing
- Documentation

Infrastructure as Code

Key DevOps Practice

Bridges the gap between development and operations

Automate infrastructure setup

Rapid deployment

Repeatable and Consistent deterministic

Easy scalability

...

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IaC in DevOps

Enabling CI/CD by seamlessly integrating into the CI/CD pipelines

Improved testing by creating necessary production-like environments

Support team collaboration as application code and IaC code can be together

Approaches

Imperative

(Task Management)

Developer describes the exact steps to set up the necessary infrastructure

Developer needs to keep track of the system state and manage any necessary changes

Declarative

(State Management)

Developers describes the necessary state of the infrastructure

Automatically handles the necessary updates and changes

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Imperative vs Declarative

Getting Dinner

Imperative	Declarative
What we need to do	What is required
Leave COM2 and go to BIZ2	Go to west coast hawker ctr
Head down Heng Mui Keng Terrace	
Cross the road and take bus 30/51/143	
Get off at west coast hawker ctr	

Imperative	Declarative
Control over every step	Requires less coding skills
Ideal for small tasks	Highly repeatable
	Adapts well to configuration drift
	Starting point does not matter
	Changes to destination handled automatically

Imperative	Declarative
Difficult to maintain	Less control
More prone to human errors	Can over complicate simple tasks
Less idempotent	

Some IaC Tools

Cloud Providers

AWS CloudFormation and CDK

Azure Resource Manager

Google Infrastructure Manager

Independent Tools

Terraform

Pulumi

AWS CloudFormation and CDK

https://aws.amazon.com/what-is/iac/

imperative

AWS Cloud Development Kit (AWS CDK)

Define cloud application resources with familiar programming languages and interactive configuration tools, all in the IDE

declarative

AWS CloudFormation

Define and manage cloud resources published in the CloudFormation registry, the developer community, and internal libraries

Azure Resource Manager

https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/overview

Deployment and management service for Azure

Manage infrastructure through declarative templates

Manage resources as a group

Redeploy resources in a consistent state

Azure RBAC is natively integrated into the management platform

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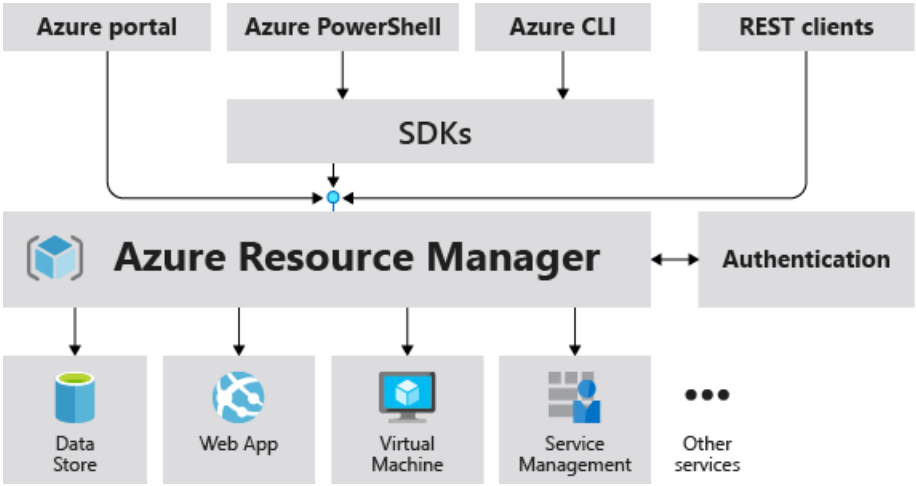
Azure Resource Manager



Google Infrastructure Manager (Infra Manager)

https://docs.cloud.google.com/infrastructure-manager/docs/overview

New! (Google Cloud Deployment Manager was discontinued on **01/04/26**)

Automates the deployment and management of Google Cloud Infrastructure

Uses . . . Terraform

- Terraform configuration defines the infrastructure
- Configuration is deployed onto Google Cloud by Infra Manager

Application deployment is done via Cloud Build and Cloud Deploy – third party tools can also be used

Pets

Servers are individually managed and cared for

Characteristics

- Named and treated with high importance
- Manual intervention is required for maintenance and recovery

E.g.: Dedicated web-server

Cattle

Servers are interchangeable resources – especially in a cloud environment

Characteristics

- Configured identically
- Easily replaced when they fail, with no need for manual repairs

E.g.: Web-server running (with a load-balancer) that can be quickly changed

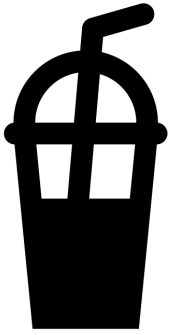
Which one to use?

Recall: IaC is about managing infrastructure

Uses **Cattle** model effectively

Allows for rapid, consistent deployment and management of cloud resources

Take a Break



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Terraform Basics

https://developer.hashicorp.com/terraform

Released in 2014 from HashiCorp

Open-source – though under a Business Source Licence from 2023

Can be used to manage complex, multi-cloud infrastructure

Uses Declarative language

Cloud Agnostic – large list of providers supported

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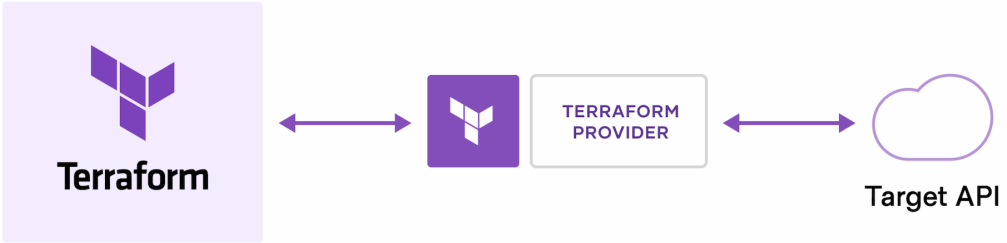
Terraform

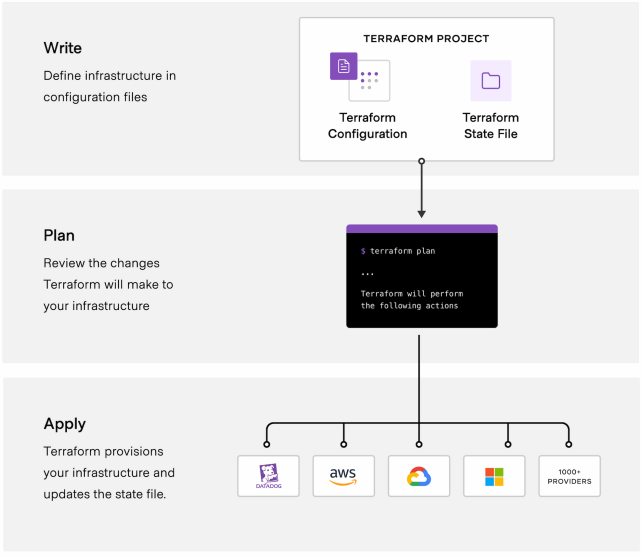
Creates and manages resources on cloud platforms through the use of APIs

Providers enable Terraform to work with virtually any platform or service with an accessible API



Terraform registry contains many thousands of providers – written by HashiCorp and the Terraform community

Core Workflow



Managing Infrastructure

Track Infrastructure

- Keeps track of infrastructure in a state file

Automate changes

- Builds a resource graph to determine resource dependencies and creates or modifies non-dependent resources in parallel

Standardize configurations

- Supports reusable configuration components called modules that define configurable collections of infrastructure
- Encourages best practices
- Can use publicly available modules from the Terraform Registry, or write your own

Collaborate

- Configuration file can be added to a version control system
- Can use HCP Terraform to effectively manage workflow across teams

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Write I

```
# Create repository
$ git init my-infra && cd my-infra

Initialized empty Git repository in /.../my-infra/.git/

# Write initial config
$ vim main.tf

# Initialize Terraform
$ terraform init

Initializing provider plugins...
# ...
Terraform has been successfully initialized!
```

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Write II

```
# Make edits to config
$ vim main.tf

# Review plan
$ terraform plan

# Make additional edits, and repeat
$ vim main.tf
```

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Plan

```
$ git add main.tf
$ git commit -m 'Managing infrastructure as code!'

[main (root-commit) f735520] Managing infrastructure as code!
1 file changed, 1 insertion(+)
```

```
$ terraform apply

An execution plan has been generated and is shown below.
# ...
```

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Apply

```
Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.
Enter a value: yes

# ...

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

```
$ git remote add origin https://github.com/*user*/*repo*.git
$ git push origin main
```

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Terraform Language

Main purpose is declaring resources (Infrastructure Objects)

Declarative

Configuration

Document that tells Terraform how to manage a given collection of infrastructure

Can have multiple files and directories

Syntax

Consists of only a few basic elements – Blocks, Arguments, and Expressions

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Example Configuration

Creating an S3 bucket

```
resource "aws_s3_bucket" "example" {
bucket = "my-tf-test-bucket"

tags = {
  Name        = "My bucket"
  Environment = "Dev"
  }
}
```

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Pulumi

https://www.pulumi.com/

Open sourced in 2018

Can write infrastructure code using familiar programming languages – TypeScript, Python, Go, and C#

Flexibility for more complex logic

Available with Agentic AI now ☺

Pulumi Neo - you can now use Natural Language!

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Creating an S3 bucket

```
import pulumi
from pulumi_aws import s3

# Create an AWS resource (S3 Bucket)
bucket = s3.Bucket(`my-bucket')

# Export the name of the bucket
pulumi.export(`bucket_name', bucket.id)
```

Learning Curve

Large number of Tools

Code Complexity

Version Control

Configuration Drift configuration drift is reduced, but still not 0, e.g. Git merge conflits

Security and Compliance

Are these enough?

Terraform
Pulumi
OpenTofu
Terragrunt
Spacelift
Puppet
Vagrant
...

`https://spacelift.io/blog/infrastructure-as-code-tools`

Misconfigurations – needs continuous monitoring to ensure no security issues (such as access to sensitive data, incorrect deployment, etc.)

Security – need to protect the code and credentials or access keys, else can lead to data breaches

Compliance – configuration drift may lead to such standards being broken, risking a regulatory violation

Using AI – deploy so and so at so and so with so and so ☺

Policy-as-Code (PaC) – making IaC pipelines policy-aware compliant

Rise of GitOps – automatically deploy changes from Git to system

Self-healing IaC – automatically detect anomalies and update, as necessary

...

Self-healing IaC

Issue

Infrastructure is growing increasingly complex

Three Steps!

Monitor – collect data on workload and current state of servers, etc.

Analyze – use analytics to compare and contrast, use baselines and prediction models as necessary etc.

Remedy – create a remedial plan and action it

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Next Week

Pricing Models & Modeling TCO

Continuous Feedback

https://forms.cloud.microsoft/r/ZJxTf2wFjr

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Pricing Models & Modeling TCO

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Outline

Pricing Models

Cost Metrics

Total Cost of Ownership (TCO)

Pricing

Providers

How do providers price cloud resources?

Consumers

What is the cost of using cloud resources?

Pricing Models

Price

Unit cost of using cloud resources, e.g., based on time, number of transactions, etc.

Charge by per hour, per minute, per second

Factors

Overhead in design, development, deployment, and operation of cloud services, and other IT resources

Opportunities to reduce expenses via IT resource sharing and data center optimisation

Market competition and regulatory requirements

Cost metrics and associated prices – depends on **on-demand** (pay-per-use) or **reserved allocation** (upfront fee + discounted hourly rate)

Discount for higher usage/volumes

Payment options and schedules

- Customisation
- Prepaid, postpaid, instalments

Examples

IaaS

Pricing based on resource allocation and usage including number of virtual servers, amount of network data transferred, and allocated storage capacity

PaaS

Price also depends on software configurations, development tools and any licensing fees

SaaS

Price is determined by number of application modules in the subscription, number of nominated cloud service consumers, and number of transactions

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AWS EC2 (https://aws.amazon.com/ec2/instance-types/)

Virtual machines (IaaS) are divided into many types to support different types of applications and consumer requirements

Types

- General purpose – M9, M8, M7, T8, ...
- Compute optimized – C9, C8, C7, C6, ...
- Memory optimized – R9, R8, X8, X2, ...
- Accelerated computing – P6, P5, G7, G6, ...
- Storage optimized – I8, I7, H1, ...
- HPC optimized – Hpc8, Hpc7, Hpc6, ...

AWS pre-created many types of machines

Each instance type is further divided into subtypes based on varying combinations of CPU, memory, storage, and network capacity

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AWS EC2

Pricing

Viewing 1,225 available instances

1 2 3 4 5 6 7 ... 62

Instance name	On-Demand hourly rate	vCPU	Memory	Storage	Network performance
t8i.nano	$0.00624	2	0.5 GiB	EBS Only	Up to 6.25 Gigabit
t8i.micro	$0.01248	2	1.0 GiB	EBS Only	Up to 6.25 Gigabit
t8i.small	$0.02496	2	2.0 GiB	EBS Only	Up to 6.25 Gigabit
t8i.medium	$0.04992	2	4.0 GiB	EBS Only	Up to 6.25 Gigabit
t4g.nano	$0.0042	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t4g.micro	$0.0084	2	1 GiB	EBS Only	Up to 5 Gigabit
t4g.small	$0.0168	2	2 GiB	EBS Only	Up to 5 Gigabit
t4g.medium	$0.0336	2	4 GiB	EBS Only	Up to 5 Gigabit
t4g.large	$0.0672	2	8 GiB	EBS Only	Up to 5 Gigabit
t4g.xlarge	$0.1344	4	16 GiB	EBS Only	Up to 5 Gigabit
t4g.2xlarge	$0.2688	8	32 GiB	EBS Only	Up to 5 Gigabit
t3.nano	$0.0052	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t3.micro	$0.0104	2	1 GiB	EBS Only	Up to 5 Gigabit
t3.small	$0.0208	2	2 GiB	EBS Only	Up to 5 Gigabit
t3.medium	$0.0416	2	4 GiB	EBS Only	Up to 5 Gigabit
t3.large	$0.0832	2	8 GiB	EBS Only	Up to 5 Gigabit
t3.xlarge	$0.1664	4	16 GiB	EBS Only	Up to 5 Gigabit
t3.2xlarge	$0.3328	8	32 GiB	EBS Only	Up to 5 Gigabit
t3a.nano	$0.0047	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t3a.micro	$0.0094	2	1 GiB	EBS Only	Up to 5 Gigabit

https://aws.amazon.com/ec2/pricing/on-demand/

Example

AWS EC2 Pricing

Charges differ across region and depends on the operating costs (energy, communication, and maintenance)

for spiky workloads

On-demand – based on usage in terms of time; charge to nearest second with minimum of 60 seconds

for steady workloads

Reserved – onetime upfront cost + pay as your go (cost is lower than on-demand)

Spot – consumers bid for unused capacity at a lower price than on-demand rate; suitable for running jobs that are not time critical and at a lower cost

for non-critical workloads

Savings Plan – flexible pricing model that offers low prices in exchange for a commitment to a consistent usage for a longer term

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you decide based on your workload!

SaaS Pricing Models

Integrated Pricing

Consumer pays SaaS provider

SaaS provider in turn pays IaaS provider for hosting the SaaS service

Separate Pricing

Consumer pays SaaS provider for SaaS service and IaaS provider for hosting the SaaS service

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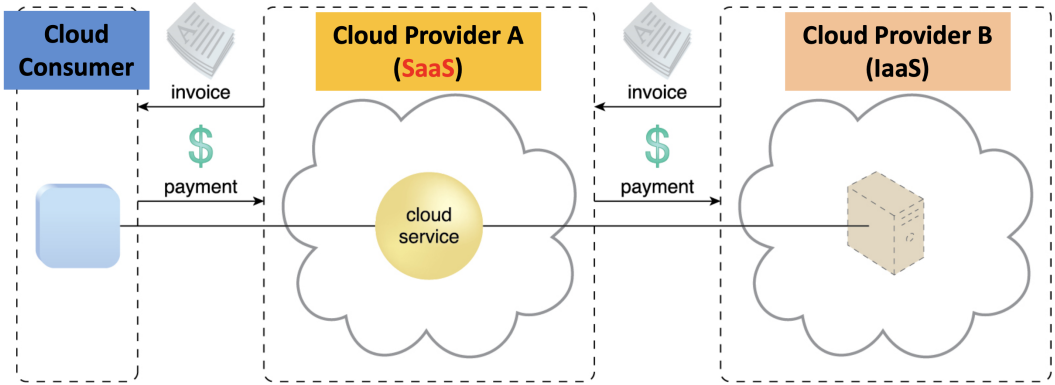
Cloud Consumer
invoice
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payment
Cloud Provider A (SaaS)
cloud service
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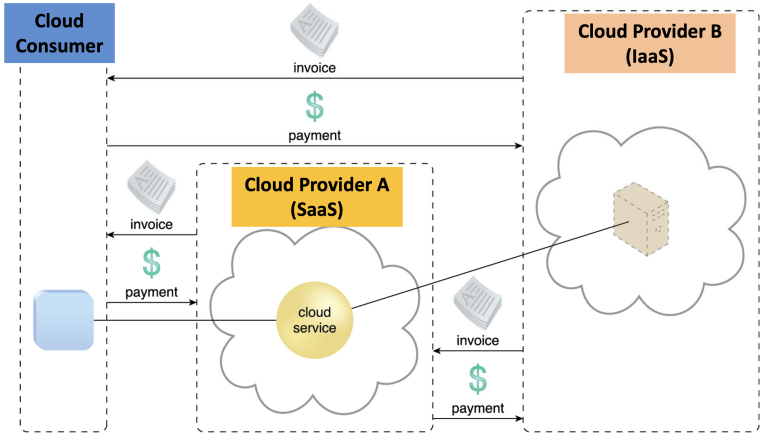
Cloud Provider B (IaaS)


Cloud Consumer
invoice
Cloud Provider B (IaaS)
payment
Cloud Provider A (SaaS)
invoice
payment
cloud service
invoice
payment

Cost Metrics

Financial analysis of cloud adoption

Compare on-premise (total cost of ownership (TCO)) vs cloud-based provisioning

Usage Scenario

Cloud consumer

Cloud provider – how to price a cloud service, etc.

Business Cost Metrics

Business costs = capital expenses + operational expenses

Up-front costs

- capital expenses (Capex)
- initial investment to fund IT resources
- obtain resource, deploy and administer
- on-premise is high – e.g., hardware, software and deployment costs
- cloud based is low – e.g., labour cost for setting up cloud environment

On-going costs

- operational expenses (Opex)
- run and maintain IT resources – e.g., licensing fee, electricity, maintenance
- higher for on-premise

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cost of borrowing

Cost of capital

Cost incurred to raise funds – if high, leasing cloud-based is justified

Sunk costs

Prior investment on existing IT resources

If significant, cannot justify cloud-based

Integration costs

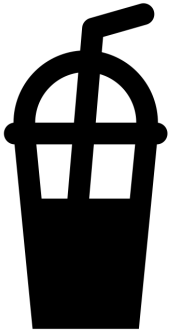
Effort needed to inter-operate IT resources on new environment – e.g., new cloud platform

High cost $\Rightarrow$ cloud-based less appealing

Locked-in costs

Movement from one cloud provider to another – high cost $\Rightarrow$ cloud-based less appealing

Take a Break



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Cloud Usage Cost Metrics

Network usage (data transfer)

Inbound, outbound and intra-cloud network traffic

Server usage (compute)

On-demand and reserved virtual machine allocation

Cloud storage device (storage)

On-demand storage allocation, I/O data transfers

Cloud service (service)

Subscription duration, number of users, number of transactions

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Network Usage Metrics

Total network traffic in bytes

Accumulated over a pre-defined period – e.g., up to 1GB free, S$ 0.01/GB till 1TB and S$ 0.005/GB after 1TB per month

Inter-Cloud

- Data replication, synchronization
- May have different rates for in-bound and out-bound (many providers do not charge for in-bound)

Intra-Cloud

Some providers may not charge for it

because that's necessary for the cloud to work

Server Usage Metrics

Range of performance attributes – CPU, RAM, dedicated storage, ...

On-demand

Pay-per-usage fee short-term

Cumulative from server start to stop date

Reserved

Upfront cost for reservation

Start date to expiry date

Costs: daily/monthly/yearly

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Cloud Storage Device Metrics

On-demand storage space

Size of storage space in bytes

Cumulative sum over date of allocation to release

I/O data transferred metric

I/O data in bytes

Some providers may not charge

Cloud Service Usage Metrics

Application subscription duration

Cumulative sum from subscription start to expiry date

Number of nominated users

Number of users – e.g., S$ 0.90/additional user per month

Number of transactions

Number of request-response message exchanges

E.g., S$ 0.05 per 1,000 transactions

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Total Cost of Ownership of a Datacenter I

Costs

capital expenses (Capex) + operational expenses (Opex)

Capex

Upfront investment depreciated over a certain time frame such as construction cost of datacenter and purchase price of servers

Opex

Recurring cost of running datacenter such as electricity costs, repairs and maintenance, salaries, etc.

Total Cost of Ownership of a Datacenter II

TCO

Cost of owning the hardware resources (hardware + infrastructure to keep hardware working)

`datacenter depreciation` + building
`datacenter Opex` +
`server depreciation` + IT equipment
`server Opex`

Comparing TCOs

Case A

High-end servers

Case B

Lower-cost, higher-power servers

on average case B lowest cost
because cost of higher power doesn't
offset cost of depreciation

Case C

Case B with 50% datacenter utilisation

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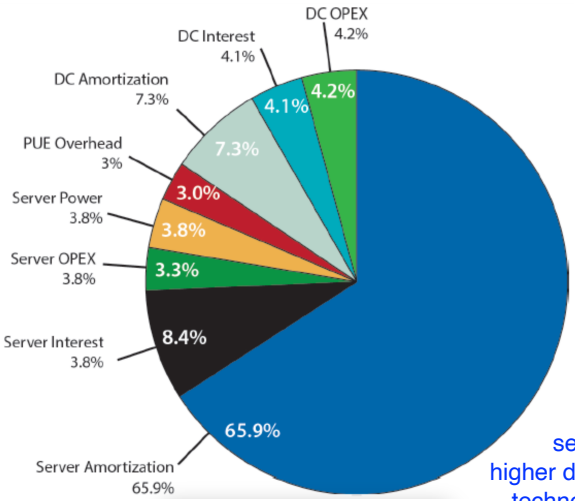
Case A

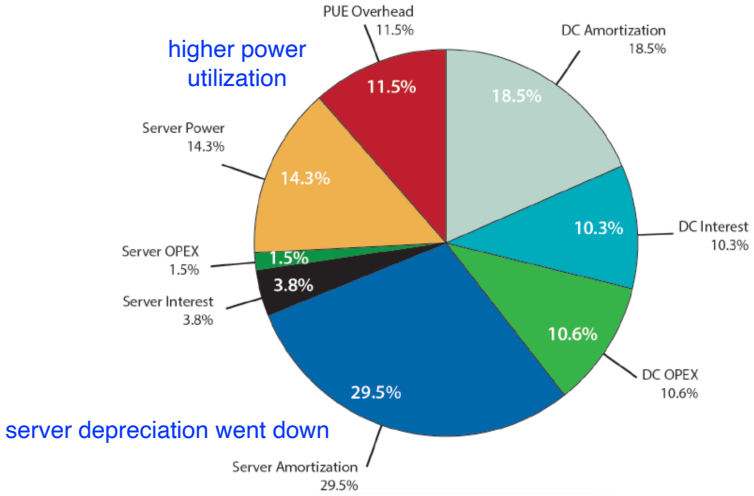
Cost Distribution



server amortization:
higher depreciation due to rapid technological obsolescence

Case B

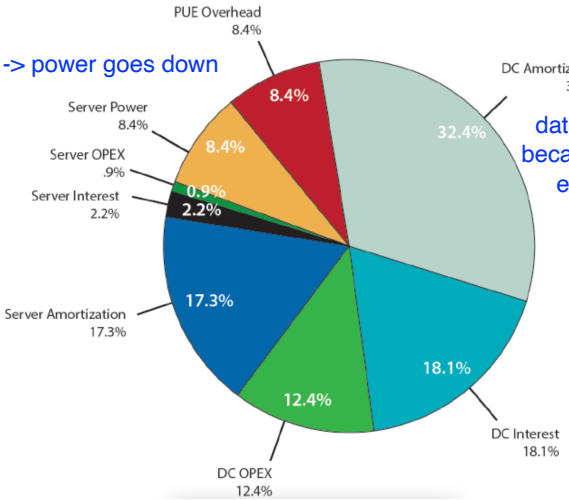
Cost Distribution



Case C

Cost Distribution

50% utilization -> power goes down



data center depreciation goes up because you're only using 50% of it even though it has been built

Test

Date – **November 3rd, 2026**

Time – **18:30 – 20:00**

Venue – LT 17

Format – Examplify (Make sure your laptops are **charged** and can last the time!)

Open-book

- We will be using the **Non-secure Block Internet** option on Examplify

Consists of MCQs, MRQs and structured questions

- Few book work
- More about applying your knowledge ☺

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Next Week – Guest Lecture

EVENT DETAILS

27 October 2026

6:30pm – 8:00 pm
NUS, LT 17

Eugene Swee

Principal AI Engineering Solutions Architect
APJ AI Innovation Hub,
Dell Technologies

"Over 20 years of experience across AI Engineering, Software Development, Operations, DevOps and MLOps, Presales and post-sales."

GUEST LECTURE

From Cloud Native to AI-Native: Building Modern Applications

What you'll explore

In today's fast-paced digital landscape, technology is a key differentiator, helping businesses deliver value faster, smarter, and at scale. This session explores how Cloud Native approaches create agile, resilient applications and infrastructure ready for AI-driven innovation.

Join Dell Technologies for an insightful session on the transformative shift in software and AI Engineering. Learn about how **microservices**, **AI Infrastructure**, **MLOps**, and **AI Engineering** work together to build, deploy, and manage modern intelligent applications.

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DELL Technologies

References

Chapter 15, Cloud Computing: Concepts, Technology & Architecture, Thomas Erl, et al., Prentice-Hall, 2013

Chapter 9, Essentials of Cloud Computing, CRC Press, 2015

Chapter 6, Modeling Costs in "The Datacenter as a Computer", 2019

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Continuous Feedback

https://forms.cloud.microsoft/r/ZJxTf2wFjr

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NATIONAL UNIVERSITY OF SINGAPORE

CS5224 – Cloud Computing

(Semester 2: AY 2024/2025)

Time Allowed: 1.5 Hours

Sample Answers

INSTRUCTIONS TO STUDENTS

1. Please write your Student Number only in the answer sheet. **Do not** write your name.

2. This assessment paper contains **TWENTY FIVE (25)** questions and comprises **EIGHT (8)** printed pages (including this cover page).

3. Total marks in this question paper is **30**.

4. Students are required to answer **ALL** questions.

5. Write your answers in the space provided **ONLY** on the answer sheet.

6. This is an **OPEN BOOK** assessment.

7. Only printed/written materials are allowed. Electronic devices are not allowed.

Examiner's Use Only

Question No.	Maximum Marks
MCQs 1 – 20	20
21	2
22	2
23	2
24	2
25	2
Total	30

Question 1

Which one of the following would be the best use case for SaaS?

(a) Applications with highly confidential and sensitive data
(b) A mobile application
(c) An application used by an organisation with people working around the world, where some have erratic or unreliable internet connections
(d) Organisations looking to reduce on-premises IT infrastructure and capital expenditure
(e) Applications that need a lot of user data

Option (d). For the rest, SaaS is not a practical application to use.

Question 2

Which of the following statements is false?

(a) Containers and virtual machines have similar resource isolation and distribution advantages
(b) Elastic load balancing automatically distributes access traffic to multiple cloud servers, expands the external service capabilities of application systems and achieves higher levels of application fault tolerance
(c) Moving a SaaS deployment from one cloud provider to another is an easy proposition
(d) Hypervisors separate VMs logically and assigns each a share of the underlying computing power, memory, and storage
(e) The NIST definition of cloud computing includes five essential characteristics, four deployment models, and three service models

Option (c). It is not easy as there may be many elements of interoperability, not least with the infrastructure.

Question 3

Which level of cloud service is usually provided for the application to run the development environment?

(a) PaaS
(b) IaaS
(c) SaaS
(d) FaaS
(e) None of the above

Option (a).

Question 4

What cloud benefits make it lower risk for enterprises to adopt the cloud?

(a) The pay-as-you-go model allows enterprises to experiment with technologies as opposed to making long-term decisions based on little or no trial
(b) Data security associated with loss or unavailability of data causing business disruption
(c) The advantage of not managing infrastructure
(d) Standardized APIs and integration tools simplify interoperability, reducing complexity and adoption risks
(e) None of the above

Option (a). (b) and (d) are actually issues. (c) is only a benefit for some types of cloud deployments and not all.

Question 5

Is it possible to run completely different operating systems on virtual machines (VMs) hosted on a single physical host with an OS? If yes, what makes this possible?

(a) Yes, virtualization makes it possible—VMs can run different operating systems on a single host
(b) No, it is not possible – VMs are software-based versions of a single host. They cannot have different environments from one another
(c) Yes, containerization makes it possible – to have VMs that are unique physical entities, so you can have completely different operating systems on them
(d) No, it is not possible – A single host can only work as one single virtual environment, and can, therefore, have only one environment
(e) Yes, it is possible – this can be done through a Type 1 Hypervisor

Option (a). Containers share host OS, while a Type 1 hypervisor is directly on H/W

Question 6

A company which recently migrated to the cloud is noticing that its customers are facing slow data access times. What could be the main cause?

(a) Virtualisation
(b) Dedicated server allocation
(c) Elastic load balancing
(d) Automatic scaling
(e) Low Bandwidth

Option (e). The rest actually help alleviate slowness in some ways.

Question 7

Which of the following is an example of Software as a Service (SaaS)?

(a) Microsoft Azure
(b) Amazon EC2
(c) Google Cloud Platform
(d) Google Photos
(e) Amazon EMR

Option (d). Rest are all cloud provisions – either PaaS or IaaS.

Question 8

An organisation *O* (cloud service consumer), accesses a cloud service offered by public cloud *A*. Based on this, which of the following statements is false?

(a) *O* has its own organisational boundary
(b) *A* has its own organisational boundary
(c) *O* extends its trust boundary to encompass the organisational boundary of *A*
(d) *A* extends its trust boundary to encompass the organisational boundary of *O*
(e) One of the above statements is false

Option (d). Both *O* and *A* define their organisational boundaries, but cloud consumer must also extend its boundary to trust the cloud provider, and not the other way around.

Question 9

Which of the following scenarios is best suited for using a hybrid cloud model?

(a) A government agency hosting sensitive data
(b) A startup hosting a website
(c) A user hosting a blog
(d) A large enterprise with varying workloads
(e) A startup having a lot of data

Option (d). Hybrid clouds allow balancing between private and public clouds.

Question 10

Your company uses virtual machines (VMs) and is experiencing resource contention (e.g., CPU, memory, disk I/O, or network). Which solution is most effective to address this issue comprehensively?

(a) Add more physical servers
(b) Add more memory
(c) Use faster storage devices
(d) Increase your network speeds
(e) Optimise VM resource allocation

Option (e). Adding more physical servers may help but it depends more on how the virtualisation is.

Question 11

A company provides users with a SaaS to file their taxes. Close to the deadline, there is a huge spike in demand. Which strategy is most effective for ensuring performance and availability for this SaaS during the spike?

(a) Design a robust load balancing architecture
(b) Design a system for the highest probable peak load
(c) Upgrade network connection to deal with potential spike
(d) Lease additional hardware during peak period, as necessary
(e) Ask the users to wait in an online queue

Option (a). We can never estimate the real peak and leasing expensive hardware is not an option due to delays in provisioning. (c) and (e) are not really options.

Question 12

What is the advantage of a multitenancy cloud environment over a single tenancy environment?

(a) Easy to customize
(b) Cost savings
(c) Faster performance
(d) Higher data security
(e) Higher data governance

Option (b). Multiple users share the same infrastructure and resource, thus leading to better resource utilisation and hence cost savings.

Question 13

Your company has built a SaaS that is meant to serve around 5000 customers. However, your SaaS has become very popular (yay!) and suddenly there is a demand from about 25000 users. This will lead to an increase in resource utilisation. What are the incremental maintenance costs (mainly management overheads such as configuring servers, patching, etc.) of adding these new resources?

(a) About the same as before
(b) Twice the original cost
(c) Three times the original cost
(d) Four times the original cost
(e) Five times the original cost

Option (a). Not much extra maintenance since the extra resources come from the cloud, which is managed by the cloud provider. There are likely extra running costs.

Question 14

What is the advantage of a single tenancy cloud environment over a multitenancy environment?

(a) Lower costs due to shared resources
(b) Enhanced data security and isolation
(c) Faster scalability for user spikes
(d) Simplified resource management
(e) Reduced need for compliance controls

Option (b).

Question 15

What is one key advantage of using virtual machines over containers?

(a) Requires less memory space
(b) More secure
(c) Faster startup time
(d) Easy to scale
(e) More cost-effective

Option (b). Each VM is isolated, while containers can be impacted by OS being compromised.

Question 16

Which of the following statements about cloud computing is always true?

(a) It cannot be used in environments where data needs to be stored within a particular region/boundary
(b) It helps protect your data through encryption
(c) It enables users to access systems and resources regardless of location
(d) It introduces latency as servers are geographically distributed
(e) It can improve web server response time by having servers closer to the user

Option (c). For (a), you can use a local provider or a sovereign cloud. (b) only in certain cases based on the cloud provider and your settings. (d) is not correct, as latency is only an issue in some cases in this scenario. (e) may or may not help.

Question 17

Your company operates a multi-cloud environment (e.g., AWS and Azure) and is experiencing data synchronization issues between the clouds, such as incomplete transfers or data loss. What is the most likely cause?

(a) Slow storage performance on one cloud
(b) Limited virtualisation on one cloud
(c) Incompatible cloud APIs
(d) Uneven resource provisioning (e.g., storage, compute, or network)
(e) Inconsistent security configurations

Option (d). You could end up with less space for the data on one of the clouds.

Question 18

Out of the following five compute models, which one requires the most maintenance?

(a) PaaS and VMs
(b) FaaS and Containers
(c) PaaS and Containers
(d) Containers and VMs
(e) PaaS and FaaS

Option (d). PaaS is managed by the provider, while FaaS requires low maintenance.

Question 19

You wish to migrate to the cloud to run a simple server side computation which maintains no state about clients. Which of the following is best suited for this purpose?

(a) Amazon EC2
(b) REST
(c) SOAP
(d) AWS Lambda
(e) Amazon EMR

Option (d). Use FaaS – computations are simple, and no state is maintained.

Question 20

What is a primary challenge when migrating legacy on-premise systems to the cloud?

(a) Decreased performance
(b) Increased data compliance
(c) Increased management
(d) Increased costs
(e) Application compatibility

Option (e). (b) is not an issue depending on choosing the right cloud.

Question 21

You are a cloud developer who is deploying your application (which has a steady-state workload) to AWS. Which pricing option would you choose and why?

Answer: You will choose the reserved option as this is typically used to run 24/7 over the long term and the cost is lower than on-demand.

Question 22

The company you work for is responsible for a large web server farm. You are tasked with increasing the availability of this farm to ensure little to no downtime. List two ways in which you can achieve this.

Answer: You want to ensure high probability that your system is operational at a given time. You can use scaling (auto-scaling in particular and have redundancy, where your server instances are in multiple zones/locations.

Question 23

Your company is using a public cloud environment but is facing some security concerns regarding its data. As the lead cloud engineer, you are asked to come up with the solution, while keeping costs down. How would you do it? Explain your answer.

Answer: Moving to a hybrid cloud will help as sensitive data can be stored on a private cloud and less sensitive workloads can be performed. You can also move everything to a private cloud but that would be more expensive overall.

Question 24

The time it takes to back up your data on the cloud depends only on the amount of data you have. Is this statement true or false? Explain your answer

Answer: False. Performance = Function of (Server Location, Latency, Bandwidth, Data Amount).

Question 25

Your workplace wishes to make the work environment a bit more fun ☺. They are going to provide advanced collaboration services, such as video, chat, and an online whiteboard. Given the cost of hosting the infrastructure for this, they have decided that a cloud-based approach is the way to go. Which cloud computing deployment* model would best suit your company's needs? Explain your answer.

Answer: It depends! ☺. The main issue is data security and privacy and whether you require it or not. If it is not as required, you would go with a public cloud and if it were, a hybrid or private model would work fine.
*corrected during the test

– END OF PAPER –

NATIONAL UNIVERSITY OF SINGAPORE

CS5224 – Cloud Computing

(Semester 1: AY 2025/2026)

Time Allowed: 1.5 Hours

Sample Answers

INSTRUCTIONS TO STUDENTS

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3. Please **ALL** questions.
4. This is a **NON-SECURE BLOCK INTERNET** assessment.
5. This is an **OPEN BOOK** assessment. You are allowed to get reading materials as well as use study materials on your laptop.

Examiner's Use Only

Question No.	Maximum Marks
MCQs 1 – 20	20
21	2
22	2
23	2
24	2
25	2
Total	30

Question 1

What is a key benefit of using cloud-based databases compared to traditional on-premise non-cloud databases?

(a) Increased security
(b) Lower upfront costs
(c) Increased accessibility and scalability
(d) Cheaper storage
(e) Lower latency

Option (c). (a) and (d) are not guaranteed.

Question 2

You are the founder of a start-up with very limited personnel. Your company handles the input and processing of infectious disease clinical trial data on the cloud. The data processing needs to be real-time. The data contain highly sensitive, personally identifiable information. However, you want to reduce as much manual work as you can due to your small team. Which service should you choose for this application?

(a) IaaS
(b) FaaS
(c) CaaS
(d) PaaS
(e) SaaS

Option (d). SaaS gives the least amount of manual work, but the data processing cannot be real-time, and data privacy is also a concern. Therefore, PaaS is the best solution for minimizing work while meeting privacy and real-time requirements.

Question 3

Which one of the statements below about the number of workers in a MapReduce framework is correct?

(a) Too many workers can reduce overall efficiency because it overloads the master node's task and increases data transfer overhead between workers
(b) The number of workers has no effect on performance, because MapReduce automatically balances tasks perfectly
(c) Using fewer workers than available CPUs is always better, because it reduces network usage and eliminates the risk of data skew
(d) MapReduce performance depends only on the size of the input data, not on the number of workers assigned
(e) Having too few workers is not efficient because total worker idle time increases

Option (b) is incorrect as more workers usually reduces latency up to a certain point. (c) is incorrect because some CPUs will be idle and network is not the only overhead. (d) is incorrect because number of workers does determine latency. (e) is incorrect because

total worker idle time does not increase a worker's idle time in most cases. Fewer workers mean each will get more tasks, which decreases their idle time.

Question 4

You have an idea for an app that finds local festivity celebration parties. During the few festivities throughout the year (e.g. New Year, National Day, etc.) many users will use your app to find the most suitable party for them to go to based on users' criteria. Which AWS service is the most suitable to the usage profile and also cost effective?

(a) EC2: Virtual server to run your app
(b) S3: Store static app contents (party flyers, images) and serve directly to users
(c) Elastic Beanstalk: Managed EC2 with auto-scaling
(d) CloudFront: Cache contents and reduce latency for app users
(e) Lambda: App logic as a serverless function

Option (e). Traffic will be bursty during festive periods but not otherwise. Lambda does not incur continuous cost when app is idle, so it is cost-effective. EC2 and EBS incur continuous cost even when app is idle. CloudFront and S3 are only for static content delivery but does not handle app logic or compute (e.g. which one is the best party for me?)

Question 5

According to the MapReduce execution flow described in the slides, which statement most accurately describes the data flow and the division of responsibility between the developer and the system?

(a) The developer codes the Map function, which writes intermediate data to a shared storage server, and the Reduce function, which reads from that shared storage to aggregate results
(b) The developer codes the Map and Reduce functions, but must also explicitly implement the "Shuffle and Sort" logic to ensure intermediate data is correctly grouped for the reducers
(c) The Master instance partitions the input data, assigns segments to Map instances, and then directly sends the intermediate output from each Map instance to the correct Reduce instance
(d) The developer's Map function writes intermediate results to the local disk of its worker node; the MapReduce system is then responsible for shuffling, merging, and sorting this data before feeding it to the developer's Reduce function
(e) Map instances run in parallel, and Reduce instances run in parallel. To optimize performance, a Reduce worker is always placed on the same node as the Map worker that generated its data, eliminating network shuffling

Answer follows directly from bookwork

Question 6

Which one of the following statements most accurately distinguishes the architectural difference between Virtualization (using a Type 1 Hypervisor) and Multitenancy?

(a) Virtualization and Multitenancy are identical concepts, both referring to a single software instance serving multiple tenants
(b) Virtualization involves a hypervisor running on a Host OS, while Multitenancy involves a hypervisor running directly on the physical infrastructure
(c) Virtualization is an enabling technology that abstracts physical hardware (e.g., CPU, memory), allowing a single server to run multiple, isolated guest OSs. Multitenancy is a software architecture where a single instance of an application serves multiple tenants, keeping their data and configuration isolated
(d) Virtualization is a security risk due to Guest OS attacks, whereas Multitenancy is inherently secure because it always guarantees tenant isolation and data security
(e) Virtualization is used for IaaS, while Multitenancy is an architecture used exclusively for SaaS

Answer follows directly from bookwork

Question 7

Assuming you are moving from an on-premise deployment to a cloud deployment, which is the most important area of concern?

(a) Scalability
(b) Security
(c) Load Balancing
(d) Storage
(e) Bandwidth

Answer: Security. Rest are dependent on SLAs and what you can get, but by going to the Cloud, your data is no longer under your control to some extent

Question 8

You are running a cloud environment and realise you need a load balancer. What is its primary purpose?

(a) To ensure a high bandwidth between your application and the client
(b) To automatically adjust the necessary amount of computing resources that are used by your applications
(c) To distribute incoming service requests across multiple servers
(d) To ensure that data in the cloud is synchronised across multiple servers
(e) To manage the various computing resources within a datacenter

Option (c)

Question 9

A datacenter operator is analysing their energy consumption using the "Energy Usage and Power Efficiency" pie chart. They want to improve their Power Usage Efficiency (PUE), which is calculated as:

$$\left(\frac{\text{total power used by datacenter}}{\text{power used by IT equipment}}\right)$$

Total power used by datacenter includes both equipment and overhead power. To achieve an ideal PUE of 1, their efforts should focus on reducing the power consumption of which components?

(a) CPUs, DRAM, and Disks
(b) Power Overhead and Cooling Overhead
(c) Networking and Miscellaneous
(d) CPUs
(e) The Electrical Yard and Main Server Hall

Option (b). Options (a), (c), (d) are IT equipment, so reducing their power will reduce both the numerator and denominator, making it impossible to reach PUE of 1. (e) is wrong because main server hall contains CPUs, RAM, Disks, etc. too

Question 10

A cloud-native consulting firm offers a new product to its clients. This product integrates a SaaS-based Cloud Resource Management from Provider A, a PaaS-based database from Provider B, and a Storage-as-a-Service solution from Provider C. The firm combines these into a single, new, unified service with one dashboard and one bill, and it manages the secure data movement between the three providers. According to the NIST Cloud Computing Reference Architecture, which specific role and service is this firm fulfilling?

(a) Cloud Auditor, performing a Security Audit
(b) Cloud Carrier, providing transport of cloud services
(c) Cloud Broker (Service Intermediation)
(d) Cloud Broker (Service Arbitrage)
(e) Cloud Broker (Service Aggregation)

Option (e)

Question 11

An organization has deployed a redundant service (Service A) on three virtual servers (Server A, Server B, Server C). To manage load, all consumer requests are sent *directly* to Server A. Server A contains application logic that allows it to assess its own load and forward a portion of the incoming requests to Server B and Server C as needed. This differs from a model where an external device intercepts all traffic. Which specific architecture does this scenario illustrate?

(a) Independent Load Balancer
(b) Sibling Resource Pool
(c) Dynamic Vertical Scaling
(d) Built-In Load Balancer
(e) Cloud Bursting Architecture

Option (a) is incorrect because independent load balancer handles all incoming requests before it reaches server.

Question 12

You are part of an e-commerce start-up and wish to understand the shopping patterns arising from last year's sales events. In order to do this, you have been tasked with doing the data analysis where you have a lot of data. Which cloud deployment will you choose as the best option?

(a) Public cloud only
(b) Private cloud only
(c) Hybrid cloud
(d) Either Public or Private cloud
(e) Multi-cloud

Option (d) as either would be fine depending on your needs.

Question 13

Who amongst the following are primary users of PaaS?

(a) Clients looking to use a software online
(b) Software developers
(c) System administrators
(d) Hardware engineers
(e) Network engineers

Option (b)

Question 14

A company wants to move its deployment to the cloud but is worried about the security threats in the cloud. Which one of the following is the major security threat that the company needs to worry about?

(a) Unauthorized physical access
(b) Data breaches due to multi-tenancy
(c) Outdated antivirus software
(d) Limited user authentication methods
(e) Data leaks due to bad programming practices

Option (b)

Question 15

Which one of the following statements is NOT true about Cloud Computing?

(a) Opex includes expenses like electricity and maintenance for on-premise IT resources
(b) Detecting spam in social media is a good example of a use case for MapReduce
(c) PaaS is the cloud service that is usually provided to run the development environment
(d) Elastic scaling can dynamically increase or decrease server resources based on a specific period
(e) Higher security threats are a disadvantage of using public PaaS for SaaS development

Option (d). There is no specific time period as this depends on the load

Question 16

Your company has a policy of encrypting all data that is stored. Your data is hosted in a virtual private cloud and attached to it are multiple storage devices which has the data that must be encrypted. You have the following options available. Based on this, which one of the following options will allow you to do this? (if multiple options are possible choose the one that allows you the most choice)

(i) Ask clients to encrypt their data before they give it to you
(ii) Use third party encryption tools to encrypt the data
(iii) Use native encryption tools available in the Operating System to encrypt the data
(iv) Encrypt your data within the application before storing

(a) (i) or (iii)
(b) Only (iv)
(c) (ii) or (iii) or (iv)
(d) (i) or (ii) or (iii)
(e) (iii) or (iv)

Option (c). All of these are fine. Asking clients is risky in case they forget ☺

Question 17

In a cloud environment, which type of scaling adjusts the number of resources based on the load and traffic?

(a) Load scaling
(b) Vertical scaling
(c) Horizontal scaling
(d) Network scaling
(e) Elastic scaling

Option (e).

Question 18

You realise that using containerised applications in a datacenter environment gives you certain advantages. Which one of the following could be one of them?

(a) Increases the physical size of the server infrastructure
(b) Eliminates the need for cloud-based applications
(c) Improves isolation between applications and operating system
(d) Reduces the number of physical servers needed
(e) Reduces the overall cost

Option (c).

Question 19

When expecting a huge variation in traffic, as a cloud engineer, we need to be able to ensure the application can scale dynamically based on demand. Which cloud computing service model would we use to implement scalability?

(a) Elastic load balancer
(b) SaaS
(c) Auto-scaling
(d) PaaS
(e) None of the above

Option (d) as it provides scaling. (a), (c) are not service models.

Question 20

Which of the following best distinguishes Dynamic Scalability Architecture from Elastic Resource Capacity Architecture?

(a) Dynamic Scalability relies on manual scaling operations, whereas Elastic Resource Capacity is fully automated and static
(b) Dynamic Scalability only involves vertical scaling, while Elastic Resource Capacity only involves horizontal scaling
(c) Dynamic Scalability focuses on scaling within a single resource pool, whereas Elastic Resource Capacity focuses solely on scaling between sibling pools
(d) Dynamic Scalability requires an Automated Scaling Listener, while Elastic Resource Capacity does not use any monitoring component
(e) Dynamic Scalability adjusts resources based on predefined conditions, while Elastic Resource Capacity uses intelligent automation to allocate or reclaim resources in real time

Option (e) – from the notes

Question 21

AWS provides instances that are optimized for different workloads. Some instances are memory-optimized, others are I/O optimized. You are assigned a project to be run on AWS. The project requires 60 Peta FLOPs and 800,000 I/O operations to complete. Assume that compute and I/O operation are sequential. The details of each instance are provided below. Which instance should you choose to optimize your project cost? Please show your working.

Tera = 10^{12}. Peta = 10^{15}. FLOP = Floating Point Operations. ms = 10^{-3} seconds.

Name	Computation Speed (TeraFLOP/s)	I/O Latency (ms/IOops)	Price ($/sec)
m7 (General purpose)	1	3	1
i8 (I/O optimized)	1	1	2
c5a (Compute optimized)	3	2	2
r7 (Memory optimized)	1	3	2
g4 (GPU)	30	10	5

Answer:

Name	TeraFLOP/s	I/O Latency (ms)	Price ($/sec)	N_ops (TeraFLOP)	N_I/O_ops	Total Time (s)	Price ($)
m7	1	3	1	60000	800000	62400	62400
i8	1	1	2	60000	800000	60800	121600
c5a	**3**	**2**	**2**	**60000**	**800000**	**21600**	**43200**
r7	1	3	2	60000	800000	62400	124800
g4	30	10	5	60000	800000	10000	50000

Question 22

As part of an annual review in your company, you realise you have been with your cloud provider for many years and start to worry about the potential risks of 'vendor lock-in'. List **two** methods in which you could help mitigate this risk. Briefly (in one or two sentences) explain your answer.

Answer: less use of proprietary technologies and hence potentially easier to migrate, spread deployment across multiple providers, using technologies such as Docker to make it easier to move workloads across different environments, etc.

Question 23

It is the upcoming holiday season and your application has gained a lot of traction – yay! However, the downside is that your cloud infrastructure is struggling with this load. You have looked into auto-scaling and load-balancing already. List **two** other methods that you can use to help with this issue.

Answer: Use caching, CDNs, optimise database for faster query processing, etc.

Question 24

You are responsible for the cloud infrastructure in company WeAreAwesome Inc. As awesome as you are, you have however noticed that your VMs that are running on the cloud are costing more than they should. Given your understanding of Cloud Computing, you wish to optimise the cost of running the VMs. Give **two** ways in which you can achieve this.

Answer: Try to eliminate over-provisioning, use reserved instances for predictable workloads, or spot instances for non-critical workloads , can do auto-scaling to scale-down as necessary

Question 25

While setting up a new cloud deployment, you are tasked with ensuring that data availability must be high and that you must also comply with data protection guidelines which mandate that you must store Singapore customer data only within Singapore. How can you ensure compliance in such a scenario while keeping costs down? Briefly explain your answer.

Answer: Use a sovereign cloud or select specific regions for data storage and ensure that data replication occurs only within those regions.

– END OF PAPER –

NATIONAL UNIVERSITY OF SINGAPORE

CS5224 – Cloud Computing

(Semester 2: AY 2025/2026)

Time Allowed: 1.5 Hours

Sample Answers

INSTRUCTIONS TO STUDENTS

1. This assessment paper contains **TWENTY FIVE (25)** questions.
2. Total marks in this question paper is **30**.
3. Please answer **ALL** questions.
4. This is a **NON-SECURE BLOCK INTERNET** assessment.
5. This is an **OPEN BOOK** assessment. You are allowed to get reading materials as well as use study materials on your laptop.
6. You are not allowed to use any AI tools in this assessment and this includes the use of any locally installed Large Language Models (LLMs).

Examiner's Use Only

Question No.	Maximum Marks
MCQs 1 – 20	20
21	2
22	2
23	2
24	2
25	2
Total	30

Question 1

How does an Application Load Balancer enable reliability?

(a) By routing traffic to the least busy instances
(b) By routing traffic away from failed instances
(c) By caching frequently accessed content
(d) By replacing failed instances
(e) By routing traffic to new instances

Option (b).

Question 2

You wish to build your own application and are operating in a highly competitive market where being first is more important than being perfect, and rapid iteration is critical. Which cloud service model is most appropriate?

(a) IaaS
(b) FaaS
(c) CaaS
(d) PaaS
(e) SaaS

Option (d). PaaS enables the fastest building and deployment. IaaS takes time while SaaS is too restrictive since you wish to build your own application here.

Question 3

Which one of the following statements is correct regarding virtualization?

(a) Virtualization increases compatibility among hardware that use different standards
(b) Operating System virtualization is better suited for an application which requires direct communication with hardware resources
(c) In hardware-assisted virtualization, the guest operating system directly communicates with hardware resources
(d) Virtualization technology allows the isolation of applications running on the same server
(e) Virtualization allows multitenancy only when it is Type 2

Option (d) – comes directly from the definition in some ways

Question 4

A company wants to use MySQL as the database for its web application while selecting the PaaS cloud service model. Which of the following is the most appropriate?

(a) The company can use AWS RDS (Relational Database Service) and configure a MySQL database
(b) The company needs to install a MySQL server
(c) The company should use MongoDB instead of a MySQL server
(d) The company should use the LAMP stack, which contains Linux, Apache web server, MySQL, and Python
(e) It is not possible to use MySQL with PaaS

Option (a). For a PaaS, you configure MySQL. LAMP is an IaaS stack.

Question 5

A public-sector platform must keep both data and metadata within one country's borders and prevent foreign access under all circumstances. Which cloud deployment model is the best fit?

(a) Public Cloud
(b) Private Cloud
(c) Hybrid Cloud
(d) Sovereign Cloud
(e) Multi Cloud

Option (d). Answer follows directly from bookwork

Question 6

A retailer normally runs their infrastructure on-premise but wants excess holiday traffic to spill over into pre-deployed cloud resources only when local thresholds are exceeded, then return to on-premise when the load drops. Which architecture best matches this?

(a) Resource pooling architecture`
(b) Service load balancing architecture
(c) Cloud bursting architecture
(d) Elastic resource capacity architecture
(e) Using horizontal or vertical scaling architecture

Option (c). Answer follows directly from bookwork

Question 7

Which requirement is the clearest sign that a workload is a poor fit for FaaS?

(a) It is triggered by file uploads
(b) It requires long-lived stateful sessions and persistent in-process connections
(c) It performs short request-driven transformations
(d) It runs sporadically and scales by event volume
(e) It requires long-lived stateless sessions

Option (b). Rest all work fine for FaaS

Question 8

A company has already made significant investments in on-premise infrastructure. Migrating to the cloud would require substantial additional costs for integration. Which of the following is the most appropriate conclusion based on this scenario?

(a) Cloud is always cheaper because it reduces upfront capital expenditure
(b) Reserved cloud pricing eliminates migration and integration costs
(c) On-premise solutions always have lower operational costs than cloud
(d) Past investments should not be considered when evaluating future decisions
(e) Cloud adoption may be less attractive due to high migration and integration costs

Option (e). (a) ignores sunk costs – may be cheaper later, but no guarantee. (b) and (c) are not really options. (d) can't ignore sunk costs

Question 9

"Great power and control come with greater responsibility."

Which cloud service model best reflects this statement?

(a) SaaS
(b) FaaS
(c) CaaS
(d) IaaS
(e) PaaS

Option (d). (IaaS) provides the highest level of control over computing resources (VMs, storage, networking). However, this also means the user is responsible for managing the OS, updates, scaling, and security.

Question 10

Which of the following features is the **primary distinguishing factor** between Virtual Machines (VMs) and Docker containers?

(a) Isolation
(b) Security
(c) Hardware virtualization
(d) Command-line Interfaces
(e) Separate namespaces

Option (e). Virtual Machines rely on hardware virtualization, which introduces additional overhead. Docker containers, on the other hand, share the host OS kernel and avoid this overhead, making them lighter weight.

Question 11

You are tasked with designing a datacenter. Your manager asks you to prioritize components based on increasing energy consumption (from lowest to highest).

Which of the following orders is correct?

(a) Networking → Disk → DRAM
(b) CPU → Disk → Networking
(c) Disk → DRAM → Cooling
(d) Cooling → Networking → DRAM
(e) Disk → Cooling → CPU

Option (e). Based on the lecture's energy breakdown, disks consume the least energy among the listed components, followed by cooling infrastructure, while CPUs typically account for the highest energy consumption.

Question 12

Which of the following best describes a serverless application?

(a) A serverless application is usually developed in the context of PaaS
(b) The developer of a serverless application needs to pay a small fee for the idle periods
(c) A serverless application does not need a server/host to run on
(d) The developer of a serverless application does not need to provision or manage any servers
(e) The developer of a serverless application has lower costs

Option (d).

Question 13

A company manages development, staging, and production environments using Terraform. Which one of the following approaches best avoids duplication and ensures consistency?

(a) Copy-paste configurations per environment
(b) Use separate state files
(c) Use only one environment
(d) Hardcode environment values
(e) Keep separate code repositories

Option (b). Copy-paste can lead to issues while having one environment is prone to error due to its 'monolithic' nature. Hardcoding isn't an option and keeping separate code repositories sometimes makes it harder to manage.

Question 14

Your company has recently gotten to grips with Terraform and wish to use it. However, they also wish to have automatic drift detection during continuous integration (the practice of merging code changes into a central repository frequently). Which of the following is the best approach to achieve this?

(a) Run `terraform apply` daily
(b) Disable drift checks
(c) Use manual audits
(d) Periodically recreate the infrastructure
(e) Run `terraform plan` and provide an alert on changes

Option (e)

Question 15

Which of the following are signs of a highly available application?

(i) Applications are protected behind multiple layers of security
(ii) Spikes in user demand are met through automatically increasing resources
(iii) A failure in one geographic region will trigger an automatic switch to resources in a different region
(iv) Virtualized hypervisor-driven systems are deployed as mandated by company policy
(v) Application developers ensure the necessary software updates are applied

(a) (i) and (iii)
(b) (ii) and (iv)
(c) Only (v)
(d) (ii) and (iii)
(e) (i) and (v)

Option (d). Security and keeping software updated doesn't directly translate into availability of the application.

Question 16

What is the primary challenge when migrating legacy systems to the cloud?

(a) Increased cost
(b) Decreased oversight
(c) Application compatibility
(d) Decreased performance
(e) Increased performance

Option (c). Cloud environment and set-up may be different

Question 17

Your users are reporting increased levels of latency. On checking you notice that CPU usage is actually low. What may be the primary cause then?

(a) Servers are overprovisioned
(b) Too many instances have been set up
(c) Horizontal scaling
(d) Network bottleneck
(e) Usage of CDNs

Option (d). Most probably cause for increase in latency.

Question 18

A company, GreatestCloudOfAll Inc, have noticed that their cloud infrastructure frequently scales up and down, which is causing great instability and hence extra costs. You have been called in as an expert to advise them. What would your advice be?

(a) You just need to increase your number of instances and it'll be fine
(b) You can re-check your thresholds for scaling and also add a cooldown period which prevents further scaling during this period
(c) You can monitor it and over time just disable scaling as that will help in the long run in this case
(d) Best is manual scaling, nothing beats it!
(e) Reduce the overall cost by reducing your number of instances

Option (b). Rest are really not practical. By having a cooldown period, you provide time for resources to adjust before making any changes, as needed.

Question 19

A company wishes to migrate its legacy application to the cloud quickly with minimal changes. Which approach from the following would be most suited (you can assume that all options listed are technically feasible)?

(a) Refactor the application to use serverless functions and then deploy it
(b) Get a cloud provider to build a SaaS equivalent solution for you and then use it
(c) Rewrite the application using a PaaS platform which allows for easy deployment
(d) Use IaaS and move application as is from on-premise to cloud
(e) Keep it on-premise

Option (d). (b) may not be a choice depending on your application. Rest all take much more time and effort. (e) is not really an option.

Question 20

Which one of the following is an optimal strategy to employ when you have a very large number of reads and write operations to a scalable cloud database?

(a) Horizontal scaling
(b) Vertical scaling
(c) Increasing storage
(d) Cloud bursting
(e) Migrate to a faster database

Option (a). Horizontal scaling allows you to distribute these reads and writes across multiple nodes.

Question 21

Assume that you are responsible for organising your cloud resources globally as the leader of an enterprise. Discuss at least **two** key factors that you should consider when deciding on cloud resource hosting.

Answer: Availability, Cost of cloud based on region, Location for latency, Location for legal requirements. This questions is about hosting cloud resources globally and what to look for, and less about models (whether it be deployment or service).

Question 22

Design a cloud strategy for a healthcare analytics SaaS used by multiple hospitals. Patient data must be tightly governed, demand can spike during outbreaks, and the system must still scale nationally. What deployment model(s) and scaling architecture(s) would you choose, and why?

Answer: can use hybrid cloud or private cloud. Can use cloud bursting, or controlled dynamic scaling – scaling should be up and down.

Question 23

You are responsible for the cloud infrastructure in company TheGreatCloud.sg. You have been told that the new system you are designing will be holding sensitive data which must comply with data residency laws across multiple countries. How will you design your data storage to be able to manage this cross-border access while being compliant to the data residency laws? Give **two** ways in which you can achieve this.

Answer: Ensure data in for a region stays within that region and route users to region-specific services, use access controls and encryption, as necessary. Can also add any data auditing, as necessary. Data storage will be fragmented but it has to be compliant. A sovereign cloud is one option but you also need to make sure requests are routed accordingly.

Question 24

A bank wants to move away from its legacy application and modernise it with strict compliance and some sensitive data. Which cloud service and cloud deployment model would you use and why?

Answer: Hybrid-cloud: sensitive workloads on private cloud, and non-sensitive on public cloud. Build your own through PaaS for the modern services while any legacy infrastructure could be on IaaS. Key here is that not all data is sensitive and also that you need to update your application, so build a new one, which implies PaaS.

Question 25

You are a Solutions Architect at a leading technology company. The IT division of the metropolitan MRT (Mass Rapid Transit) system has approached you to design a next-generation payment platform to support SimplyGo and phase out the legacy EZ-Link card system. A key concern is that during peak hours, the current system experiences high latency, resulting in long queues and poor commuter experience. The reason for this is that for each user, the system performs fare calculation, balance deduction, and displays the update immediately before letting the user pass through the gate.

Your task is to propose **two** ways in which this payment platform can be scalable and also have low-latency.

Answer: Primary issue is the sequence of operations before letting a user through. You can do some of these later – just do what is necessary to let user through and other operations can be performed later. Doing these operations async is key and here FaaS or any other similar element can help. As before, scalability must be elastic which allows system to scale up/down depending on user numbers.

– END OF PAPER –